
Living in the IT Era

MST 101d Lecture Notes

Rey Joseph Nieto

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Preface

These lecture notes accompany *Living in the IT Era*, taught to undergraduate biology students at Bulacan State University's San Rafael Campus. I am not trying to turn you into computer scientists or make you memorise catalogues of devices, inventors, and fashionable technologies. I want these notes to help you understand the digital systems that shape ordinary life, use those systems effectively, evaluate information critically, protect yourself and others, and recognise when specialist technical knowledge is needed.

The full course is organised into eight chapters. This edition contains the Introduction and Chapters 1–8. Each chapter is built around a central question and a practical task. Examples are drawn whenever possible from biology, health, research, and Philippine digital life.

The scope follows the general-education framework for *Living in the IT Era* in CHED Memorandum Order No. 20, series of 2013 [1]. The order places the course within the Mathematics, Science, and Technology elective domain and emphasises the science, culture, and ethics of information technology; its influence on modern life and relationships; and its use for personal, professional, and social advancement.

Reports of errors and suggestions are welcome at

`rj (at) rjnieto (dot) com`

I expect these notes to change as you use them. They inevitably reflect what I know, what I can readily bring to mind, and what I expect a class will need. You may identify a point that needs a better explanation, ask for a practical topic that I had not considered, or show that an apparently minor issue deserves more attention. I welcome those requests when they would make the course more useful.

That flexibility is deliberate. I want the module to respond both to the digital problems you face now and to those you are likely to meet later. I hope you finish the course better prepared to understand new technologies, make sensible decisions about them, and meet the IT challenges ahead.

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Introduction

Most of you already use information technology throughout the day. A phone serves as an alarm, camera, map, notebook, bank, library, entertainment centre, and means of communication. University work may involve search engines, learning platforms, shared documents, spreadsheets, online forms, biological databases, and AI assistants. Familiarity, however, is not the same as understanding. A person can use a system every day without knowing what it does with their information, why it sometimes fails, or what risks accompany its convenience.

I do not expect you to become computer technicians. I want you to know enough to make better decisions: choosing a device or internet service, improving a home network, recognising a scam, protecting an identity, behaving responsibly and civilly online, understanding online tracking, using digital tools in scientific work, and judging claims about AI and other emerging technologies.

The organising idea is information. The usual definition of information technology can sound unnecessarily dry, but the central idea is simple: there is information, and IT is concerned with handling that information electronically. For this course, we shall define information technology as the organised use of computers, networks, and software to create, process, store, communicate, and use electronic data. This working definition is adapted from the University of Cincinnati's overview of the field [2].

I use the term broadly here. IT includes the hardware on which data are handled, the software that gives instructions to that hardware, and the networks that allow devices to exchange data. It also includes data management, cybersecurity, cloud computing, and newer applications such as artificial intelligence and machine learning [2]. These areas overlap. An online laboratory record, for example, may involve a phone or computer, a browser, a wireless network, a remote database, account permissions, and automated analysis.

IT matters because it changes what people can do with information. It can make communication faster, automate repetitive work, organise large collections of records, support decisions, and protect information from unauthorised access. In biology, these functions appear in field observations, laboratory instruments, sequence databases, statistical analysis, medical records, and collaboration between researchers. None of these systems removes the need for human judgement.

A computer can process a large data set quickly without knowing whether the measurements were sound or whether the conclusion is justified.

Data are recorded representations such as numbers, text, images, audio, and measurements. They become informative when they are interpreted in context. For example, 37.8 is only a recorded number until we learn that it is a patient's temperature in degrees Celsius. Context allows the value to answer a question or support a decision.

Most digital activities can be understood through five actions. A system receives **input**, performs some form of **processing**, may retain data in **storage**, produces **output**, and often **communicates** with another device. These actions need not occur in one machine. A student may enter information on a phone while the processing and storage take place in a distant data centre.

The Course at a Glance

The course follows a route rather than a catalogue. Chapter 1 gives us a working map of devices, connections, and software. Chapter 2 goes beneath the screen to the physical and industrial systems on which that map depends. Chapters 3–4 then show how infrastructure becomes communication and useful digital work. Chapter 5 follows information into platforms and examines how those platforms affect attention, conduct, and relationships. Chapter 6 brings together security threats, identity safeguards, privacy, and legal responsibilities. Chapters 7–8 use the whole map to examine artificial intelligence and other emerging technologies.

Table 1: The eight-chapter course map.

Chapter	Question	How the course develops
1	How IT Works and Why It Matters What parts make up an IT system, and how should we choose among devices, connections, and software?	Establishes our working map: information flow, computer components, specification sheets, internet services, and the purposes, costs, licences, privacy, and security of software.
2	The Physical Foundations of Computing What material and industrial systems make digital activity possible, and how do chokepoints create power?	Goes beneath the map: electrical states, transistors, chips, infrastructure, energy, workers, and supply chains. It then uses Chinese and US leverage to ask what technological interdependence means for the Philippines.
3	The Internet, Web, and Home Networks How does information travel between devices and services?	Takes up the connection distinctions in Chapter 1 and the physical route introduced in Chapter 2. We distinguish the internet from the Web, follow a website request, interpret common browser messages, and diagnose provider, home-network, Wi-Fi, mesh, and VPN problems.

Continued on the next page

Chapter	Question	How the course develops
4	Software, Cloud Services, and Digital Work How do software and remote services turn infrastructure into useful work?	Returns to the software choices introduced in Chapter 1, now in the setting of connected systems: operating systems, applications, files, cloud storage, collaboration, reproducibility, copyright, and the control exchanged for convenience.
5	Digital Platforms, Information, and Online Life How do platforms shape what we see, what they learn about us, and how we interact with other people?	Follows information into cookies, digital footprints, tracking, advertising, search, and recommendation systems. A case involving an online health claim connects source evaluation and synthetic media with persuasive design, attention, social comparison, disinhibition, harassment, and civil conduct online.
6	Cybersecurity, Identity, and Privacy How do attackers exploit connected people and systems, and how should individuals and institutions protect identity and data?	Uses an account-takeover case to connect phishing, malicious links and downloads, scams, malware, identity theft, and social engineering with passwords, multi-factor authentication, one-time passwords, passwordless methods, and biometrics. It then distinguishes cybersecurity from privacy and examines responsible data collection, Philippine cybercrime law, and the Data Privacy Act.
7	Artificial Intelligence: How It Works and Fails Why can an AI system produce a fluent, confident, and helpful-looking answer while being wrong?	Uses the earlier chapters on hardware, data, networks, software, and platforms to explain machine learning and generative AI, including training, prediction, hallucination, cascading errors, bias, uncertainty, and sycophancy.
8	Responsible AI and the Future of Computing When should AI be used, restricted, or rejected, and which claims about future computing justify action now rather than later?	Brings the course together through a research-workflow case involving verification, privacy, copyright, academic integrity, meaningful human review, monitoring, and appeal. A short post-quantum case distinguishes prediction from prudent migration.

You are not expected to become a specialist in every field listed here. You should learn enough to ask better questions, recognise when a claim requires checking, and know when expert help is needed.

How Classes Will Work. I will not simply repeat these notes in class. Read the assigned pages before coming; I will announce them in advance. We shall use a **flipped classroom** approach: you first encounter the material through the assigned reading, and then we use class time for a bird's-eye view, questions, discussion, examples, and practical applications. The reading is part of the course, not optional preparation for it.

Chapter 1

How IT Works and Why It Matters

Learning Outcomes

By the end of this chapter, you should be able to:

1. identify the basic components of a computer and explain their roles;
2. interpret selected laptop and mobile-phone specifications without assuming that larger numbers are always better;
3. distinguish common internet connection types and interpret download rate, upload rate, latency, jitter, and packet loss;
4. classify software by purpose, delivery, payment, and licence model; and
5. select hardware, an internet service, or software using criteria suited to the user's needs rather than marketing claims alone.

Do not try to memorise every entry in the tables. I want you to learn how to read them when you actually need to compare a device, connection, or application. We will examine a few examples closely in class; you can apply the same questions to the devices and services you encounter.

1.1 Hardware Basics

A computer is a collection of components with different jobs. We begin with a basic table because you need to recognise each component's role, not memorise model numbers or become a repair technician.

Component	Main role	Plain-language description
Keyboard, mouse, camera, and microphone	Input	Allow a user or the environment to supply data to the computer.
CPU	General processing	Executes instructions and coordinates much of the computer's work. The CPU is a chip inside the computer, not the entire case.
GPU	Highly parallel processing	Produces graphics and performs many similar calculations at once; this is also useful in scientific computing and AI.
RAM	Working memory	Holds data and instructions that are being used. Its contents are normally lost when power is removed.
SSD (solid-state drive) or hard disk	Long-term storage	Retains the operating system, programs, documents, and other data when the computer is switched off.
Monitor and speakers	Output	Present visual or audio results to the user.
Ethernet and USB ports	Wired connection	Connect the computer to a network, storage device, peripheral, or other equipment.
Wi-Fi and Bluetooth	Wireless connection	Exchange data with networks or nearby devices without a cable.

The components work together. Pressing a key creates input; software assigns meaning to it; the CPU processes the instruction using data held in RAM; the result may appear on the monitor, be saved to storage, or be transmitted over a network. A more powerful component is useful only when the rest of the system and the intended task can take advantage of it.

Figure 1.1 shows why a computer should be understood as a system rather than as a single “processor.” The motherboard connects the CPU, memory, storage interfaces, expansion slots, power, cooling, and external ports; a real machine also contains cables and mechanical structures that simplified specification tables omit.

1.2 Reading Specifications

I chose two expensive flagship devices because their specification sheets contain many of the terms that buyers encounter. I am not suggesting that you buy either of them. The same method applies to a modest laptop or phone: begin with the work it must do, then ask which figures matter for that work and which are mainly selling points.

1.2.1 Laptop Specifications

A laptop specification sheet compresses a great deal of information into a small space. Before interpreting the terms, let us organise one actual example. Table 1.1 summarises specifications reported by EveryMac and Apple for a 2026 16-inch MacBook Pro [3, 4].



Figure 1.1: The interior of a desktop computer. Visible parts include the motherboard, processor cooling fan, memory slots, storage bays, power and data cables, and external connectors.

Image: Kurt Kaiser, via [Wikimedia Commons](#); CC0 1.0. Reproduced without content changes; scaled to fit.

Table 1.1: Selected laptop specifications reported by EveryMac and Apple.

Specification	Reported value
Processor and graphics	Multi-core CPU and integrated GPU; core counts require comparable tests before they can be used to judge performance.
Memory and storage	Unified RAM and a built-in SSD; both capacities must be chosen at purchase because neither is later upgradeable.
Display	A high-resolution display with an adaptive refresh rate; these features affect particular uses, cost, and power consumption differently.
Connections	USB-C-shaped ports with specified Thunderbolt capabilities, HDMI, wireless networking, and no built-in Ethernet port.
Battery and portability	Manufacturer claims stated as “up to” a given number of hours, together with the machine’s dimensions and mass.
Operating system	macOS; required applications and laboratory equipment must be checked for compatibility.

The model name tells us which machine is being described, but identifiers such as A3429 and Mac17,6 are often more reliable when checking compatibility, replacement parts, or support. Two laptops sold under similar names may belong to different generations. The date and exact identifier therefore matter.

Processor, graphics, and memory. A CPU's core count and clock speed do not by themselves establish that it is faster than another CPU. The processor generation, design, cooling, and task also matter. A **benchmark** is a standardised performance test. Its results can help, but only when the same benchmark and version are used under comparable conditions. A single-core score reflects work that depends mainly on one core; a multi-core score reflects work that can be divided among several cores. GPU core counts are likewise poor for comparisons across different designs or manufacturers. They become meaningful only with suitable tests of the work one actually intends to perform. The Neural Engine is Apple's name for hardware designed to perform certain machine-learning calculations.

RAM is the computer's short-term working space. More RAM allows more programs and larger data sets to remain readily available, but unused capacity does not make every task faster. In this machine the CPU and GPU share unified memory. The important purchasing detail is that the memory is built into the machine: its capacity must be chosen at purchase because it cannot be expanded later.

Storage and upgradeability. The 2 TB figure describes how much the built-in SSD can store, not its speed or the amount of RAM. Photographs, videos, large data sets, and applications consume storage over time. This laptop's internal storage cannot be upgraded after purchase, although external drives and network or cloud storage remain possible. A buyer should distinguish among the standard configuration, an option available when ordering, and a component that can be upgraded later; these are not the same thing.

Display and connections. Screen size is measured diagonally. Resolution gives the number of pixels, pixel density describes how closely they are packed, and refresh rate gives the maximum number of times the display can update each second. Higher values can improve particular uses, but they may also increase cost or power consumption. The software being used must be able to benefit from them.

Port descriptions require careful reading. USB-C describes the connector's shape, whereas Thunderbolt 5 describes capabilities such as data transfer, display output, and charging. Devices with the same connector need not support the same functions or speeds. The absence of built-in Ethernet does not prevent a wired network connection, but it requires a compatible adaptor or dock. Likewise, obtaining the benefits of Wi-Fi 7 requires a compatible network, not merely a compatible laptop.

Battery, portability, and software. The words "up to" mark a best-case claim measured under stated test conditions. Screen brightness, wireless activity, software, temperature, and workload can make actual battery life shorter. Mass and dimensions matter when a laptop is carried daily,

while the operating system matters because required applications and laboratory equipment may support only particular platforms. A powerful computer that cannot run the required software is the wrong computer for the task.

From specifications to a sensible choice. Begin with the work the computer must perform, the software and connections it requires, the need for portability, and the available budget. Then compare machines using consistent definitions and suitable tests. Specifications cannot fully describe keyboard comfort, fan noise, repairability, reliability, warranty service, or real-world battery life, so independent reviews and first-hand inspection remain useful. The best choice is not necessarily the machine with the largest numbers; it is the one that meets the user's needs at an acceptable total cost.

A specification table remains a collection of claims. Important details should be checked against the manufacturer's information for the exact model and, where possible, an independent test of the intended work.

1.2.2 Phone Specifications

Mobile-phone specification sheets use many of the same terms as laptop sheets, but add cellular networks, SIMs, cameras, positioning, sensors, and resistance to water and dust. Table 1.2 organises selected specifications reported by GSMArena and Apple for the Apple iPhone 17 Pro Max [5, 6, 7].

Table 1.2: Selected mobile-phone specifications reported by GSMArena and Apple.

Specification	Reported value
Regional versions	The same product name can cover models with different SIM arrangements, battery capacities, network support, and warranty terms.
Processor, memory, and storage	A multi-core processor, built-in RAM, and storage that must be chosen at purchase because there is no memory-card slot.
Display and cameras	Resolution, refresh rate, brightness, several cameras, and different optical fields of view; megapixels alone do not determine image quality.
Networks and positioning	Mobile-network support, Wi-Fi, Bluetooth, NFC, GPS, and other positioning systems whose availability depends on the exact model and local service.
Battery, charging, and resistance	Capacity and test results that vary by model and conditions, conditional fast-charging claims, and a laboratory water- and dust-resistance rating.

Start with the exact version. The product name is not enough. The model number identifies the regional version, which can affect the available SIM arrangement, battery capacity, network compatibility, and warranty service. A phone imported from another market may therefore differ from the version sold locally even when both carry the same product name. Before buying, check

the exact model against the mobile operator's supported frequency bands and the manufacturer's local support terms. A label such as "5G" does not guarantee 5G service everywhere: the phone, network bands, operator, subscription, and local coverage must all be compatible. A Nano-SIM is a small physical card, whereas an eSIM is a digital SIM profile stored by the phone.

Processor, RAM, and storage. Interpret these as you would for a laptop: compare like tests and the intended applications, distinguish working memory from retained storage, and check whether capacity can be expanded. Cloud storage does not increase the phone's built-in storage.

Display and cameras. Display size is measured diagonally, while resolution, pixel density, refresh rate, brightness, colour reproduction, and power use describe different properties. The sheet lists both advertised brightness levels and a separately measured result. Such figures should be compared only when the testing conditions and definitions match.

A camera's megapixel count describes the number of recorded pixels, not the overall quality of every photograph. Sensor size, lens, aperture, optical image stabilisation, focusing, image processing, lighting, and the photographer all matter. Multiple rear cameras provide different fields of view or optical capabilities; three 48-megapixel cameras should not be read as a single 144-megapixel camera. Optical zoom uses the camera's optics, whereas digital zoom enlarges or crops the recorded image and may lose detail.

Battery, charging, and durability. Battery capacity alone cannot predict running time across different phones. Controlled use tests are more informative, while fast-charging claims depend on the charger, cable, temperature, battery state, and protocol.

An IP68 rating indicates performance under specified laboratory conditions; it does not mean that the phone is impossible to damage with liquid. Wear, impact, repairs, depth, time, temperature, pressure, and the type of liquid can affect resistance. Durability, repairability, spare-parts availability, and warranty conditions deserve attention alongside processor and camera figures.

Software, privacy, and cost. An operating-system version on a specification sheet is a snapshot, not a promise of support for a particular number of years. Check the manufacturer's current update policy and whether essential applications will run on the device. Face ID is convenient, but biometric authentication also raises questions about how biometric data are stored and what fallback method is used; these questions return in our study of privacy and cybersecurity.

Compare total cost over the expected period of use, including storage, accessories, repairs, and services. The best phone is the one that meets the requirements reliably, not the one with the longest specification sheet.

For your work in biology, a phone may also be a field notebook, camera, location recorder, timer, reference tool, or means of entering observations. Suitability for scientific work depends not only on the hardware but also on measurement accuracy, consistent methods, file formats, data export, privacy, and whether the work can be checked or repeated.

1.3 Internet Connections

A useful device still needs a suitable connection. The internet reaches homes and other premises through several technologies. The connection type affects the speeds that are possible, the consistency of those speeds, latency, reliability, installation, and cost. It also determines which services can feasibly be offered at a particular address. The table below draws on the overview by David Anders and Sean Jackson but omits its provider-specific and United States market information [8].

Connection	How data reach the premises	Main characteristics
Fibre-optic	Pulses of light travel through thin strands of glass or plastic.	Can provide very high and often similar download and upload speeds. Building the required network is costly, so service is not available everywhere.
Cable	Electrical signals travel through coaxial cable of the kind also used for cable television.	Can provide high download speeds, but upload speeds are often lower. Capacity may be shared locally, so performance can fall during busy periods.
DSL	Electrical signals travel through copper telephone lines.	Makes use of existing telephone infrastructure and, unlike dial-up, allows telephone voice service to be used at the same time. Performance depends strongly on line quality and distance from network equipment and is generally more limited than fibre or cable.
Fixed wireless	Radio signals travel between a base station and a receiver at a fixed location.	Can serve places where laying a cable is impractical. Distance, obstructions, alignment, interference, and network demand can affect performance; some systems require a mounted outdoor receiver.
Cellular home internet	A router receives a 4G LTE or 5G mobile-network signal and shares the connection with devices in the premises.	Is often quick to install, but performance varies with coverage, indoor signal strength, network load, and the router's position.
Satellite	A dish or terminal exchanges radio signals with satellites in orbit.	Can reach remote places without nearby terrestrial cables or towers. It requires a suitable view of the sky and may involve costly equipment, data limits, and greater latency. Low-Earth-orbit systems reduce the signal's travel distance compared with geostationary systems.

In the broad technical sense, cellular home internet is also a form of fixed wireless access. The table separates it because consumer plans based on mobile 4G or 5G networks are commonly discussed apart from systems using a dedicated fixed receiver.

Use these descriptions to identify broad families, not to predict a particular level of service. Two connections using the same technology may perform differently because of the local infrastructure, distance, equipment, congestion, maintenance, and plan conditions. Parts of a network may also use different media. For example, fibre may carry data across most of a provider's network while another technology completes the final connection to a building. The exact connection installed at the premises matters more than a vague marketing label.

The internet connection is not the Wi-Fi. The service provider's connection brings internet access to a modem, optical network terminal, fixed receiver, satellite terminal, or cellular router. An optical network terminal is the box at the end of a fibre connection. Wi-Fi is the short-range wireless network that a router normally uses to share the connection within the premises. A weak Wi-Fi signal in one bedroom does not necessarily mean that the internet service entering the home is slow. Conversely, an excellent Wi-Fi signal cannot make a slow external connection faster. Testing a computer over a wired Ethernet connection, where practicable, helps distinguish an internal Wi-Fi problem from a problem with the internet service itself.

What the performance figures mean. Download rate describes how quickly data travel from the internet to the user, as when loading a page or streaming a video. Upload rate describes travel in the other direction, as when sending a file, backing up photographs, joining a video call, or submitting a large data set. A connection is **symmetrical** when its download and upload rates are approximately equal; otherwise it is **asymmetrical**.

Connection rates are normally advertised in megabits per second (Mbps) or gigabits per second (Gbps), while file sizes are commonly stated in megabytes (MB) or gigabytes (GB). Since one byte contains eight bits, a rate of 100 Mbps corresponds to at most

$$\frac{100}{8} = 12.5 \text{ MB/s}$$

before overhead and other real-world limitations are considered. Here, **overhead** means the additional data and work required to manage the transmission itself. Thus, Mbps and MB/s are not interchangeable. The time required to transfer a file can be estimated by dividing its size in bits by the transfer rate in bits per second, provided that the units are first made consistent.

Rate is not the whole story. **Latency** is the time taken for data to travel between points in a network. Speed tests commonly report the round-trip time in milliseconds; a smaller value is generally better for conversation, remote control, and online games. **Jitter** is variation in that delay, while **packet loss** occurs when some transmitted data fail to arrive and may have to be

recovered or sent again. A connection with a high advertised rate can still feel poor during a video call if latency, jitter, or packet loss is excessive.

Choosing a connection and plan. Begin with what is actually available at the exact address. Consider how many people and devices will use the connection at once, and distinguish download-heavy activities from tasks that need substantial upload capacity or low latency. Read whether an advertised rate is a maximum or an expected range and whether the service has a data allowance, a fair-use policy, or reduced speeds after a threshold. Account for installation charges, required equipment, contract length, post-promotional price, repair response, and the cost of ending the service.

No single factor decides every case: a household that streams recorded video may value different things from one that regularly makes calls or uploads large files. Compare total cost, actual shared use, both transfer directions, delay, consistency, reliability, and support under the conditions that matter.

When comparing plans in the Philippines, give the provider the exact service address and ask which connection will actually be installed. Availability can differ within the same municipality or city, and the advertised plan may not be offered at every address.

Once connected, test at different times and record download rate, upload rate, latency, and whether the test used Ethernet or Wi-Fi. A single result is only a snapshot. Several measurements under stated conditions provide better evidence of the service the household actually receives.

1.4 Software Selection

Hardware determines what a device is physically capable of doing, but software turns that capacity towards a particular task. Choosing software therefore begins with the task, not with whichever application happens to be popular. You must also consider compatibility, cost, privacy, security, support, and whether you will still be able to open or export your work later.

1.4.1 Software Types

Software can be grouped according to its principal purpose. These categories are convenient rather than absolute: one application may perform several kinds of work and therefore belong to more than one category.

Table 1.3: Common types of software classified by use.

Type	Principal use	Examples of tasks
System software	Operates the device and manages its hardware resources.	Running the operating system, controlling a printer, and managing files and memory.
Productivity software	Helps users create, organise, calculate, and present work.	Writing documents, using spreadsheets, preparing presentations, and taking notes.
Communication and collaboration software	Allows people to exchange messages or work together.	Email, instant messaging, video meetings, shared calendars, and collaborative editing.
Scientific and data software	Collects, manages, analyses, models, or visualises data.	Statistical analysis, database work, programming, bioinformatics, mapping, and laboratory data collection.
Creative software	Produces or edits visual, audio, and multimedia material.	Drawing, photography, page layout, animation, audio recording, and video editing.
Utility software	Maintains, organises, diagnoses, or extends a computer system.	Backing up data, compressing files, managing storage, and monitoring system performance.
Security software	Helps protect devices, accounts, networks, or data.	Malware detection, password management, encryption, firewalls, and secure authentication.
Educational software	Supports teaching, learning, practice, or assessment.	Learning platforms, simulations, tutorials, reference tools, and quizzes.
Entertainment software	Provides recreation or access to media.	Games, media players, streaming, and digital reading.

The intended use determines which features deserve attention. A spreadsheet may be entirely adequate for a small table but awkward or unsafe for a large, shared research data set. A sophisticated statistical package may be powerful but unnecessarily difficult for a simple calculation. Software should be proportionate to the task and the consequences of error.

1.4.2 Distribution and Pricing

The labels used for software do not all describe the same thing. Some describe how software is delivered, some describe how payment works, and others describe what its licence permits. A single product may therefore carry several labels. For example, an online service can be SaaS, subscription-funded, and partly open source at the same time. **Proprietary software** is software whose owner does not normally supply the source code or grant users permission to inspect, modify, and redistribute it.

Table 1.4: Common ways in which software is supplied, licensed, or funded.

Model	What it means	Important qualification
Perpetual licence (often sold as a one-time purchase)	The licence allows the user to keep using a particular product or version without a scheduled expiry.	The payment may not include future major versions, continuing support, cloud storage, or compatibility with future devices. The user normally buys a licence, not ownership of the code.
Subscription licence	Continued use or access requires payment at regular intervals.	Updates and services may be included, but access to the software or some features may end when the subscription ends.
Software as a Service (SaaS)	The provider operates the software as an online service, commonly accessed through a browser or application.	SaaS describes delivery, not necessarily price. It may be subscription-funded, free to the user, advertising-funded, or paid for by an institution. Internet access and the provider's continued operation may be essential.
Freeware	In this chapter, the term means proprietary software supplied without a monetary charge.	A zero price does not normally grant permission to inspect, modify, or redistribute the source code. Other costs may include advertising, data collection, or paid companion services.
Free and open-source software (FOSS)	The source code is available under a licence that grants rights to use, inspect, modify, and redistribute it subject to stated conditions.	Open source does not necessarily mean that every service, copy, or form of support is free of charge. Its licence obligations still apply.
Freemium	A basic version is free, while additional features, capacity, or convenience require payment.	The free version may impose storage, export, collaboration, usage, or advertising limits.
Institutional or educational access	A university, school, or employer pays for or negotiates access on behalf of eligible users.	Access may end when the agreement expires or the user leaves the institution. Personal files should be exported before eligibility ends.
Trial or demonstration version	Use is allowed temporarily or with restricted features so that the product can be evaluated.	The software may stop working, limit saving or export, or require payment after the trial.

Keep two questions separate: how software is paid for and where it runs. Installed software can require a subscription, while a web-based service can be provided without charge. Likewise, “freeware” is not a synonym for “open source”: in this chapter the first concerns price, whereas the second concerns licence permissions.

1.4.3 Choosing Software

Begin by stating what the software must accomplish and which file formats, devices, operating systems, and collaborators it must support. Test whether it can import existing work and export data in a form that other applications can read. A convenient tool can become a trap if all of the user's work is held in a proprietary format that cannot be moved elsewhere.

Look beyond the price shown on the first day. Add renewals, storage, additional users, paid extensions, training, accessories, and the cost of moving data to another system. A one-time purchase is not automatically cheaper than a subscription, nor is free software automatically the better choice. The answer depends on the task and the period over which the software will be needed.

Privacy and security matter whenever software handles personal, academic, financial, health, or research data. Before using it, check what data it collects, where those data are stored, who can access them, whether suitable backups and encryption are available, and how the provider responds to security problems. Software should come from a dependable source and receive appropriate updates. Open source permits inspection of the code, but that fact alone does not guarantee that a particular program is secure or well maintained.

Also consider usability, accessibility, documentation, technical support, offline operation, and the ability of collaborators to use the same files. Using an unlicensed copy of commercial software is not a sensible way to save money. An altered installer may contain malicious software, and unauthorised copying or use may violate licence terms or copyright law. A simpler legal tool that does the required job is usually the better choice.

Freeware and paid software. Price is neither a security nor a quality test. Freeware can be legitimate and sufficient, while paid software can still be insecure, poorly maintained, intrusive, or difficult to leave. Confirm the developer and official source, understand how the product is funded, review permissions and updates, and test whether important files can be exported and backed up. These checks apply to free, paid, subscription, and open-source software alike [9, 10].

Practical Work

List the digital systems you have used since waking up. Choose one activity and trace its input, processing, storage, output, and communication. Identify the hardware, software, data, and networks involved. For one piece of software, identify its principal use and how it is supplied or paid for. End by stating one thing about the system that you would like this course to help you understand.

1.5 Exercises

We will use some of these questions in class. Use the others to review the chapter and test whether you can apply its ideas.

1. A student enters observations into an online form during a biology laboratory activity. Identify a possible input, processing step, form of storage, output, and act of communication. Name the main hardware and software involved.
2. A biology student must choose a laptop for data analysis and a phone for field documentation. Explain how CPU or GPU cores, RAM, storage, upgradeability, battery claims, regional phone models, and camera specifications should be interpreted. Why are the largest numbers not automatically the best choice?
3. Ignoring overhead, estimate how long a 100 Mbps connection would take to download an 800 MB file. Show the conversion between bytes and bits before calculating the time.
4. A household regularly makes video calls, uploads large files, and has several simultaneous users. Apart from advertised download rate, what performance measures, availability constraints, and plan conditions should it examine before subscribing? Explain why the service entering the home and the Wi-Fi network inside it must be evaluated separately.
5. Explain why SaaS, subscription software, freeware, and open-source software are not mutually exclusive categories. Then assess the claim that freeware is probably malware whereas paid software is safe.
6. A biology group needs software for recording observations, analysing data, and sharing results. Recommend selection criteria covering compatibility, total cost, privacy, security, offline use, data export, and continued access.

Chapter 2

The Physical Foundations of Computing

Learning Outcomes

By the end of this chapter, you should be able to:

1. trace a familiar digital activity through the physical devices, connections, facilities, energy, and workers that make it possible;
2. explain, without unnecessary engineering detail, how electrical states, transistors, circuits, and software belong to different levels of a computing system;
3. describe the main stages through which chips and electronic devices are produced and the physical requirements that keep digital systems operating;
4. explain how concentration, chokepoints, substitution, and interdependence create both technological power and technological vulnerability;
5. distinguish participation in a global value chain from control of that chain; and
6. evaluate a technological policy or partnership according to stated Philippine interests rather than automatic loyalty to a foreign power.

How do physical and financial chokepoints become political leverage?

Digital activity can feel weightless. A photograph disappears into “the cloud”, a video call joins people in different provinces, and a search result arrives before anyone has time to wonder where it came from. Yet every one of these actions depends on matter moving, storing, or transforming energy. There are chips, batteries, radios, routers, cables, data centres, cooling systems, ports, mines, factories, and people behind the screen.

Those physical foundations have political consequences. In November 2025, the *South China Morning Post* examined China’s use of controls on rare-earth exports amid a wider dispute over access to advanced chips and chipmaking equipment [11]. A rather small amount of a specialised

material can hold up the production of cars, electronics, motors, and other much more expensive goods. The dispute is not simply a contest between a powerful China and a vulnerable United States. China has material and industrial chokepoints; the United States has financial and technological ones. Both still need capabilities supplied by other countries.

So let us begin with the exact link each side can influence. How can it apply pressure? What substitutes are available, and what would using that power cost? We need some technical knowledge to answer these questions, but not a degree in electrical engineering. We need to see the system clearly enough to judge claims about it.

Before Class

1. **Read.** *How China's Rare Earth Leverage Works—Limits, Risks and Global Responses*, by Kandy Wong, *South China Morning Post*, 12 November 2025; updated 13 November 2025 [11].
2. **Read.** *Semiconductors: A Global Supply Chain Powering Modern Life*, CGTN Europe, 30 November 2024 [12].
3. **Map.** On one page, sketch the journey of a phone, laptop, router, or data-centre server from materials to use. Label the stages and places named in the sources, circle one link that appears difficult to replace, and note one simplification that the chapter helps you correct. Bring the sketch to class.

Treat both sources as starting points rather than final authorities.

2.1 Physical Infrastructure

Suppose a biology student photographs a leaf specimen and uploads the image to a shared folder. The camera sensor responds to light. Electronics convert its output into data. The processor follows software instructions to construct an image, RAM holds material being used, and storage retains the file. The phone's battery supplies energy while its radio sends a signal to a Wi-Fi router or mobile base station.

The photograph then passes through equipment operated by several organisations. It may travel through fibre-optic cables, switching and routing equipment, and possibly submarine links before reaching a provider's data centre. There, servers check the account, storage systems retain the file, and other network equipment makes it available to authorised users. Power equipment keeps the facility running and cooling equipment carries away heat. When a classmate opens the photograph, another physical route delivers a copy to another physical device.

In Chapter 3, we will examine how networks organise and direct data. For now, keep three observations in mind. First, the user's phone is only one part of the system. Second, “wireless”

normally describes only part of the journey: a radio link often leads to extensive wired infrastructure. Third, the cloud is not an immaterial place. For most everyday cloud services, it means using computers operated by another organisation in buildings somewhere else.

Part of the route	What must work	One possible interruption
Student's device	Sensor or input, processor, memory, storage, software, battery, and network interface	Full storage, a damaged battery, an application error, or a failed radio
Local connection	Router or mobile base station, power, and a usable signal	Power failure, poor coverage, interference, or an overloaded network
Provider networks	Cables, network equipment, power, maintenance, and links to other networks	A damaged cable, failed equipment, or an upstream outage
Remote service	Servers, storage, account systems, cooling, electricity, and staff	A service fault, failed storage, loss of power, or a security incident
Other user's device	A working connection, compatible software, display, and permission to access the file	No access permission, an unsupported format, or another local outage

An online activity can therefore fail even when the student's phone is in perfect condition. It can also continue when one component fails if the system has a second route, backup power, replicated storage, or another form of **redundancy**. Redundancy can look wasteful during ordinary operation, because capacity is kept in reserve. During a failure, it becomes the reason a service survives. This tension between low cost and resilience will reappear in the discussion of supply chains.

2.1.1 Infrastructure Maintenance

A digital system has two broad kinds of physical requirement. It has an **embodied requirement**: the materials, energy, factories, transport, and labour used to make its equipment. It also has an **operational requirement**: the electricity, cooling, connectivity, maintenance, and replacement parts needed while that equipment is in use.

Data centres make the distinction easy to see. Their servers, storage devices, network equipment, buildings, generators, batteries, and cooling systems must first be manufactured. Once installed, they need reliable electricity and temperature control. Cooling may use air, water, or a combination, depending on the facility and local conditions. Efficiency per task can improve while total demand still grows. The International Energy Agency reports rapid reductions in the energy used for individual AI tasks alongside a 17 per cent increase in global data-centre electricity demand in 2025 [13]. Equipment efficiency does not by itself make total resource use disappear.

Workers are part of this physical system. Engineers design equipment; miners and refinery workers produce materials; factory workers manufacture and test components; seafarers, drivers,

and port workers move them; technicians install and repair equipment; and power, telecommunications, and data-centre workers keep services operating. A diagram that contains only boxes and arrows can hide all of them.

2.2 From Electricity to Computing

At a basic level, digital electronics distinguishes physical states that can be treated as 0 and 1. The states may be represented by controlled electrical conditions. There are no tiny printed zeroes and ones travelling through a chip. A **bit** is an interpretation of a physical state.

A **transistor** is an electronic device that can control current. It can be used as part of a circuit that represents and transforms digital states.

Many transistors can be connected into circuits that perform small logical and memory operations. Many such circuits can be organised into a processor that follows instructions, or into memory that holds information for rapid use. Software supplies organised instructions. An application such as a camera or spreadsheet is therefore several levels removed from the electrical conditions inside the machine, but it cannot operate without them.

Level	Question answered at that level
Electrical state	Which controlled physical condition is present?
Transistor and circuit	What small operation can be performed or remembered?
Processor and memory	Which instructions and data are being handled?
Operating system	How are the device's resources made available and managed?
Application	What recognisable task is being performed for the user?
Human and institution	Who chose the task, rules, data, and purpose?

These descriptions are not competing explanations. Saying that a student submitted an assignment, that an application transmitted a file, and that transistors changed electrical states can all describe the same event at different levels. Confusion results when a claim appropriate to one level is silently transferred to another. A computer can compare encoded values, for example, without understanding why a teacher imposed a deadline.

Modern chips integrate enormous numbers of transistors. Miniaturisation allows much work to be performed in a small space, but switching and moving information require energy, and some energy becomes heat. Designers must balance speed, power use, battery life, cost, and cooling. A phone becoming warm during a demanding task is therefore not mysterious. Unusual heat, swelling, a burning smell, a damaged cable, or repeated shutdowns, however, should not be ignored.

The main material in most modern chips is a **semiconductor**, whose electrical behaviour can be carefully controlled. Silicon is widely used, but a chip is not merely a polished piece of silicon. It is a manufactured arrangement of extremely small structures, layers, and connections. That

observation takes us from how a chip works to the more politically important question of how a chip comes to exist.

2.3 Global Supply Chains

A product may carry the words “Made in” followed by one country. That label does not normally tell the full history of its materials, design, machinery, components, software, assembly, or transport. A phone is better understood as the temporary meeting point of several value chains.

2.3.1 Chip Production

Chip production can be understood without memorising every chemical or machine:

1. A chip is **designed**. Engineers decide what it should do and create the detailed description needed to manufacture it.
2. Semiconductor material is purified and formed into thin **wafers**.
3. A fabrication plant repeatedly forms and connects microscopic structures on those wafers. This requires specialised equipment, chemicals, gases, exceptionally clean conditions, water, electricity, and skilled work.
4. The wafer is tested to identify the parts that work as intended. It is then divided into individual pieces called **dies**.
5. Working dies are **packaged** so that they can be protected, supplied with power, cooled, and connected to other components. They are tested again before being incorporated into a larger product.

The repeated rectangles in Figure 2.1 are candidate dies formed together on one wafer. Edge regions and test patterns help explain why the circular wafer does not become one large chip and why fabrication yield matters to cost and supply.

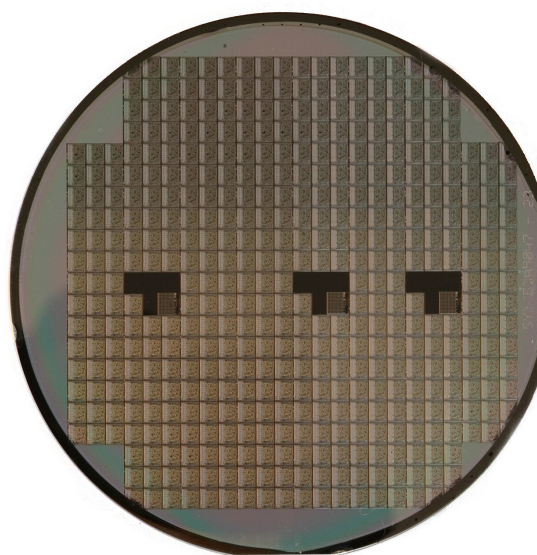


Figure 2.1: A processed silicon wafer containing repeated die patterns. After testing, usable dies can be separated and packaged as individual chips.

Image: Inductiveload, via [Wikimedia Commons](#); public domain. Reproduced without content changes; scaled to fit.

The United States Department of Commerce describes design, fabrication, assembly, testing and packaging, materials, and manufacturing equipment as distinct but connected parts of the semiconductor supply chain [14]. A firm may design chips without owning a fabrication plant. A fabrication firm may manufacture designs created elsewhere. Other firms may specialise in equipment, materials, packaging, or testing. This specialisation can improve efficiency and make extraordinary technical capability possible. It also creates dependencies.

Figure 2.2 provides a useful bird's-eye view. Design software and reusable circuit designs feed chip design; designs, manufacturing equipment, and materials then come together in wafer fabrication. Finished dies proceed to assembly, testing, and packaging before reaching customers. The geographical distribution changes from one activity to the next, and no region leads every stage. The figures are a snapshot rather than a timeless league table. Regional categories simplify multinational firms and cross-border production. Figures for different activities should be compared only when their dates, products, and measures are consistent.

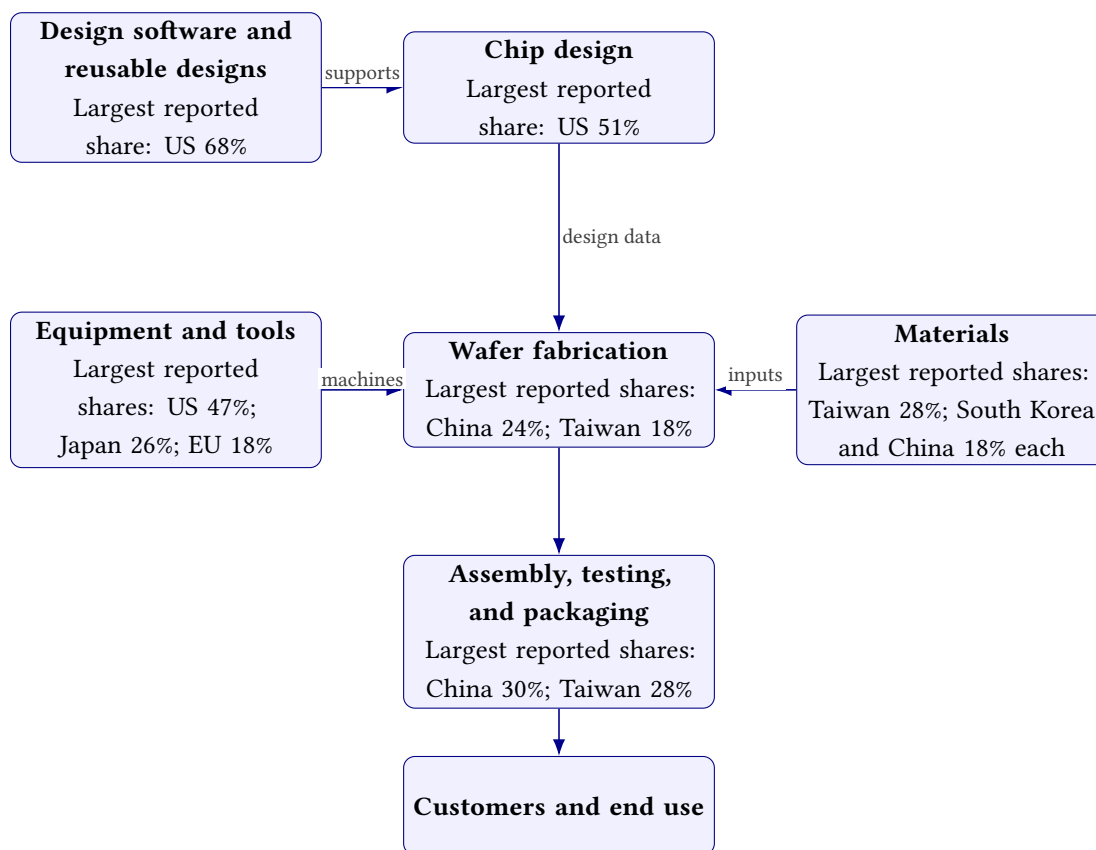


Figure 2.2: Selected geographical concentrations in the semiconductor value chain, using 2022 data from the Semiconductor Industry Association and Boston Consulting Group. Some percentages are based on company revenue and headquarters; others are based on where production capacity is located. Regions not shown complete each category [15].

Silicon is abundant in the Earth’s crust, but an ordinary pile of sand is not a tray of computer processors waiting to happen. The difficult parts include purification, precise design, specialised equipment, repeated processing, quality control, packaging, knowledge, and the coordination of suppliers. A country can possess a raw material without possessing all the capabilities needed to turn it into a competitive chip.

2.3.2 Other Components

Even the most advanced processor is not a phone. A complete device also needs working memory and storage, circuit boards and connections, a display, cameras, radios, speakers, microphones, sensors, a battery, charging electronics, a case, connectors, and software. These require different materials and manufacturing processes.

Rare-earth elements enter this story through particular materials and components, including some high-performance permanent magnets and specialised electronic, optical, and industrial applications. They are not the principal ingredient of every chip, and the seventeen elements

grouped under the term “rare earths” are not interchangeable. The relevant question is always which element, in which form, at which stage, for which use.

2.3.3 Supply Networks and Life Cycles

The word **supply chain** is useful because it emphasises connection: if an indispensable link fails, the final product may not appear. The real arrangement is more like a network. One component has several suppliers; one refinery serves several industries; a factory depends on electricity, water, finance, transport, software, standards, and workers; used products may supply recycled material.

The broader **value chain** asks where useful capabilities and economic value are added. Its stages can be summarised as follows:

Stage	Examples	Question to ask
Extraction	Mining and recovery of raw materials	Who has usable deposits, and what are the environmental and community costs?
Processing	Separation, refining, and production of usable materials	Where does the knowledge and industrial capacity exist?
Components	Chips, batteries, displays, circuit boards, magnets, and sensors	Which parts are difficult to substitute or expand quickly?
Assembly and testing	Combining parts, checking quality, and preparing products for sale	How much capability and value are retained locally?
Distribution	Ports, shipping, warehouses, retailers, and service providers	Which routes and markets are necessary?
Operation and support	Electricity, networks, data centres, maintenance, updates, and repair	What keeps the product useful and secure?
End of life	Reuse, parts recovery, recycling, and safe disposal	Can value be recovered and harm reduced?

Participation at one stage matters, but it is not control of the whole. A country may assemble many devices while importing important components. Another may mine a mineral but send it elsewhere for refining. Another may design advanced chips while relying on foreign fabrication, packaging, customers, and materials. “We are part of the chain” and “we can command the chain” are very different claims.

Nor does price reveal strategic importance. A component costing only a few pesos can halt the sale of a product worth thousands if the component is indispensable and no qualified replacement is available. Conversely, an expensive component may provide little leverage when buyers can obtain an equivalent product from several suppliers. Power arises from the structure of dependence, not from national pride or the size of an object.

2.4 Supply-Chain Leverage

Four concepts help explain why one participant can exert pressure on others:

1. **Concentration.** A large share of a required material, component, or capability is supplied from a small number of places.
2. **Chokepoint.** A link is sufficiently important and difficult to bypass that its interruption constrains the wider system.
3. **Substitution.** Another material, supplier, design, or process can perform the required role. Substitution may nevertheless require redesign, testing, approval, new equipment, and time.
4. **Interdependence.** A powerful supplier still depends on other participants for some combination of inputs, equipment, knowledge, finance, transport, security, and customers.

Concentration alone does not prove that an interruption will be effective. A buyer may have inventories. Other producers may possess unused capacity. The controlled input may be important in one product but irrelevant to another. Smuggling or indirect trade may weaken a restriction. A substitute may be inferior but adequate. To assess leverage, we must ask how much is controlled, at which stage, for how long, with what alternatives, and at what cost to both sides.

2.4.1 China's Rare-Earth Leverage

The advantage is not simply geological. China's position is not explained merely by the location of mines. For the magnet rare earths it examines, the International Energy Agency reports that in 2024 China accounted for 60 per cent of mined production, 91 per cent of refined output, and 94 per cent of permanent-magnet production [16]. Processing and manufacturing capability are therefore at least as important to the case as possession of deposits.

The process is difficult. Mining produces an intermediate material, not a finished magnet. The minerals must be concentrated, chemically separated, converted into metals and alloys, and manufactured to an accepted standard. Other countries can perform these activities, but reproducing the connected system quickly, competitively, and at the required quality is difficult [16].

An industrial base. China's advantage comes from capacity across these stages: scale, skilled workers, specialised suppliers, infrastructure, customers, and production knowledge reinforce one another. A competing chain needs permits, equipment, finance, environmental safeguards, qualified output, and buyers. Announcing a mine therefore does not create a mine-to-magnet industry.

Controls are specific. Export controls apply to stated materials, products, destinations, licences, and dates. “China banned rare earths” is therefore too broad a summary; the particular measure and affected stage must be identified.

This industrial depth is what turns concentration into leverage. A manufacturer that cannot obtain a qualified magnet cannot replace it with a bag of unprocessed ore or an announced mining project. Even a licensing delay can therefore cause uncertainty, higher prices, or interrupted production.

Leverage has limits. The leverage is not unlimited. Controls may reduce income for Chinese suppliers and impose costs on firms in China that need foreign markets or components. Customers may build inventories, support competing producers, redesign products, use substitutes, recover material through recycling, or accept higher costs in exchange for more secure supply. Aggressive use of a chokepoint can motivate other participants to weaken it. Leverage is therefore real, but conditional and capable of changing.

2.4.2 US Financial and Technology Leverage

Finance. The US dollar is widely used as a reserve asset, a currency for trade and payments, and a unit for international banking. This makes access to US banks, markets, and dollar settlement valuable even to firms and governments outside the United States [17].

US financial sanctions draw power from this position and from American legal authority over institutions and transactions connected to its financial system. Exclusion can make payments, borrowing, investment, and trade more difficult. The dollar is not, however, a switch that Washington can turn off for the entire world. The effect of a measure depends on its legal scope, enforcement, cooperation from other governments and financial institutions, and the target’s alternatives. The US Treasury warns that alternative payment systems, digital assets, and reduced exposure to the dollar could erode the long-term effectiveness of sanctions; it also stresses clear objectives, multilateral coordination, and attention to unintended harm [18].

Technology. As Figure 2.2 illustrates, American firms occupy strong positions in chip design and electronic design automation and intellectual property. Export rules can also restrict specified advanced chips, manufacturing equipment, software, and technology, including some items made abroad with covered American inputs [19]. The exact products and rules change; the durable point is that control over a narrow but difficult-to-replace input can influence production elsewhere.

This is substantial leverage, but it is not ownership of “technology” in general. Advanced semiconductor production also relies on equipment and materials from Europe and Japan, fabrication in Taiwan and South Korea, and assembly, testing, packaging, manufacturing, and markets across Asia. China has developed major capabilities of its own and continues to invest in alternatives.

The comparison here concerns particular chokepoints, not a ranking of the total scientific or technological ability of two countries.

2.5 Competing Chokepoints

China and the United States can therefore place pressure on different parts of the same international system. We can compare these forms of leverage without pretending that the two countries' capabilities, policies, or conduct are identical.

Dimension	China	United States
Concentrated positions	Rare-earth separation, refining, and permanent-magnet production for important industrial uses	Dollar-centred finance; advanced chip design; EDA and IP; selected advanced chips, equipment, software, and technology
Means of pressure	Export licensing, delays, restrictions, and the possibility of withholding difficult-to-replace materials or components	Financial sanctions, restrictions on access to the US financial system, and export controls on specified technologies and products
Continuing dependencies	Foreign semiconductor capabilities, equipment, customers, transport, some international payment channels, and other internationally supplied inputs	Foreign fabrication, specialist equipment and materials, assembly and testing, manufacturing capacity, allied cooperation, and overseas markets
Ways leverage may weaken	New suppliers, stockpiles, recycling, substitution, redesign, and reduced reliance on Chinese processing	Alternative payment channels, reduced dollar exposure, new suppliers, technological substitution, domestic development, and weaker cooperation by other countries

China's advantage in rare-earth processing does not give it command of every electronic product. American strength in finance and semiconductor technology does not allow the United States to manufacture the whole chain alone. In both cases leverage is strongest when the controlled link is indispensable, substitutes are costly or slow, and other participants cooperate with or comply with the restriction. Using a chokepoint can also encourage customers to escape it.

When a headline says that a country "controls technology", rewrite it more precisely. Identify the stage, product, share, period, legal measure, affected users, and available alternatives. The same standard applies whether the pressure comes from Beijing, Washington, or another government.

2.6 A Philippine Strategy

Technology policy cannot be reduced to choosing a foreign team. Materials, equipment, factories, workers, finance, routes, markets, and standards cross political boundaries. The Philippines can

cooperate with a country in one area without endorsing every policy or accepting every contract from that country.

The supply chains in this chapter point to a prior Philippine interest: global political stability. Philippine households and firms cannot enjoy reliable access to technology if major economies repeatedly interrupt trade, finance, shipping, or production. A conflict among them would travel through the same networks on which the Philippines depends. Interdependence gives countries practical reasons to preserve peace and predictable exchange, but it does not make peace automatic. A chokepoint can be used precisely because others depend on it. Stability therefore still requires diplomacy, predictable rules, peaceful settlement of disputes, functioning trade routes, and enough resilience that no participant can easily hold the rest hostage.

That wider stability is the condition within which the Philippines can try to tilt particular bargains in its favour. The sensible objective is neither automatic alignment nor a refusal to choose. The United States and China do not have identical institutions, capabilities, conduct, or relationships with the Philippines, and particular actions need not deserve identical judgements. The relevant test is what a proposal does to Philippine welfare, capability, security, and freedom of action. Cooperation with the United States in one area and China in another is not necessarily inconsistent. Dependence so deep that the Philippines cannot refuse either partner is the more serious problem.

“Pro-Philippines” sounds reassuring, which is precisely why it must not be allowed to do the work of an argument. Filipinos can disagree honestly about what a policy should prioritise. Consumers may want lower prices, workers may want secure employment, firms may want reliable inputs, communities may oppose pollution, and government agencies may worry about security or excessive dependence. A serious national-interest argument identifies these interests and explains its trade-offs.

The Philippines is not merely a spectator in electronics. According to the Philippine Statistics Authority, electronic products accounted for 54.3 per cent of the country’s merchandise export earnings in 2025 [20]. The Board of Investments describes Philippine strengths in semiconductor assembly, testing, and packaging and in electronics manufacturing services, alongside efforts to develop design and higher-value capabilities [21]. These activities provide jobs, exports, experience, and infrastructure. They do not mean that the Philippines controls chip design, fabrication equipment, raw materials, transport, or overseas demand. Participation matters, but it does not amount to complete control.

When you consider a technological investment, restriction, contract, or partnership, give “pro-Philippines” some content by asking these questions.

Table 2.1: Questions for evaluating a technological proposal from a Philippine perspective.

Concern	Questions to investigate
Access and cost	Will useful technology, components, or services remain available at a price Philippine households, schools, firms, and government can afford? Who receives a subsidy, guarantee, or exclusive advantage?
Jobs and capability	What work and income will be created, and will Filipino workers, researchers, and firms gain durable knowledge, equipment, design ability, or decision-making power?
Resilience and future choice	Does the proposal diversify suppliers and build alternatives, or create a dependency that will be costly to leave?
Security, law, and control	Who owns, operates, inspects, repairs, and can access the system or its data? Are Philippine law, rights, procurement rules, and accountable remedies effective?
Environment and communities	Who uses the land, water, and electricity? Who is exposed to pollution, waste, unsafe work, displacement, or other costs? What obligations remain after a facility closes?
Distribution	Which Filipinos gain, which Filipinos pay, and are the promised national benefits actually shared?

No partner will score perfectly on every question. A proposal may create jobs but consume scarce electricity; reduce immediate prices but deepen dependence; improve security but restrict access; or transfer some knowledge while leaving the most valuable decisions elsewhere. Compare the benefits, costs, alternatives, safeguards, and time horizons openly.

Diversification is often part of the answer, but it is not magic. Maintaining several suppliers can cost more. Stockpiles can become obsolete or prove too small. Domestic production may require subsidies and still depend on imported equipment. Recycling cannot immediately meet all demand. Substitution may reduce performance. Resilience is not the absence of trade; it is the ability to continue, adapt, and choose when one relationship or link is disrupted.

2.7 Supply-Chain Lessons

Computing joins many layers of technology and many parts of the world. Control of an indispensable link can create leverage, but the supplier normally remains dependent on other links and can encourage customers to seek alternatives.

Strategic independence does not require the Philippines to manufacture every input. It requires understanding vulnerabilities, developing useful capability, maintaining alternatives where failure would be serious, and judging each proposal by Philippine interests rather than a foreign partner's identity.

2.8 Exercises

We will work through selected exercises in class. Use the remaining questions to check your understanding of the chapter.

1. Trace what happens physically when a student uploads a laboratory photograph and a classmate opens it. Include the two devices, local connections, provider networks, remote service, electricity, and cooling. Identify two points at which the activity could fail even if both phones work.
2. Explain how the statements “the student submitted an assignment”, “an application transmitted a file”, and “transistors changed electrical states” can all describe the same event. Why would it be a mistake to say that the transistors understood the deadline?
3. A phone is labelled “Made in Country A”. Construct a plausible value chain involving at least four countries or regions. What can the label establish, and what can it not establish? Use concentration, substitution, and interdependence to explain how one inexpensive link could nevertheless become a chokepoint.
4. Compare China’s leverage in rare-earth processing and magnet production with US leverage in finance and advanced semiconductor technology. Identify the mechanism, dependencies, and limits on each side, and explain why the comparison does not establish that either country controls technology as a whole.
5. A foreign company proposes to build a large data centre in the Philippines. Use at least six concerns in Table 2.1 to recommend acceptance, rejection, or renegotiation. State what evidence is missing and which safeguards you would seek.

Chapter 3

The Internet, Web, and Home Networks

Learning Outcomes

By the end of this chapter, you should be able to:

1. distinguish the internet, the Web, Wi-Fi, mobile data, and cloud services;
2. explain in practical terms how a website name, DNS, an IP address, a router, and a server work together;
3. interpret HTTPS indicators and common browser messages such as 403, 404, and 503;
4. use simple comparisons and status-checking tools to distinguish a device or home-network problem from a provider or website problem;
5. design and test a local network whose wired and wireless parts suit the access connection, building, users, and activities; and
6. recommend a usable and reasonably secure home-network arrangement while explaining what a virtual private network can and cannot protect.

How does information travel between devices and services?

Imagine a household with two students, a parent who sometimes works from home, four phones, one laptop, and a television. The internet plan is advertised as fast, yet a video call freezes in a bedroom while streaming works in the living room. One person recommends a faster plan, another wants a new router, and a third says that a virtual private network will solve the problem.

Each proposal could be useful, but only for the right problem. Before buying anything, locate the failure along the route. Is the laptop struggling? Is the Wi-Fi signal weak in the room? Is the household filling the connection? Is the provider congested? Is the remote service unavailable?

Chapter 1 introduced internet connection types and measures such as download rate, upload rate, latency, jitter, and packet loss. Chapter 2 followed a digital action through physical devices,

cables, data centres, energy, and workers. We now examine how networks organise that physical route and how a household can make sensible choices about it.

Before Class

1. **Map.** Sketch the network through which you most often reach the internet. Include your device, any Wi-Fi access point or router, the equipment that connects the premises to a provider, and one remote service that you use. If you rely on mobile data, draw the mobile base station instead of inventing a home router.
2. **Observe.** Mark where performance is good and where it is poor. Record the time, place, device, activity, and whether you used Wi-Fi, Ethernet, or mobile data. Do not record or share a network password.
3. **Diagnose.** Write two competing explanations for the poor result and one safe test that would help distinguish them. Bring the map to class.

3.1 Internet Basics

Everyday speech often compresses several systems into the word “internet”. That is convenient until something fails. The distinctions in Table 3.1 tell us which organisation, device, or connection to examine.

Term	What it describes	What it is not
Internet	Interconnected networks that exchange data using shared technical rules, called protocols.	A single company, one cable, or the collection of websites.
Web	A system of interlinked resources that browsers request from web servers, normally using HTTP or HTTPS.	The whole internet; email, online games, and many other services also use the internet.
Wi-Fi	A family of technologies that connects nearby devices to a local network by radio.	Internet service. A Wi-Fi network can exist when its internet connection has failed.
Mobile data	Internet access supplied through a mobile operator’s radio and core networks.	Wi-Fi, even though a phone can share mobile data by creating a Wi-Fi hotspot.
Cloud service	Storage, processing, or an application supplied through remote computers operated by a provider.	An immaterial place or an automatic backup of everything on a device.

Table 3.1: Related but distinct parts of connected computing.

The internet is commonly described as a *network of networks*. A university, mobile operator, household, government agency, and cloud provider can each operate a network. Agreed technical rules allow data to cross the boundaries between them. No single organisation owns the whole

internet, even though a small number of organisations may control important facilities or services within it.

The physical routes matter. Figure 3.1 maps a 2015 snapshot of submarine communications cables and landing points. It is not a current inventory, but it makes two durable ideas visible: international data often travels through physical cables, and many routes converge at particular coasts and hubs.

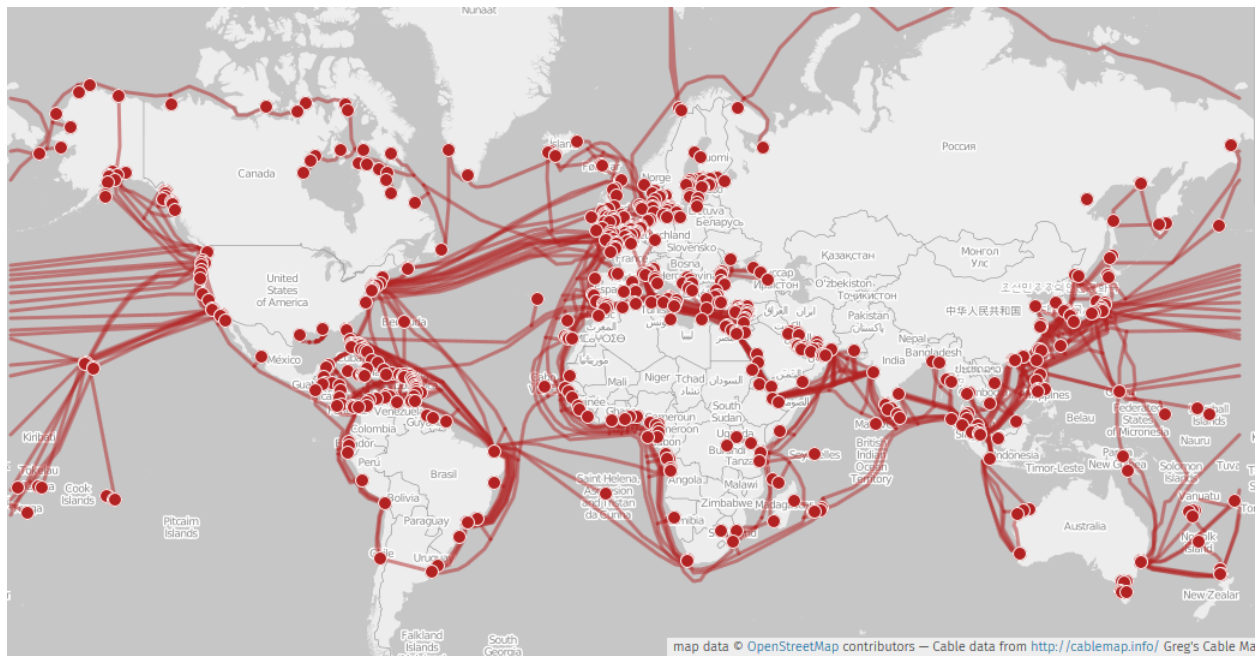


Figure 3.1: World submarine-cable routes and landing points as mapped in 2015. The map is historical and should not be used as a current network inventory.

Image: Greg Mahlknecht and OpenStreetMap contributors, via [Wikimedia Commons](#); CC BY-SA 2.0; cable data [GPLv3/OSM data ODbL](#). Reproduced without content changes; scaled to fit.

The Web is one useful system built on that foundation. A browser requests a resource identified by a web address, and a web server returns a response. Following a link is therefore a Web action carried over the internet. Sending email and joining an online game also use the internet, but neither action is simply “browsing the Web”.

3.2 How Web Requests Travel

Suppose a student enters a course website’s domain name in a browser and opens a laboratory guide. The details vary among networks, but the following route provides a useful model.

1. The device first needs a working connection through Wi-Fi, Ethernet, or mobile data.
2. The browser asks DNS to resolve the website name into an IP address. This lets people enter a name such as `example.com` instead of remembering a numerical address [22].

3. The device sends the request through the home router or mobile network and then the internet provider. The request travels in small pieces called *packets*; routers use the address to move them towards the website.
4. The website's server receives the request and sends a response. When HTTPS is working correctly, it protects the information exchanged while it travels between the browser and that website.
5. The browser puts the response together and displays the requested content.

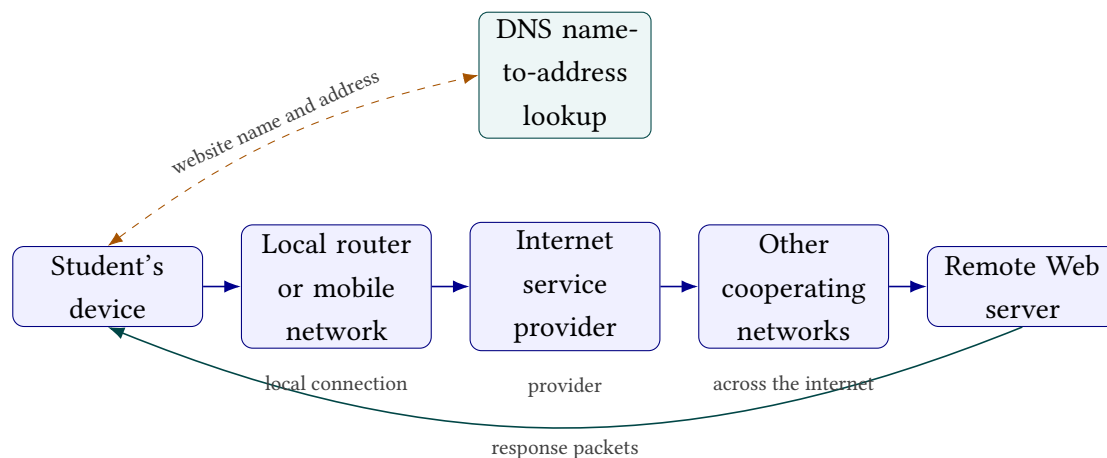


Figure 3.2: A simplified route for a Web request. DNS helps obtain an address; routers then forward request and response packets across several networks.

This is a useful model rather than a fixed itinerary. The networks may choose a different path when conditions change, but users normally do not need to manage that choice.

3.2.1 Names and Addresses

A domain name is the readable part people normally use, such as `example.com`. DNS *resolves* that name to one or more IP addresses: the numerical addresses computers use to deliver the request. It works rather like looking up a contact's name before placing a call, although the technical arrangement is different. If DNS fails, the website may be operating normally but the browser cannot learn which address to contact.

An IP address is a numerical address that networks use to deliver data. It should not be treated as a permanent name for a human being. A household's public address may change; several people and devices may share one public address; and a person may use different addresses at home, on campus, and through mobile data.

Devices on a home network commonly receive addresses used only inside that network. The router lets them share one or more public addresses supplied by the provider.

The address returned for a name can change when a service moves, distributes work among several servers, or sends users to a nearby copy. Users normally do not need to manage this process. They do need to recognise that a “server address not found” or similar DNS message is different from a 404 page: in the first case the browser did not find an address; in the second, it reached a server and received a reply.

3.2.2 HTTPS

HTTP is the set of rules a browser and website use to exchange requests and responses. HTTPS adds a protected connection. The practical point of the **S** is that data travelling between the browser and the named website is protected against reading or alteration while in transit, and the browser checks a certificate for that website. This matters when sending passwords, messages, payments, or personal information.

HTTPS protects the *connection*; it does not guarantee that the website is honest, that its claims are true, or that it will handle information responsibly. Check the domain name in the address bar. Do not enter sensitive information after a certificate or “connection is not private” warning, and do not bypass such a warning merely because the page looks familiar. Browser symbols vary, so learn the security indicator used by your browser. Google Chrome’s guidance similarly distinguishes a private connection from trust in the site itself ([Check if a site’s connection is secure](#)).

3.2.3 Common Browser Messages

An error message is a clue about how far the request travelled. The exact wording varies, but the distinctions in Table 3.2 are useful in ordinary browsing.

Message	What it usually tells you	A sensible first response
404 Not Found	The browser reached a server, but that server could not find the requested page.	Check the spelling and the end of the address. Open the site's home page and search for a page that may have moved.
403 Forbidden	The server understood the request but refuses access.	Check that you are signed in to the correct account and have permission. Repeatedly refreshing the same request usually changes nothing.
500 Internal Server Error or 503 Service Unavailable	The website reports a problem on its side; 503 often indicates a temporary outage, maintenance, or overload.	Wait and try again, check the service's official status page, or use another channel to contact the organisation.
Server address not found or DNS error	The browser could not turn the domain name into an IP address.	Check the spelling, try unrelated websites, compare another device or connection, and check the provider's status.
Connection timed out	No useful reply arrived in time; the cause could be the device, connection, route, or website.	Compare another site, device, time, and connection before deciding which part failed.
Certificate or private-connection warning	The browser could not establish the expected protected HTTPS connection.	Do not enter sensitive information or bypass the warning. Check the address, device date and time, and network; then wait or contact the site owner.

Table 3.2: Common browser clues and proportionate first responses.

The numbers are HTTP *status codes*: short replies from a server about a request. A 403 or 404 therefore shows that some connection to a server worked; it does not mean that the whole internet is down. The Internet Engineering Task Force defines these codes in the HTTP standard ([official HTTP standard](#)).

3.3 Slow Internet

A browser message can show how far one request travelled. Slowness, freezing, and intermittent failure are less tidy. To investigate them, compare activities and parts of the route rather than treating every symptom as “bad internet”.

3.3.1 Calls and Downloads

Messages, videos, updates, and other users share the same connections. A download can wait for missing data and continue when it arrives. A live call cannot wait long without freezing or becoming difficult to understand.

Limited capacity, delay, interference, or lost data can therefore disturb a call even when messages and downloads still work. The comparison does not identify the cause by itself, but it shows why one successful activity does not prove that the connection suits every other activity.

3.3.2 Distance and Delay

The student's provider is only the first large network in the route. Data may then pass through networks operated by other organisations before reaching the website. A provider can manage its own connection, but it cannot guarantee every part of the journey or the availability of the website itself.

Physical distance is therefore only part of performance. A nearby service may still be reached through a busy route, while a service that keeps a copy of popular material inside the Philippines may respond quickly. Current demand, the route, the website's own capacity, and its design all matter.

Record the exact message, address, and time, then compare another site, device, and connection. An official status page or independent checker can add evidence, but no single observation proves where a regional, intermittent, or account-specific failure lies.

3.4 Inside a Home Network

The comparisons above move from the remote service back towards the household. The local network is the first part of that route the household can inspect, reposition, or replace directly. Its equipment is often combined into one box, which makes the separate jobs easy to confuse.

Part	Main job	Important qualification
Modem or optical network terminal (ONT)	Connects the provider's line to the network equipment inside the premises.	Fibre commonly uses an ONT; other services use different equipment. The provider may combine this function with a router.
Router	Connects the local network to other networks and forwards traffic between them.	A home router commonly also assigns local settings, lets devices share a public address, and includes security functions.
Wi-Fi access point	Connects nearby wireless devices to the local network.	It may be built into the router or supplied by separate mesh units.
Ethernet switch	Connects several wired devices within the local network.	A small switch is often built into the home router; extra switches can add wired ports.

Table 3.3: Common parts of a home network.

Equipment can be arranged in several valid ways. A provider might install an ONT and a separate household router. Another provider might supply one all-in-one device, often called a

gateway, that connects the provider's line, wired devices, and Wi-Fi devices. A more complex home might use the provider's equipment plus separate access points. When troubleshooting, identify functions rather than assuming that every box with antennas performs the same job.

An authorised household member can find the router or default-gateway address in a connected device's network details or consult the provider's instructions and the model's manual. The administration login may differ from the Wi-Fi password. Record settings before changing them, protect the administration account, and do not factory-reset provider equipment without knowing how the service will be restored.

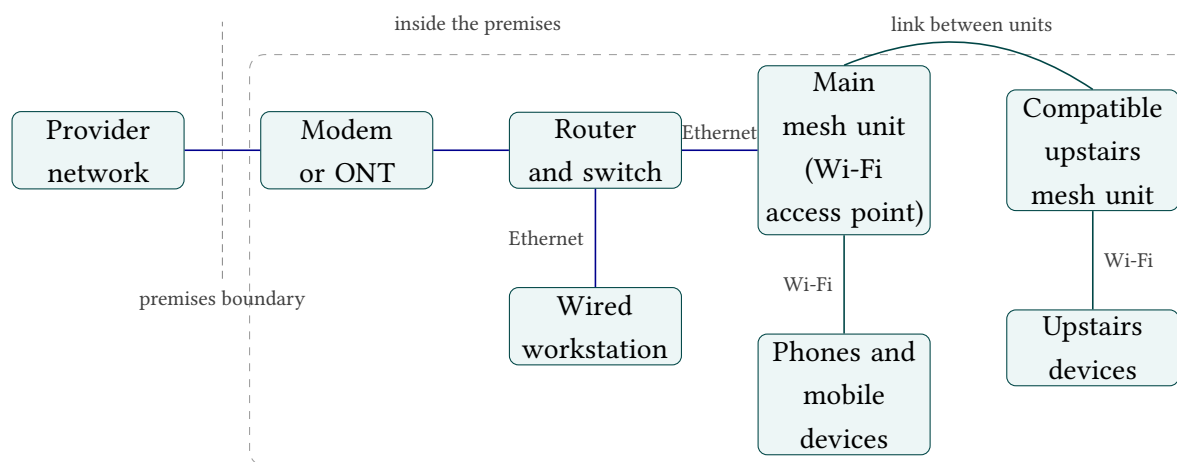


Figure 3.3: One possible home-network arrangement. The functions may be combined in fewer boxes. The two mesh units must be compatible, and their link remains distinct from wired device links and Wi-Fi coverage within each room.

3.4.1 Router Indicator Lights

Indicator names, colours, and blinking patterns vary by model, so the label and manual matter more than a generic colour rule. Before unplugging anything, photograph the lights and note which ones are steady, blinking, or off. Power, provider-link, Wi-Fi, and Ethernet indicators can help distinguish loss of power, loss of the incoming service, a disabled radio, and a local cable problem. A loss-of-signal warning on fibre equipment commonly points towards the provider link rather than Wi-Fi coverage. Check the model's manual, record the evidence, and contact the provider when the incoming signal is absent. Do not stare into, disconnect, or attempt to repair a fibre connector.

3.4.2 Ethernet and Wi-Fi

Chapter 1 used Ethernet as a point of comparison because a suitable cable removes radio distance and interference from the test. Ethernet also has a design role: it can give a fixed computer a stable link or provide the link that carries traffic between an access point and the main router.

Wi-Fi provides mobility and avoids running a cable to every device, but its capacity is shared and its radio conditions vary. Walls, floors, furniture, distance, neighbouring networks, and other equipment can affect performance. The highest number printed on a router's box is not a guaranteed internet rate in every room. It may combine theoretical capacities under conditions that a particular device, channel, and internet plan cannot reproduce.

The 2.4 GHz Wi-Fi band usually reaches farther and passes through obstacles better, but it can be crowded and offers less capacity than newer alternatives. The 5 GHz band can provide higher rates with compatible equipment but generally has shorter reach. The United States Federal Communications Commission's home Wi-Fi guidance similarly recommends central router placement and notes that a direct Ethernet connection can remove Wi-Fi congestion from a test [23]. These physical trade-offs apply in the Philippines even though radio regulations and available channels differ by country.

3.4.3 Wi-Fi Coverage

Begin with placement before buying more equipment. A router hidden on the floor behind a television, placed at one end of a concrete house, or surrounded by obstacles must overcome conditions that a more central and elevated position could avoid. Place it where important devices can reach it, while respecting the provider cable, power, safety, and household layout.

Avoid enclosing a Wi-Fi router in a metal cabinet: the metal can act like a partial Faraday cage and weaken the radio signal the household is trying to use.

Keep Wi-Fi equipment in an open, ventilated position away from metal, water, thick concrete, and unnecessary sources of radio interference. Test before and after moving it rather than assuming that every nearby appliance is the cause.

A range extender receives and sends Wi-Fi traffic again. It may improve coverage cheaply, but repeating traffic over the same radio connection can reduce usable capacity and add delay. Its own link to the main router must be good; placing it inside the dead zone merely gives it a weak signal to repeat.

A mesh system uses several coordinated access points. It can make movement and management easier and may cover a difficult layout better. Mesh is not a synonym for unlimited speed. Wireless mesh units still require good links to one another, and a wired Ethernet link between units can provide a stronger connection where cabling is practical. A small home with good central coverage may gain little from an expensive mesh kit.

An additional access point or mesh unit belongs where it can receive a good link, not at the centre of a dead zone. Use Ethernet between units where practical and test a real activity in each important room. These devices redistribute the incoming connection; they cannot repair a provider outage or make a slow access plan faster.

3.5 Network Diagnosis

Names matter when they help locate a fault. Change one factor at a time and record the conditions. Running many speed tests without noting the room, device, connection method, server, or time produces numbers without a diagnosis.

Figure 3.4 gives the overall logic. Each answer identifies where to investigate first; it does not prove that the named part is faulty.

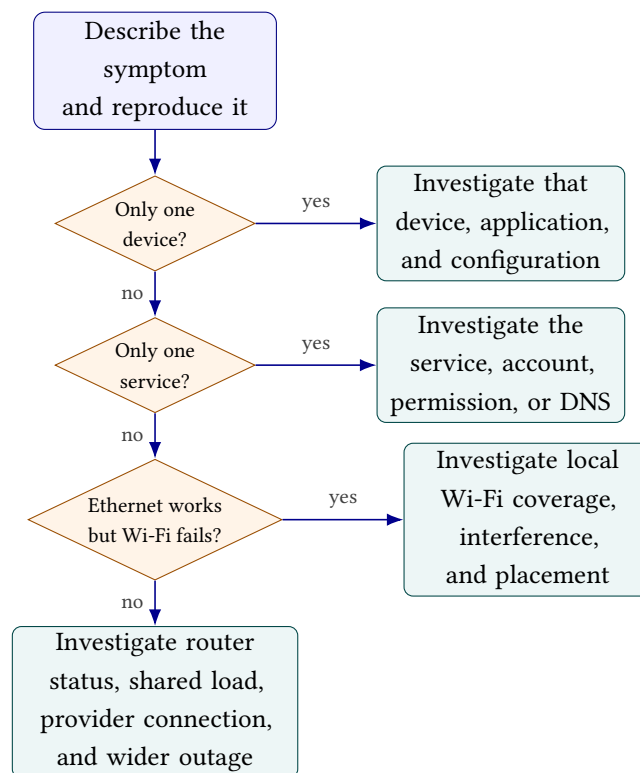


Figure 3.4: A diagnostic flow for deciding which part to investigate first. Comparisons narrow the possibilities; repeated measurements and supporting evidence are still required.

1. Define the exact symptom, place, time, device, and activity.
2. Compare another device and another service on the same connection.
3. Compare the problem room with a position near the access point and, where possible, compare Wi-Fi with Ethernet.

4. Check simultaneous activity, equipment indicators, and provider or service status information.
5. Make the smallest change supported by the comparisons, then repeat the same test and record the result.

Restarting equipment can clear a temporary fault, but it is not a diagnosis. Repeatedly needing to restart it is itself evidence to record. Do not press a factory-reset button casually: that operation can erase provider and household settings and may require passwords, provider details, or assistance to restore.

3.6 Avoiding Bottlenecks

Diagnosis narrows the problem; design matches the parts to the work. Chapter 1 explained how to compare access technologies and plans. Once a household has selected or inherited a service, the next problem is to connect its users to that service without confusing the incoming connection with the local network. A fast access link cannot compensate for a weak radio link in a bedroom, while excellent local Wi-Fi cannot increase the capacity entering the premises.

The place where the provider's line enters may not be the best place for Wi-Fi coverage. An ONT might have to remain near the fibre entry, while equipment that receives fixed-wireless service by radio may need the part of the building with the best external signal. Where the provider's setup permits, Ethernet can carry the connection from that entry equipment to a centrally placed router. Ethernet from the router can also feed a separate access point closer to users. Where cabling is impractical, a mesh system can extend coverage, provided that its units still have good links to one another. The design should follow measurements rather than the convenience of putting every function in one box.

The incoming connection and local network must also share their capacity among the household's activities. A stable wired connection may suit a fixed workstation or a device making important calls. Mobile devices need useful coverage in the rooms where they are used. Equipment that keeps remote copies in sync or uploads large files can affect other traffic if it fills a limited link. The purpose is not to give every device its theoretical maximum at once, but to keep the household's important activities usable together.

A backup works only if its route remains available during the failure being planned for. Mobile data may back up a fixed connection, but only if it has a usable indoor signal, sufficient data, available power, and a working way to share the connection. If both services depend on the same damaged provider facility or unpowered household equipment, the apparent backup may fail with the main connection. Test the changeover before an emergency.

3.7 Home Network Security

Good coverage does not make a network secure or easy to maintain. Once the basic arrangement works, concentrate on avoidable weaknesses that the household can manage:

1. Change the router's default administration password and protect the new one. Change the default administration username too if the router supports it. The administration password controls settings; it need not be the same as the password that devices use to join Wi-Fi. The United States Cybersecurity and Infrastructure Security Agency includes changing default login details among its basic protections because published or widely known defaults allow unauthorised changes [24].
2. Use the strongest current Wi-Fi security mode supported by the router and household devices, with a strong, non-default passphrase (a long Wi-Fi password). Consult the model's manual if the choice is unclear.
3. Install router and device security updates. If the router no longer receives updates, replacement may be justified even when its advertised radio speed remains adequate.
4. Disable the option to manage the router from outside the home unless it is genuinely needed and can be configured safely. Making its settings page reachable from the internet creates a different risk from allowing a household device to browse the Web.
5. Use a guest network for visitors or less-trusted connected devices when the equipment can actually keep them separate from household devices. Check the manual rather than assuming that a differently named Wi-Fi network does so.
6. Record essential settings and know who can contact the provider. Do not share passwords or publish photographs that expose labels, quick response (QR) codes, account numbers, or administrative information.

These precautions do not replace security on the devices and accounts. A safe router cannot stop a user from giving a password to a convincing fake site. Likewise, a device that has been taken over or no longer receives security updates can create risk inside a carefully configured network. The full course returns to these issues in its cybersecurity and privacy units.

3.7.1 Public Wi-Fi: Useful, but Not Yours

A public hotspot is operated by someone else and shared with people the user does not know. Someone may join a fake hotspot with a convincing name, send data through an unencrypted service, bypass a certificate warning, or leave file sharing and automatic reconnection enabled. A login page for the hotspot—called a *captive portal*—can also be imitated to collect email addresses, passwords, or payment details.

Modern HTTPS protects most ordinary website traffic much better than in the early years of public Wi-Fi. The United States Federal Trade Commission notes that public Wi-Fi is usually safe when websites use encryption, while still advising users to check HTTPS, keep devices updated, and recognise fake sites ([Are public Wi-Fi networks safe?](#)). Practical precautions remain worthwhile:

1. Confirm the exact network name with the venue. A name such as “Free Airport Wi-Fi” is not proof that the airport operates it.
2. Disable automatic joining and unnecessary file or printer sharing. Mark the connection as public or untrusted if the operating system asks.
3. Prefer mobile data or a personal hotspot for especially sensitive work when it is available. On any connection, check the domain and HTTPS before signing in, and never dismiss a certificate warning casually.
4. If an employer or school supplies a VPN, follow its instructions. A reputable consumer VPN can also keep the hotspot operator or nearby users from reading traffic carried through the VPN tunnel, but it creates a new provider that must be trusted.

The safest summary is not “public Wi-Fi is always dangerous” or “a VPN makes it safe”. HTTPS, careful choice of network and website, current software, and a suitable VPN each address different risks.

3.7.2 VPNs

A virtual private network (VPN) creates a protected path, often called a *tunnel*, over existing networks. An employee may use one to reach an organisation’s internal systems. A consumer VPN instead commonly sends device traffic through a commercial provider’s server before it continues towards other destinations.

On an untrusted local network, a correctly configured VPN protects traffic between the device and the VPN server. People and organisations carrying data on that part of the route cannot read the protected contents. The tunnel ends at the VPN server. Remote services may then see the VPN provider’s public address rather than the household’s usual one.

A VPN does not make the device anonymous or the user invulnerable. Its provider becomes a new intermediary that handles the traffic and must itself be trusted. Accounts, browser cookies, identifying details from the device, payments, and behaviour can still connect activity to a person. A VPN cannot make a fraudulent website honest, correct false information, remove malware already on the device, or prevent a user from disclosing a password. It also cannot repair weak Wi-Fi; the encrypted traffic must still cross the weak radio link and may incur extra delay.

Many customers may share a VPN address, and its apparent location may differ from the user's. A service may therefore request an additional check or refuse the connection. Comparing the same trusted service with and without a consumer VPN can identify this cause, but an organisation-required VPN should remain in use while accessing its internal systems.

HTTPS and a VPN protect different portions of a route. HTTPS protects data exchanged between the browser and the HTTPS website. A consumer VPN protects traffic between the device and the VPN server; after that server, protection such as HTTPS still matters. Neither technology guarantees that the recipient will use the disclosed data responsibly.

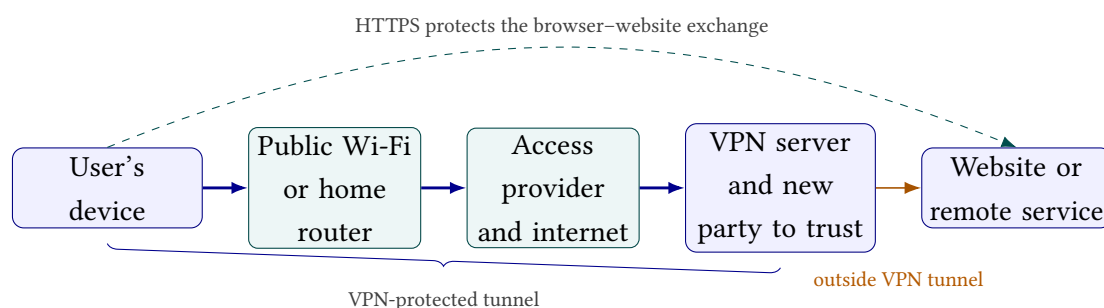


Figure 3.5: A consumer VPN changes one portion of the route. By itself, it does not protect traffic beyond the VPN server, replace HTTPS, or make the destination trustworthy.

3.8 Network Recommendations

Return to the household at the beginning of the chapter. Good Ethernet and nearby Wi-Fi but poor distant coverage justify investigating placement, cabling, an access point, or mesh. Poor Ethernet and Wi-Fi together point farther along the route. A VPN answers a protection or access question, not either performance problem.

A sound recommendation should state:

1. the activities, users, rooms, devices, and budget that matter;
2. the evidence about the present bottleneck;
3. the proposed arrangement, expected benefit, and remaining limitation; and
4. the security, maintenance, and test procedures that will show whether the change worked.

The household should end up with an arrangement that suits its work and is simple enough to investigate when something fails.

3.9 Exercises

We will work through selected exercises in class. Use the remaining questions to check your understanding of the chapter.

1. A login page uses HTTPS. Explain what the S changes, what students should check in the address bar, and why HTTPS does not prove that the page is honest.
2. A university website will not open and the browser reports either a DNS error, 403, 404, or 503. Explain what each clue means and design a short checking procedure using another site, device, connection, or official status page. State what the student should record and what no single test proves.
3. A fibre service performs well through Ethernet and near the router, but calls fail in an upstairs room of a concrete home. Compare repositioning, a range extender, wireless mesh, and a wired access point. Recommend an order of tests and changes suited to the layout, cabling options, and budget.
4. Several devices lose internet access together while the local Wi-Fi network remains visible. Explain how equipment indicators, another service, another connection, and provider status information can help locate the fault. Why are a factory reset and a faster plan poor first responses without further evidence?
5. Give a practical security and maintenance checklist for a home network. Include administration access, Wi-Fi protection, updates, guest access, records, and one tested backup connection where the household genuinely needs it.
6. A café lists one public Wi-Fi name, but a student's phone shows two similar names. What should the student check and disable before connecting? Then assess this claim: "A VPN makes public Wi-Fi safe, hides everything I do, blocks scams and malware, and fixes a slow signal." What can a VPN protect, where does that protection end, why might a service challenge or refuse the connection, and what requires other safeguards?

Chapter 4

Software, Cloud Services, and Digital Work

Learning Outcomes

By the end of this chapter, you should be able to:

1. distinguish an operating system, application, file format, local file, cloud service, and online account without treating them as one product;
2. explain what a cloud service moves to a provider's computers and what responsibilities remain with the user;
3. organise files, names, versions, metadata, and backups so that work can be found, checked, and continued;
4. choose sharing permissions and collaboration practices suited to the people, information, connection, and purpose involved;
5. design a simple reproducible workflow for a student research project; and
6. distinguish access, ownership, copyright permission, and attribution when using or sharing digital material.

When does moving work into the cloud help, and what control or responsibility does the user give up?

A biology group records mangrove observations at three field sites. Four students take photographs and measurements on different phones. One member enters results into a spreadsheet, another prepares graphs, and another writes the report. Some files are in a shared online folder, some were sent through a messaging application, and some exist only on one phone. Two files are both called `final.xlsx`. A corrected value appears in one copy but not the other. The student who created the shared folder is about to graduate, and the group cannot say whether deleting a synchronised file from a laptop will also delete it online.

The difficulty is not confined to one device or application. The group has not agreed where the authoritative files are kept, who may change them, or how a result can be checked. Hardware,

software, accounts, files, permissions, naming rules, internet access, and human decisions all affect the work. Clear procedures are especially important when students move between personal phones, shared campus computers, institutional and personal accounts, and connections paid for through limited mobile data.

Chapter 1 introduced operating systems, applications, software delivery, and licences. Chapter 3 followed information between a device and a remote service. We now examine what happens when the service becomes the place where people write, calculate, store, and collaborate.

Before Class

Choose one school project whose files you are allowed to inspect. Do not expose personal, confidential, or unpublished information.

1. **Map.** List the devices, applications, accounts, and storage locations used for the project. Mark which files exist only once.
2. **Find.** Select one important file. Can another authorised group member identify the newest approved version without opening every copy?
3. **Test.** Check whether the file can be exported in a common format and whether earlier versions or deleted files can be recovered. Do not delete the working copy merely to test recovery.
4. **Explain.** Write one improvement that would make the project easier to continue if a device failed or one member became unavailable.

Bring the map and explanation to class.

4.1 Devices, Software, Files, and Accounts

Students often say that they “made the file in Google”, “saved it in Word”, or “put it on the internet”. Those expressions may be sufficient in casual conversation, but they hide distinctions needed when work must be moved, recovered, shared, or checked.

Layer	Practical question	Example
Operating system	What manages the device and provides services to applications?	Android, iOS, Linux, macOS, or Windows.
Application	What program performs the task?	A spreadsheet, word processor, image editor, statistical package, or browser.

Layer	Practical question	Example
File format	How are the contents represented so that software can read them?	PDF, CSV, JPEG, DOCX, ODT, or a program-specific project format.
Storage location	Where is the current copy?	Internal storage, removable drive, institutional server, or provider's data centre.
Account and permission	Who can enter the service, and what may each person do?	Owner, editor, commenter, viewer, or public visitor.
Workflow	How does work move from collection to checking, analysis, approval, and preservation?	Raw observations are copied, cleaned, analysed, reviewed, and released according to agreed steps.

A browser is not the same thing as a cloud service. The browser is an application on the device; it can provide an interface to many remote services. Likewise, an application and its file format are not identical. Two applications may open the same format, while one application may support several formats. Features can be lost when a file moves between them, so “it opens” does not prove that formulas, fonts, comments, references, or accessibility information survived intact.

The operating system still matters even when the work appears to happen in a browser. It controls local files, accounts, permissions, updates, peripheral devices, and whether an application can run. A required application may not be available on every operating system, and a browser-based service may offer only part of its functionality offline. Before adopting a tool, the group should test it on the devices its members actually use. It should also test export, printing, accessibility, and reopening the exported file in other suitable software. These checks extend the software-selection questions from Chapter 1 to a shared project.

The choice should begin with the work. A field group may need offline data entry, reliable export, photographs linked to specimen identifiers, and a way to prevent later editing of raw observations. A writing group may care more about comments, simultaneous editing, citations, and an approval history. No single list of popular applications answers both cases.

4.2 Cloud Services

Cloud computing makes remotely operated computing resources available over a network as needed [25]. The equipment remains physical, but another organisation operates much of it.

In ordinary student work, the most visible model is **software as a service** (SaaS). The provider operates the application and much of the underlying infrastructure; the user reaches it through a

browser or client application. Cloud storage is related but narrower: its main purpose is to keep and exchange files. In either case, the provider operates servers, storage, and much of the software. The user still manages the account, content, permissions, exports, backups, and decision about whether the service suits the work.

Chapter 2 described the data centres, networks, electricity, cooling, and workers behind remote services. Sharing those resources can make collaboration and access from several devices practical, but it also makes the work dependent on the provider, account, and connection.

Figure 4.1 replaces the cloud metaphor with some of the physical equipment it hides. Rows of racks still require power distribution, network cabling, cooling, maintenance, security, and replacement parts.



Figure 4.1: Server and network-equipment racks in a data centre. A cloud service runs on physical systems operated in facilities such as this one.

Image: Carl Lender, via [Wikimedia Commons](#); CC BY 2.0. Reproduced without content changes; scaled to fit.

4.2.1 Examples for Student Work

The word *cloud* becomes useful when we can say what crosses the network, what happens on the provider's computers, and what comes back. The following services illustrate three different jobs. They are examples, not endorsements; features, limits, and terms can change.

Example	Useful job	What leaves the device	Check before and after
ChatGPT	Questioning, explaining, drafting, or analysing permitted material	The prompt, conversation context, and any uploaded file needed for the request	Data controls and institutional rules; factual accuracy, cited sources, and whether the answer follows the evidence
Dropbox	Storing, synchronising, and sharing a controlled project folder	Files placed in the service and the account, device, and sharing information needed to operate it	Named access, offline availability, sync status, version or recovery limits, and an independent backup
iLovePDF	Merging, splitting, compressing, rearranging, or converting PDFs	For an online operation, the document itself and the chosen operation	Whether upload is permitted; current retention terms; page order, legibility, file size, and whether the downloaded result opens correctly

Table 4.2: Three cloud services perform different jobs and expose different information.

ChatGPT

ChatGPT is a chat-based AI service. In ordinary cloud use, a student sends a prompt—and sometimes a file—to remotely operated systems and receives a generated response. Opening it in a phone or desktop application does not mean that the processing occurs entirely on that device.

A useful request for a biology student might be: “Using only the abstract below, explain the difference between the study’s measured result and the authors’ interpretation. List any term I should check in the paper.” The student must compare the response with the paper because generated summaries, references, and calculations can be wrong. Identifiable records, unpublished research, examination material, and text the student is not permitted to share should not be submitted. A chat history is not the authoritative project file.

Dropbox

Dropbox is an example of cloud storage with synchronisation. A biology group might create a project folder containing 01_raw, 02_working, 03_analysis, and 04_report. Named members may view the preserved raw files, while only assigned members edit the working copy. A dated

approved report can be shared read-only. Google Drive, Microsoft OneDrive, and Apple iCloud Drive belong to the same broad category, although their permissions and recovery features differ.

Dropbox distinguishes files available offline from files that appear in the local folder but must be downloaded when opened. It also explains that edits to synchronised files propagate across devices [26]. Before fieldwork, students should make required files available offline and test them. After reconnecting, they should check sync status and conflicting copies. Sync can spread a mistaken edit or deletion, so a separate milestone backup still matters.

iLovePDF

iLovePDF is an example of an online utility. Suppose a student has three public or authorised PDF handouts and must submit one file below a stated size limit. The student can upload the documents, arrange them, merge them, download the result, and compress a copy if necessary [27]. The whole document is sent to another organisation, even when the operation takes only a minute. Students should check current retention terms and must not upload completed consent forms, identifiable records, unpublished data, or other restricted material merely because the tool is convenient.

After processing, open the downloaded file and check the page count, order, orientation, legibility, links, and file size. Keep the original files until the result is verified.

4.2.2 Sync, Backup, and Version History

Synchronisation attempts to keep selected locations consistent. If a file changes on one device, the change may be copied to the remote service and then to other devices. This is useful, but it can faithfully synchronise a bad change or deletion.

A **backup** is a separate recoverable copy kept for the possibility that the working copy is lost, damaged, overwritten, or attacked. A backup is useful only if the required file can actually be restored. Keeping the only local copy inside a synchronised folder does not automatically create an independent backup.

Version history retains earlier states of a file or document. It can help reverse an unwanted edit, identify when something changed, or compare contributions. Its retention period and detail depend on the service and plan. It is therefore a valuable recovery layer, not a promise that every past state will remain forever.

A Useful Failure Test

Ask four separate questions: What if the device is lost? What if a file is deleted or overwritten? What if the account is inaccessible? What if the provider or internet connection is unavailable? A design that answers only one of these does not yet protect the work as a whole.

4.3 File Organisation

File organisation can reduce mistaken analysis, accidental overwriting, and dependence on one person's memory. An authorised classmate should be able to identify and interpret the correct file later, even if its creator is absent.

4.3.1 Names, Folders, and Versions

A useful filename describes the content without requiring the file to be opened. For a field project, a pattern such as `mangrove_siteA_observations_2026-09-21.csv` is more informative than `data.csv`. Use a consistent date form such as `YYYY-MM-DD` so that names sort chronologically. Avoid relying on `final`, `final2`, and `really-final`; those labels record confidence, not history.

The group should distinguish at least:

- **raw material**, preserved as received from the instrument, form, camera, or observer;
- **working material**, in which errors are checked and transformations are performed;
- **outputs**, such as tables, figures, and reports produced from the working material; and
- **documentation**, explaining variables, units, methods, changes, software, and responsibilities.

One modest structure might be:

```
mangrove-project/  
  README.txt  
  01_raw/  
  02_working/  
  03_analysis/  
  04_outputs/  
  05_report/
```

The numbers preserve the intended order. They are not compulsory; consistency and explanation matter more than this particular scheme. Raw files should normally be copied before cleaning and then treated as read-only. If a value is corrected, the group should retain the original and record what changed, why, when, and by whom.

4.3.2 Formats and Documentation

An editable working format may preserve formulas, comments, layers, or references that a simple exchange format cannot. A common exchange format may be easier to inspect with different software and years later. A sensible workflow can keep both: for example, the spreadsheet used for calculation, a CSV export of each rectangular data table, and a PDF of the approved report.

A README need not be elaborate. It can state the project's purpose, folder structure, responsible people, collection dates, software requirements, and the file that a new reader should open first. A data dictionary can define each column, unit, code, allowable value, and representation of missing data. Without that context, a column called length may be unusable: length of what, measured how, and in millimetres or centimetres?

The FAIR principles describe research data in terms of being findable, accessible, interoperable, and reusable [28]. FAIR does not mean that every dataset must be public. Sensitive or restricted data can be described and governed so that authorised people know what exists and under what conditions it may be used.

4.4 Sharing, Permissions, and Continuity

Cloud collaboration can replace a chain of attachments with one shared working location. Several people may edit, comment, or review without producing a new attachment at every turn. That convenience depends on deliberate permissions and roles.

Permission	Appropriate use	Risk if granted too broadly
Viewer	Reading an approved source, protocol, or report	Material may still be copied or photographed; viewing is not a confidentiality guarantee.
Commenter or reviewer	Suggesting changes without directly replacing the approved text	Comments may expose confidential discussion or remain unresolved.
Editor	Entering data or changing a working document	Accidental deletion, incorrect values, and changes whose authority is unclear.
Manager or owner	Changing access, moving or deleting the shared material, and transferring control	One account can remove everyone else's access or leave the institution while retaining ownership.
Public link	Releasing material intended for unrestricted access	The link can travel beyond the expected audience and may be indexed or copied.

Use the least permission that allows the assigned work, but do not mistake least privilege for an excuse to make collaboration impossible. A data encoder needs somewhere appropriate to enter data; a reviewer may need comments rather than editing; a public reader normally needs neither.

The group should identify one authoritative working location, assign who may change and approve material, record important decisions, export milestones, and ensure that another authorised person can take control if the owner becomes unavailable.

Simultaneous editing does not eliminate conflicts. Someone may work offline, edit an exported copy, paste over another contribution, or change a formula without explanation. Version history may show that a change occurred, but not whether the scientific reason was sound. The group must therefore review important changes and record the reason for accepting or rejecting them.

4.5 Reproducible Analysis

A result is reproducible in the practical sense used here when another authorised person can start from the documented inputs, follow the recorded steps, and obtain the same output, allowing for any stated source of randomness or software difference. Reproducibility is not proof that the study design was good or that the conclusion is true. It makes the path to the result visible enough to inspect.

Spreadsheets are useful, but they become fragile when data, formatting, calculation, and commentary are mixed without rules. Practical guidance for spreadsheet data includes keeping one rectangular table per sheet, one variable per column, one observation per row, consistent missing-value codes, and no important information encoded only through colour or formatting [29].

Weak practice	More reproducible practice
Typing a corrected value over the raw value	Preserve the raw table; make the correction in a working copy and record the reason.
Using colour alone to mean “diseased”	Store the category in a documented column; colour may then help display it.
Writing -, blank, NA, and 0 interchangeably	Choose and document a missing-value convention; reserve zero for a measured zero.
Copying totals into a report by hand	Link or regenerate outputs from the checked data where practical, then verify the released result.
Sending <code>final.xlsx</code> through several chats	Keep one controlled working location, use version history or dated releases, and name the approved export.
Remembering the method	Record formulas, filters, exclusions, units, software, and settings in the workbook, script, or accompanying documentation.

Table 4.4: Small workflow choices can determine whether a result can be checked.

Automation can improve reproducibility when a script or recorded sequence performs the same declared transformation each time. It can also repeat a mistake perfectly. Whether using formulas, menu commands, scripts, or a hosted notebook, the group still has to check units, assumptions, exclusions, and the meaning of the output.

A beginning biology project should preserve the untouched inputs, define its variables and units, record cleaning and analysis steps, explain manual decisions, and connect each released table or figure to the data that produced it. It should also retain a dated approved version of the report.

Personal, health, location, and unpublished research data require additional controls. Reproducibility does not authorise a student to distribute restricted data. A group can preserve a checkable process while limiting access to the people and environment approved for that work.

4.6 Limits of a Cloud-Based Workflow

Cloud services can make work available across devices and reduce local maintenance, but they also create dependencies. Internet failure, account lockout, service changes, incorrect permissions, and synchronised mistakes can affect the whole group. Outsourcing the equipment does not outsource decisions about privacy, access, recovery, or lawful use [30].

The appropriate arrangement may be local, cloud-based, or hybrid. A field team may collect data offline, synchronise when a reliable connection returns, retain an institutionally controlled working copy, and export milestone backups in common formats. In a Philippine classroom, the plan may also need to account for prepaid mobile data, uneven signal coverage, shared computers, and the loss of an institutional account after graduation. The choice should reflect data sensitivity, available connections, institutional rules, cost, and the consequences of failure.

4.7 Access, Ownership, and Attribution

Digital material is easy to copy, but technical ease is not legal or ethical permission. Four questions should be kept separate:

1. **Access:** Can I open or download it?
2. **Ownership:** Who owns the account, file, device, or intellectual property?
3. **Permission:** What does the law, licence, agreement, or owner allow me to do with it?
4. **Attribution:** How should I identify the creator and source?

A photograph visible on a public page is accessible; that does not by itself place the photograph in the public domain. A citation identifies a source but does not automatically grant permission

to reproduce an entire protected work. Conversely, a licence may allow reuse but still require attribution, a link to the licence, an indication of changes, or compliance with other conditions.

The Philippine Intellectual Property Code protects specified works while also providing limitations and exceptions, including fair use [31, 32]. Educational purpose can be relevant, but it does not make everything used for class free to copy. Fair-use analysis considers the purpose, nature of the work, amount used, and market effect. Seek qualified guidance when the consequences matter.

Open licences make permissions more explicit, but the exact licence matters: “Creative Commons” names a family of licences, not one universal permission [33].

For a student report, record enough information to relocate and identify a source: creator, title, date where available, publisher or site, stable address or DOI, and licence when relevant. Place attribution near a reused figure or in the form required by the licence and discipline. Do not strip a creator’s name or licence merely because a service makes copying effortless.

4.8 Mangrove Project Workflow

Return to the mangrove group. A defensible arrangement could include the following:

1. Give each observation and photograph a stable site or specimen identifier.
2. Copy original exports and photographs into a restricted raw-data folder; work on documented copies using consistent units and missing-value codes.
3. Give named members only the permissions required for their roles, with a second authorised person able to maintain continuity.
4. Record cleaning and analysis steps, export milestones in common formats, and keep a backup independent of the synchronised folder.
5. Before release, have another member follow the documentation from the raw inputs to one result and confirm that only permitted data and media are shared.

Several combinations of applications and services could support this plan. The group should assess each combination against the stated requirements: meaning must be preserved, responsibility must be clear, files must be recoverable, and another authorised person must be able to inspect how a result was produced.

4.9 Exercises

We will work through selected exercises in class. Use the remaining questions to check whether you can apply the chapter's distinctions.

1. A student using both a personal phone and a campus computer says, "My report is in Chrome and saved in Google." Identify the distinctions that must be clarified before another person can find, open, or preserve the report.
2. Explain the difference between synchronisation, backup, and version history. Give one failure that each can address and one failure it may not address.
3. A water-quality project contains `data.csv`, `data-new.csv`, `final.xlsx`, `final2.xlsx`, and `IMG_4821.jpg`. The spreadsheet uses colour to mean "contaminated" and treats blanks and zero as the same. Propose clearer names, folders, and data-recording practices. State what you would need to know first.
4. Design a workflow for four students collecting field data where coverage is intermittent and some members depend on prepaid data. Include offline work, identifiers, raw and working files, permissions, synchronisation, backup, and a simple test that another member can reproduce one result.
5. A biology group plans to ask ChatGPT to summarise a paper, keep its field project in Dropbox, and use iLovePDF to merge the final submission. For each service, state a useful input, the remote operation, the expected output, one verification step, and information that should not be uploaded. Then explain why chat history, synchronisation, and a processed download are not backups of the project.
6. A publicly accessible article contains a photograph that would improve a class report. Explain why access, permission, attribution, and fair use are separate questions. What should the students check before reproducing it?

Chapter 5

Digital Platforms, Information, and Online Life

Learning Outcomes

By the end of this chapter, you should be able to:

1. distinguish information that a platform receives directly, observes from activity, and infers from patterns;
2. explain how search results, recommendations, and targeted advertisements are selected without treating their visibility as proof of truth or quality;
3. evaluate an online health or science claim by tracing its source, evidence, context, and independent corroboration;
4. assess synthetic and AI-generated media using provenance, context, and external evidence rather than appearance or a detector score alone;
5. examine how notifications, feed design, and social comparison can shape attention and experience without assuming that every user is affected in the same way; and
6. respond to online disagreement and harassment with appropriate evidence, civil conduct, platform tools, and institutional support.

How do platforms shape what we see, what they learn about us, and how we interact with other people?

A short video claims that a herbal drink prevents dengue infection. A person in a laboratory coat points to a polished chart and says that “a laboratory study proves it.” The post includes enthusiastic testimonials and a link to a shop. After watching several dengue-related videos, a biology student sees the claim repeatedly. Friends begin forwarding it, and the comments become hostile when one person asks to see the study.

This is a fictional example, and the claim should not be treated as medical advice. It brings together several problems that occur in ordinary online life. Why was the video shown to this student? What might the platform know or infer? Does repetition make the claim more credible? Could the presenter, voice, chart, or comments have been generated or altered? How should a student check the evidence and respond without amplifying a doubtful health claim?

A **digital platform** brings people, information, organisations, and services together through software and rules. Search engines, social networks, video-sharing services, marketplaces, and discussion forums are different kinds of platform. Each makes choices about what information to collect, how to rank material, what conduct to permit, and which actions to make easy. Users, advertisers, creators, institutions, and governments also influence what happens. A platform is therefore neither a neutral pipe nor an all-powerful controller of behaviour.

Before Class

Choose one public science or health claim that appeared in a search result, feed, advertisement, or message. Do not share the claim further merely to complete this task, and do not expose another person's account or private data.

1. **Record.** Write the exact claim, where it appeared, and whether it was a search result, recommendation, advertisement, or message from a person.
2. **Trace.** Follow the claim one step towards its original source. Record the author or organisation, date, and evidence named. Do not rely only on a screenshot or repost.
3. **Check.** Open two independent sources outside the original page. Note what they say about the source or claim.
4. **Protect.** If you bring a screenshot, remove usernames, notifications, location, and other unnecessary personal information.

Bring your notes and links to class. You are evaluating the information, not diagnosing or treating a person.

5.1 Platform Data

When the student opens the video, the platform needs some information simply to provide the service: which item was requested, where to send it, and whether the student is signed in. It may also record what was displayed, how long it remained visible, whether the sound was turned on, and what the student liked, shared, hid, searched for, or purchased. The details vary by service, device, settings, law, and provider practice.

A **cookie** is a small value that a website asks a browser to store and return with later requests to the relevant site. Cookies can keep a user signed in, remember a shopping cart, or maintain

other state between requests. They can also contribute to tracking, especially when activity from several pages is connected to the same browser or identifier. The web standard therefore describes both useful state management and privacy concerns, including third-party tracking [34]. Cookies are not the only tracking method. Applications may use account identifiers, advertising identifiers, device information, pixels, server logs, and links containing identifying parameters.

The resulting traces form part of a person's **digital footprint**. Some are active: the person deliberately posts a photograph, completes a profile, or submits a form. Others arise through use: a service records searches, locations, viewing time, purchases, or device details. A footprint is not a perfect record of the person. It can contain errors, shared-device activity, outdated information, and actions that do not reveal the user's intention.

Kind of information	Example	Important limit
Provided	Age range entered in a profile or an email address used to register	It may be incomplete, false, old, or supplied by someone using the account.
Observed	A search, pause, click, approximate location, or purchase	The action is real evidence of activity, but its purpose may be unclear.
Inferred	A predicted interest in dengue news or membership in an audience category	It is a probabilistic conclusion, not a fact disclosed by the user.
Obtained from others	An advertiser's customer list or data supplied by a partner	The person may not know its origin, accuracy, or route to the platform.

Table 5.1: Platforms can receive, observe, and infer different kinds of information. None gives a complete view of a person.

An **inference** is a conclusion drawn from available evidence. If a student watches several dengue videos, a system may predict that related material will hold the student's attention. It does not thereby know whether the student is ill, preparing a report, helping a relative, or correcting misinformation. Inferences can be useful for ranking and advertising, but they can also be sensitive, mistaken, or unfair.

Philippine data-protection principles require personal information processing to follow transparency, legitimate purpose, and proportionality. The Data Privacy Act also gives data subjects specified rights and places obligations on organisations that control or process personal information [35]. Those principles do not mean that all collection is forbidden. They make purpose, notice, necessity, accuracy, security, retention, and accountability questions that should be asked.

Privacy controls can reduce some collection or personalisation, but no single button makes a person invisible. A sound review checks application permissions, browser and account settings, advertising controls, connected services, public profile fields, and old posts. The user should also ask whether the service is needed and what information the task actually requires.

5.2 Search, Recommendation, and Advertising

The dengue video may appear through several routes. A search engine responds to words entered by the user. A recommender chooses items for a feed or a “next video” position. An advertising system selects a paid placement. One screen may contain all three, so the label and surrounding interface matter.

Route	Common basis for selection	What the position does not establish
Search result	The query, estimated relevance, quality signals, location, and other context	That the first result is correct, complete, or suitable for a health decision.
Recommendation	Past and present activity, similarities among items or users, subscriptions, recency, and predicted response	That the platform endorses the claim or that every user sees the same feed.
Targeted advertisement	An advertiser’s chosen audience, bid, campaign goal, available profile or context, and platform rules	That payment, repetition, or precise targeting proves the product works.

Table 5.2: Visibility answers how an item reached a screen, not whether its claim is true. Exact ranking systems differ and change over time.

A recommendation system must choose an objective that can be measured, such as the probability of a click, continued viewing, purchase, or a user rating. It then uses signals that may help predict that outcome. The objective is not the same as truth. A surprising or frightening claim may attract attention even when its evidence is weak. Conversely, a careful correction may be important but less immediately engaging.

Recommendations also create a feedback loop. What the system ranks affects what the user can notice; the user’s response supplies new data for later ranking. This does not force a person to believe or select an item, and it does not imply that a single click permanently determines the feed. It does mean that platform design and user action can shape one another over time.

Targeted advertising uses information or context to select an audience thought more likely to respond. The targeting may be broad, such as language and region, or may use inferred interests and previous activity. A highly relevant advertisement can therefore feel as though the platform “heard” a private conversation even when more ordinary observations and inferences are enough to explain it. That possibility does not rule out improper collection; it is a reason to inspect permissions and evidence before deciding what happened.

5.3 Evaluating Health Claims

This section concerns **health misinformation**: false, inaccurate, or misleading information about health, disease, prevention, diagnosis, or treatment. The problem is broader than a fabricated news story. It can appear in a testimonial, advertisement, short video, reposted graphic, podcast, message, or genuine study whose findings have been exaggerated.

High view counts, repeated recommendations, confident delivery, and long comment threads are evidence of circulation. They are not evidence that a herbal drink prevents dengue. Begin by converting the health post into a statement that could be checked: What exactly is claimed, for whom, under what conditions, and compared with what alternative?

A practical check has four parts:

1. **Claim.** Write the claim accurately. Distinguish prevention, treatment, symptom relief, association, and personal testimony.
2. **Source.** Identify who first made or published it, not merely the account that reposted it. Look for relevant expertise, conflicts of interest, contact information, and a record that can be examined.
3. **Evidence.** Locate the named study or data. Ask what was measured, in what system or population, against what comparison, with what uncertainty, and whether the result supports the public claim.
4. **Context and corroboration.** Check the date, missing qualifications, later corrections, and what independent expert bodies or studies report.

Do not remain inside a doubtful page and judge it only by its design. Open new tabs and investigate what other sources say about the site, author, and claim. Professional fact-checkers in one study used this **lateral reading** strategy more effectively than participants who concentrated mainly on the original page [36]. The study does not make every outside source reliable; it shows why independent context matters. For a health claim, that context should include the original research when available and current guidance from appropriate clinical or public-health authorities.

Biology students should pay close attention to the distance between an experiment and a public health claim. An effect in a test tube does not by itself establish prevention in humans. An animal result may justify further study without establishing a safe or effective human dose. An observational association does not alone show that the exposure caused the outcome. A human trial still requires attention to design, sample, comparison group, outcome, uncertainty, replication, and relevance to the population.

Testimonials can suggest a question but usually cannot separate the effect of a product from recovery, expectation, simultaneous treatment, selective reporting, or other explanations. A sales

link creates a possible conflict of interest; it does not automatically make the claim false. The response is to seek better evidence, not to replace uncritical trust with automatic dismissal.

For an uncertain or consequential health decision, consult an appropriate health professional and current guidance from responsible public-health authorities. A class investigation can assess the information; it should not turn classmates' symptoms or medical histories into public content.

5.4 Synthetic Media and AI-Generated Content

Text, voices, photographs, video, graphs, and profile images can be generated or altered with software. **Synthetic media** is a broad term for such machine-generated or substantially manipulated material. Some uses are ordinary and disclosed, such as an illustration or translated narration. Other uses impersonate a person, fabricate evidence, or hide commercial production behind apparently independent testimonials.

The dengue video still requires evaluation if every pixel is authentic: a real person can misstate a study, and a genuine image can carry a false caption. Conversely, an artificial illustration is not automatically false. Visual oddities and automated detector scores may justify closer checking, but neither proves authenticity or deception. Tools for recording origin or detecting manipulation also have limits [37].

For a consequential post, find the earliest available version, the speaker's official channel, the study or event allegedly shown, and independent reporting. Preserve the link if it must be reported, but do not redistribute harmful material unnecessarily. The decisive question is what evidence supports the claim, not whether the media look polished or artificial.

5.5 Attention by Design

Notifications can signal a direct message, deadline, emergency, or result that the user genuinely wants. They can also interrupt an experiment, study session, meal, or sleep. Autoplay removes the decision to begin the next item. An infinite feed removes a natural stopping point. Counts, streaks, and prompts make social response continuously visible. These are design choices, not unchangeable properties of technology.

The term **persuasive design** refers to design intended to influence attention or action. Persuasion is not necessarily harmful: a laboratory form can remind a student to record a unit, and a health service can make an appointment easier to keep. The ethical questions concern the objective, evidence, transparency, user interest, vulnerability, and ease of refusing or changing the design.

Attention is also affected by workload, habit, social expectations, sleep, anxiety, device access, and the importance of a message. It would therefore be too strong to say that one feature causes

the same outcome for everyone. A 2024 consensus report on social media and adolescent health found both benefits and harms, substantial variation among people and uses, and insufficient evidence for broad population-level causal conclusions [38]. That uncertainty is not a reason to ignore design. It is a reason to state claims at the strength the evidence permits.

A student can conduct a modest attention audit: record which notifications arrive for two days, which required prompt action, what task was interrupted, and whether the alert came from a person or an automated system. The student might then silence non-urgent categories, schedule a focus period, move a tempting application off the first screen, or disable autoplay, and compare the result. This is a practical self-observation, not a clinical experiment or proof that the same setting will help everyone.

5.6 Social Comparison and Online Disinhibition

A feed is a selected sample, not a random window into other people's lives. People often share achievements, attractive images, unusual fieldwork, or strong opinions while leaving ordinary failures and uncertainty unseen. Comparing one's complete experience with another person's selected posts can therefore be misleading. The direction and effect of comparison vary. A post may inspire one student, discourage another, or do both at different times.

Online communication also changes the conditions of interaction. Physical distance, delayed replies, reduced social cues, perceived anonymity, and the sense that an audience is not fully present can lower restraint. This **online disinhibition effect** can support honest disclosure and mutual help, but it can also contribute to insults, threats, and impulsive sharing [39]. It is an explanation of conditions that may influence conduct, not an excuse and not proof that online behaviour reveals a person's "true self."

Before posting in a disagreement, pause long enough to ask:

- Am I addressing the claim and evidence, or attacking the person?
- Would I use these words if the person and my instructor were present?
- Am I exposing private information or sending an audience towards an individual?
- Will a correction clarify the issue, or will quoting the original merely spread it further?
- What would change my mind, and have I represented uncertainty honestly?

5.7 Responding to Harassment

Disagreement, criticism, harassment, and a credible threat are not the same. A classmate may reject an argument sharply without committing harassment. A pattern of targeted abuse, sexual

remarks, impersonation, non-consensual image sharing, doxxing, stalking, or threats requires a different response. The affected person need not continue a public debate to prove civility.

Where it is safe, preserve the URL, date, username, relevant messages, and context. Use block, mute, audience, and report controls appropriate to the situation. Tell a trusted person and use university procedures when the conduct affects the class or campus. An immediate or credible threat should be brought to appropriate emergency or law-enforcement channels rather than handled as a comment-section argument. Evidence should be stored carefully and not reposted widely, especially when it contains private or sexual material.

The Philippine Safe Spaces Act specifically addresses forms of gender-based sexual harassment in streets, public spaces, workplaces, educational settings, and online. Its online provisions include specified unwanted sexual remarks, threats, non-consensual sharing of sexual media, cyberstalking, and certain forms of impersonation [40]. Not every rude or false comment falls under that Act. Platform rules, university policies, civil or criminal law, and the facts of the case may differ. Seek qualified guidance when legal consequences matter.

A civil correction of the dengue post could say: “The video claims that this drink prevents dengue, but the linked material does not show that result in humans. Please do not use or share it as medical guidance. Here is the current advice from the responsible health authority.” This addresses the claim, states the evidential gap, avoids diagnosing the poster’s motives, and offers a better source. If the thread has become abusive, reporting and disengaging may be more useful than supplying another reply.

5.8 Evaluating the Dengue Scenario

The student can now use a short procedure:

1. Identify whether the video arrived as a search result, recommendation, advertisement, or message.
2. Write the exact health claim and trace it past reposts to its original source and named evidence.
3. Check whether the evidence concerns cells, animals, or people and whether it actually supports prevention in humans.
4. Compare the claim with independent guidance outside the platform.
5. Decide whether to ignore, correct, or report the post without spreading it further; preserve evidence if the surrounding conduct becomes harmful.

Repeated recommendations do not show that the platform diagnosed the student, and proof that the presenter is real would not validate the health claim.

5.9 Exercises

We will work through selected exercises in class. Use the remaining questions to practise tracing information and choosing a proportionate response.

1. After watching and searching for dengue videos, a student sees the same herbal-drink claim as a search result, feed recommendation, and sponsored post. Distinguish information the platform may have observed from interests it may have inferred. Explain the three routes and why none proves either that the claim is true or that the phone recorded a private conversation.
2. Evaluate the claim, “A laboratory study proves that this drink prevents dengue.” State the exact questions you would ask about the claim, source, study, biological system, comparison, and independent corroboration.
3. A polished video uses a laboratory-coated presenter, synthetic narration, and a chart. Explain why neither its appearance nor an AI-detector result establishes whether the health claim is true. Give a practical checking procedure.
4. Two students follow the same biology pages but receive different feeds. Explain how recommendation and feedback can produce this result without claiming that the system fully controls either student.
5. Assess the statement, “Social media lowers every student’s self-esteem.” Identify possible benefits, harms, and other explanations. Then propose a small personal notification audit and explain why its result would not prove a universal causal rule.
6. A student who questions the dengue post receives repeated sexual insults, an edited image, and a message threatening to publish the student’s address. Recommend immediate and later actions. Distinguish a civil evidence-based reply from the response appropriate to harassment or a credible threat.

Chapter 6

Cybersecurity, Identity, and Privacy

Learning Outcomes

By the end of this chapter, you should be able to:

1. distinguish assets, threats, vulnerabilities, controls, and residual risk in a connected student or research system;
2. analyse social-engineering and malware risks across messages, links, attachments, QR codes, downloads, authentication prompts, and active sessions;
3. plan proportionate prevention, detection, response, recovery, and evidence preservation after suspected account compromise;
4. distinguish identification, authentication, authorisation, and accountability and evaluate common authentication and recovery methods;
5. apply purpose limitation, proportionality, access, retention, and breach response to personal and sensitive personal information; and
6. distinguish technical capability, legal permission, ethical justification, and evidential support when assigning security and privacy responsibilities.

How do attackers exploit connected people and systems, and how should individuals and institutions protect identity and data without blaming the victim or treating one safeguard as sufficient?

A biology student receives a text message saying that updated laboratory results are available. The message uses the university's name, warns that the link will expire, and displays a QR code. The student scans it. The page looks like the university sign-in screen, and its address begins with `https://`. The student enters a password. An authentication request then appears on the student's phone. Thinking that it completes the sign-in, the student approves it.

The page never shows laboratory results. Soon afterward, the student's account sends messages that the student did not write. The account includes email, a shared field-project folder, specimen

locations, classmates' contact details, and files stored under an institutional account. Some messages can be used to reset other accounts. The group does not know whether the intruder only read files, copied them, changed them, or created a way to return later.

The incident began with deception, but its consequences depend on several systems and decisions. The attacker imitated a trusted institution. The service accepted a password and an approved prompt. The account connected email and shared files. The group placed information there, and the university selected the service, permissions, logging, recovery process, and support arrangements. Recognising these dependencies avoids two errors: treating the student as the sole cause, and assuming that one additional security feature would have made the whole system safe.

Chapter 3 distinguished a domain name from the page displayed in a browser and explained that HTTPS protects a connection without proving that the site is honest. Chapter 4 separated devices, applications, files, accounts, permissions, backups, and version history. Chapter 5 showed that a QR code can conceal an address and that observed activity is not the same as a person's intention. We now use those distinctions to protect connected work and respond when a protection fails.

Before Class

Use a school or personal account that you are authorised to inspect. Do not change university-wide settings, test another person's account, scan a suspicious code, or expose recovery codes in class.

1. **Map.** List what the account can reach: email, files, contacts, payments, devices, or other accounts. Mark one item whose loss, alteration, or disclosure would cause significant harm.
2. **Check.** From the service's official page or application, inspect recent sign-ins, connected devices or sessions, recovery methods, and multi-factor authentication. Record what can be checked; do not copy secrets.
3. **Recover.** Find the official reporting and account-recovery route. Ask what would happen if both the password and phone became unavailable.
4. **Limit.** Identify one permission, old session, shared link, or unneeded recovery method that deserves review. Do not remove institutional access without understanding its purpose. Bring a small account map with private details removed.

6.1 Assets and Risks

Cybersecurity is the protection of connected systems and information against unauthorised access, use, disclosure, disruption, alteration, or destruction. It is a continuing risk-management problem, not a condition in which nothing bad can happen. A useful analysis separates five terms.

An **asset** is something worth protecting: an account, device, file, service, reputation, research record, or ability to continue work. A **threat** is a circumstance or actor capable of causing harm. A **vulnerability** is a weakness that a threat can exploit. A **control** is a safeguard intended to reduce likelihood or impact. **Residual risk** is the risk that remains after controls are applied.

Part of the case	Example	Decision it changes
Asset	Institutional email, raw observations, specimen locations, contact details, and continuity of the field project	Which losses matter, who could be harmed, and what needs a separate recovery copy.
Threat	An attacker seeking account access or information that can support fraud, resale, extortion, or a further attack	Which behaviour and access patterns should be anticipated.
Vulnerability	A convincing counterfeit page, reusable password, approval-only prompt, broad sharing, or weak recovery route	Where safeguards can interrupt the attack rather than relying on one person's suspicion.
Control	Link filtering, user verification, password manager, phishing-resistant authentication, limited permissions, alerts, logs, and tested backups	Which preventive and recovery measures work together and who operates them.
Residual risk	A stolen active session, compromised recovery channel, malicious insider, configuration error, or new deception	What must still be detected, contained, documented, and accepted or reduced.

Table 6.1: Risk terms describe different parts of the laboratory-results case.

Harm also has several forms. **Confidentiality** concerns who can see information. **Integrity** concerns whether information remains accurate, complete, and traceable. **Availability** concerns whether authorised people can obtain the information or service when needed. Copying classmates'

details threatens confidentiality; changing specimen coordinates threatens integrity; locking the group out before a deadline threatens availability. A backup can improve availability after deletion but cannot undo disclosure of a copied file. Encryption can protect some data but cannot correct a false value entered by an authorised account.

Risk depends on context. A public field guide and an unpublished map of a threatened species may use the same file format, yet disclosure has different consequences. A control should therefore match the asset, plausible threat, likely harm, cost, access needs, and people who must use it. The NIST Cybersecurity Framework organises institutional outcomes around governing, identifying, protecting, detecting, responding, and recovering; it does not prescribe one product for every organisation [41].

6.2 Cross-Channel Attacks

Social engineering uses deception or pressure to induce a person to perform an action useful to an attacker. Phishing is one form: a message or page impersonates a trusted party to obtain information, money, software execution, or account access. The same technique can arrive by email, text, chat, voice call, social-media account, advertisement, shared document, or QR code. A QR code is a carrier, not proof of legitimacy. It makes the destination harder to inspect before scanning.

The fraudulent message combines plausible context with urgency. The university name supplies authority; “updated results” fits the student’s work; an expiry warning discourages independent checking. Awkward grammar can be a warning, but fluent wording is not evidence of safety. Sender names can be imitated, real accounts can be taken over, and an attacker may know enough about a course to write a specific message.

Figure 6.1 is an older fictional example, but its structure remains instructive: a trusted identity, alarming account activity, urgency, and a link chosen by the sender. Modern messages may be more polished and may use different channels, so spelling errors are not a dependable test.



Dear valued customer of TrustedBank,

We have recieved notice that you have recently attempted to withdraw the following amount from your checking account while in another country: \$135.25.

If this information is not correct, someone unknown may have access to your account. As a safety measure, please visit our website via the link below to verify your personal information:

<http://www.trustedbank.com/general/custverifyinfo.asp>

Once you have done this, our fraud department will work to resolve this discrepancy. We are happy you have chosen us to do business with.

Thank you,
TrustedBank

Member FDIC © 2005 TrustedBank, Inc.

Figure 6.1: A fictional phishing message. The displayed bank name and apparent link are claims made by the message, not independent proof of its origin.

Image: Andrew Levine, via [Wikimedia Commons](#); [public domain](#). Reproduced without content changes; scaled to fit.

The safest useful question is not “Does this look professional?” but “Can I verify this request through a route I selected independently?” Instead of using the code or link, the student can open the known university portal from a bookmark or official directory, contact the laboratory through a known number, or ask the responsible office whether it sent the notice. The exact domain matters. A padlock or `https://` means the browser has a protected connection to that domain; it does not mean the domain belongs to the university.

Links, attachments, and downloads can also deliver **malware**: software intended to perform a harmful or unauthorised action. Malware may steal data, monitor activity, alter files, display unwanted advertising, encrypt data for ransom, or give an attacker remote control. A catalogue of malware names is less useful than asking what action occurred. Did the user only view a message, enter a secret, install an application, enable a document feature, grant a permission, or run a downloaded file? Those facts guide response. Current software, restricted privileges, application controls, filtering, and backups can reduce risk, but no scanner guarantees that a device is clean [9].

In the case, the counterfeit page captured the password. The unexpected prompt was a second opportunity to stop the attack, but its meaning was unclear to the student. Repeated unsolicited prompts can also form an **authentication-fatigue** attack: an attacker already has one factor and hopes that annoyance or confusion will produce an approval. A person who receives a prompt without starting a sign-in should deny it and report the event. A well-designed system

should communicate enough transaction context, limit repeated attempts, and offer stronger authentication rather than making the user the only control [42].

6.3 Incident Response

Once compromise is suspected, speed matters, but unrecorded or destructive action can remove evidence and make recovery harder. The student should stop interacting with the suspicious page and use a known clean device and an independently reached official service or university support channel. If the university has an incident procedure, follow it promptly. Do not keep testing the attacker, forward the message to classmates, or wipe a device before the responsible support or investigation team advises what to preserve.

The first report should state what happened, when, which account and device were involved, what was entered or approved, and what unusual activity has been seen. Preserve the original message where possible, sender and recipient, full address behind the QR code or link if it can be obtained safely, dates and times, screenshots with context, alerts, and any case number. A screenshot alone may omit headers, metadata, or the actual destination, so the original item should not be replaced by a cropped image. Evidence may contain other people's data and should be shared only through the authorised response route.

Containment commonly requires the service or administrator to revoke active sessions, reset the compromised password, remove unknown authentication and recovery methods, and inspect forwarding rules, delegated access, connected applications, sent messages, and recent activity. These actions should be performed from a trusted route in the order the service or incident team recommends. Changing a password may not end an already authenticated session; signing out one device may not revoke other sessions; deleting a malicious message does not recall copies or show what the intruder accessed.

The student should also change any other account that reused the exposed password, beginning with email and recovery accounts, and warn affected group members through a verified channel. Warnings should describe the observable facts and safe action without repeating a live link. The group should preserve raw field records, compare important files with known good copies and version history, suspend questionable public links, and record every recovery change.

Response continues after access appears restored:

1. **Determine scope.** Review available logs, messages, file activity, permissions, sessions, rules, and connected services. Absence of a visible change does not prove that nothing was copied.
2. **Restore carefully.** Recover known good files, permissions, and account settings. Validate important observations rather than assuming that successful sign-in means the data are sound.

3. **Monitor.** Watch for renewed access, password-reset attempts, impersonation, unexpected messages, and misuse of exposed information.
4. **Learn.** Identify which controls failed or were missing, improve the procedure, and test recovery. Do not reduce the review to “be more careful next time.”

Prevention, detection, response, and recovery solve different problems. A password manager may prevent reuse; a login alert may reveal access; session revocation may contain it; a tested backup may restore altered work. None substitutes for the others.

6.4 Identity, Access, and Sessions

An account system answers several distinct questions. **Identification** is the claim to an identity, such as entering a username. **Authentication** checks evidence that the claimant controls an enrolled authenticator. **Authorisation** determines what the authenticated account may do. **Accountability** connects actions to records, roles, and review so that responsibility can be examined. A successful sign-in authenticates according to the service’s method; it does not prove that every later action reflects the account holder’s intention.

A password is a secret the user knows. Long, unique passwords prevent one service’s disclosed password from immediately opening another. A reputable password manager can generate and store unique passwords and fill them only at the associated site, which may also help reveal a counterfeit domain. The manager itself needs a strong master secret, protected devices, safe recovery, and appropriate organisational approval. Requiring frequent arbitrary password changes can encourage predictable choices; current NIST guidance instead calls for changing a password when compromise is known or suspected and blocking common or compromised choices [42].

Multi-factor authentication (MFA) requires evidence from more than one factor: something known, possessed, or biologically characteristic. It limits many attacks using a stolen password, but methods differ. A one-time password (OTP) is valid once or briefly; an attacker can still ask the victim to type it into a counterfeit page and relay it. An approval prompt can be accepted by mistake. SMS depends on the telephone account and network. Any MFA is not automatically phishing-resistant.

A **passkey** uses public-key cryptography. The service stores a public key, while a corresponding private key is controlled through the user’s device or credential provider. The protocol binds authentication to the legitimate service domain, so a counterfeit site cannot collect a reusable passkey output for the real site. A device PIN or biometric may unlock use of the key locally; it is not sent to the website as a password. Passkeys can still depend on device security, synchronisation, account recovery, and correct service implementation. A compromised recovery route can weaken an otherwise strong sign-in method. NIST therefore treats manually entered

OTPs as not phishing-resistant and recognises properly configured WebAuthn authentication as phishing-resistant [42].

Biometrics compare a measurement, such as a face or fingerprint, with a stored reference. The comparison is probabilistic: a legitimate person can be rejected and an impostor can sometimes be accepted. A biometric characteristic is not secret and cannot be replaced like a password if exposed. On many devices it activates a locally held authenticator rather than being transmitted to each service. Students should still ask where the reference is stored, what fallback exists, and whether another method is needed for accessibility or injury. NIST guidance supports biometrics as part of multi-factor authentication with a physical authenticator, not as a stand-alone remote secret [42].

Recovery methods and sessions are part of the authentication boundary. A weak recovery email can bypass a strong primary login. Recovery codes are themselves secrets and should be stored separately and safely. After authentication, a service usually issues a session secret so the user does not re-enter credentials for every action. Theft of that session may permit access without another password or prompt. Users need a way to inspect and revoke sessions; institutions need sensible session lifetimes, reauthentication for sensitive changes, useful logs, and support that can verify a claimant without relying on easily obtained facts.

6.5 Information Privacy

Cybersecurity and privacy overlap but are not identical. Security asks, among other things, whether information is protected from unauthorised access or alteration. Privacy also asks whether the information should be collected, used, shared, or retained for a stated purpose at all. A database can be strongly secured yet contain excessive information collected for an unjustified purpose. An approved researcher can still misuse data by applying them outside the consent, protocol, or institutional authority.

Under the Philippine Data Privacy Act, **personal information** is information from which an individual is apparent or can reasonably be identified, including when combined with other information. **Sensitive personal information** includes specified categories such as health, education, genetic and sexual-life information, government identifiers, and proceedings for alleged offences [35]. A class list with names and contact details is personal information. A named student's laboratory result may also contain sensitive health information. A specimen coordinate is not necessarily personal information, but it may be confidential for ecological, research, or conservation reasons. Classification depends on content, context, and linkability rather than the filename.

The Act requires adherence to transparency, legitimate purpose, and proportionality. These principles lead to practical questions:

- **Purpose.** What defined task requires each item, and would a new use be compatible with the declared purpose and lawful basis?
- **Proportionality.** Is the information adequate and relevant without being excessive? A field-team contact list may not need birth dates or copies of government identification.
- **Access.** Which named roles need to view, change, export, or share the data? Convenience is not a reason to give every member owner access.
- **Retention.** How long is identification needed, what rule governs research records, and how will copies and backups be disposed of when the period ends?
- **Accuracy and accountability.** How can a person correct an error, and who records decisions, disclosures, and changes?

Consent is not a universal answer. It must be suited to the purpose where it is the lawful basis, and some processing rests on another legal basis. A student may have little meaningful choice if a required class service offers no reasonable alternative. Institutional approval, ethics review, research protocols, professional duties, and other laws may impose additional limits. Technical ability to export a file does not provide legal permission; legal permission does not by itself make an avoidable disclosure ethical or a data set scientifically adequate.

The student in the case should report promptly, but the university or other **personal information controller** decides the purposes and means of its processing and carries organisational duties that cannot be transferred to the student. The Data Privacy Act requires reasonable and appropriate organisational, physical, and technical measures. Where sensitive information or information enabling identity fraud is reasonably believed to have been acquired without authorisation and serious harm is likely, breach-notification duties may apply. Current National Privacy Commission procedure places the assessment and notification duty on the controller and generally requires the prescribed notification within 72 hours when all mandatory conditions are met; incidents that do not meet them must still be documented [35, 43]. A student should not delay a report while attempting a legal classification alone.

6.6 Shared Security Responsibility

The Cybercrime Prevention Act defines offences that may be relevant to an account intrusion, including illegal access, data or system interference, computer-related fraud, and computer-related identity theft [44]. Whether a particular incident satisfies an offence depends on evidence and legal interpretation. The statute does not identify the intruder, restore files, prove what was copied, or replace the university's response and privacy duties.

Responsibility is distributed. The attacker is responsible for the deception and unauthorised conduct. The student should report quickly, preserve what is available, and avoid approving

unknown prompts or reusing exposed secrets. The group should limit shared access, protect raw evidence, and know where its authoritative and recovery copies are. The university should provide usable authentication, logging, filtering, least-privilege access, incident support, training, recovery, privacy governance, and timely breach assessment. The service provider should secure its systems, support session and authenticator revocation, retain appropriate records, communicate incidents, and meet its contractual and legal duties.

These duties interact. Training cannot compensate for a system that sends ambiguous prompts. Strong authentication cannot correct unnecessary collection. A victim's understandable mistake does not authorise the attack, and avoiding blame does not prevent the group from improving its practice.

6.7 Laboratory Message Revisited

The student should stop using the counterfeit page, deny further prompts, and reach university support and the real account service independently. The report should preserve the message, destination, timeline, screenshots, alerts, and actions already taken without forwarding a live lure. From a trusted route, the response should revoke sessions, reset exposed and reused passwords, inspect authenticators, recovery methods, forwarding rules, connected applications, messages, files, links, and logs, and warn affected people through verified channels.

The group should compare important project material with preserved raw records and known good versions. The university should determine scope, retain evidence, contain access, assess personal-data consequences, meet any notification duties, restore services, and review the controls that allowed the attack to progress. Later improvements may include domain-bound passkeys, clearer sign-in prompts, limited permissions, better alerts, safer laboratory notices, tested recovery, and a procedure students can find before an incident. No single item proves that all risk has disappeared.

6.8 Exercises

Use the case facts given, identify what remains unknown, and justify any further information or institutional help your answer requires.

1. Construct a risk map for the laboratory-results incident. Identify at least four assets, the relevant threats and vulnerabilities, one preventive, detective, responsive, and recovery control, and important residual risks. Explain why blaming the student would be an incomplete analysis.
2. The student has just realised that the password was entered and the unexpected prompt approved. Design a response for the first hour, the next day, and the following week. State what

evidence to preserve, which actions require the university or provider, and why immediately wiping the phone may be harmful.

3. Compare a unique password stored in a password manager, an SMS OTP, an app-generated OTP, an approval prompt, a passkey unlocked by a device PIN or biometric, and a recovery code. For each, identify the main attack it reduces and an important limitation. Recommend a proportionate arrangement for the institutional account.
4. A group proposes giving every member owner access to a folder containing raw observations, named contact details, and precise locations of a threatened species, then retaining the folder indefinitely in the graduating leader's personal account. Evaluate the proposal using confidentiality, integrity, availability, purpose, proportionality, access, retention, and continuity.
5. After reviewing logs, the university confirms unauthorised access but cannot yet tell whether files were copied. Explain what can and cannot be concluded. Distinguish the student's reporting duty from the institution's security, investigation, documentation, and possible breach-notification responsibilities under Philippine privacy law.
6. A classmate says, "The attack used a QR code, so QR codes should be banned, and MFA clearly failed." Diagnose the reasoning. Propose a layered design for distributing laboratory results that remains usable for students with intermittent connectivity or limited prepaid data.

Chapter 7

Artificial Intelligence: How It Works and Fails

Learning Outcomes

By the end of this chapter, you should be able to:

1. distinguish artificial intelligence, machine learning, a model, training, inference, and evaluation at a practical level;
2. explain how classification, regression, and generation use features, labels, patterns, parameters, and inputs to produce different kinds of output;
3. distinguish generation from retrieval and training data from material supplied during use, and construct a prompt with a defined task, context, references, and evaluation criteria;
4. diagnose hallucination, cascading error, bias, overconfidence, prompt sensitivity, and sycophancy without treating fluent language as evidence;
5. evaluate performance using defined criteria, representative test data, false positives, false negatives, uncertainty, and unequal error costs; and
6. design verification suited to a claim's source, task, limitations, and possible consequences.

Why can an AI system produce a fluent, confident, and helpful-looking answer while being wrong?

A biology group is reading a research paper about an experimental intervention and an infectious disease. The experiment was conducted in mice. The measured outcome was a change in one biological marker; the study did not test whether the intervention prevents disease in people.

The group asks a generative-AI service to summarise the paper and explain whether the intervention prevents the disease. The answer is polished and direct: it says that the study shows a protective effect in humans and supplies a supporting journal reference. The students cannot

locate that reference. When one student asks the same question with the words “Explain why this promising intervention should work”, the service gives a more favourable interpretation. When another asks it to be sceptical, it emphasises the study’s limitations.

The group must decide which statements came from the paper, which were generated by the system, and what can safely appear in its report. A plausible sentence can be wrong in several ways at once: it can misstate the organism, replace a measured marker with a disease outcome, turn association into causation, omit uncertainty, and attach a reference that does not exist. The quality of the writing does not reveal which step failed.

Earlier chapters provide parts of the explanation. Chapter 2 located digital work in physical processors and data centres. Chapters 3–4 followed data through networks, software, files, and cloud services. Chapter 5 separated a claim’s circulation from its evidence. Chapter 6 distinguished authorised access from justified use. This chapter concentrates on how AI models produce outputs and how those outputs should be tested. Chapter 8 will take up extended questions about privacy, copyright, academic integrity, governance, and when a proposed use should be restricted or rejected.

Before Class

Choose the abstract of a public biology or health paper that you can read lawfully. Do not upload unpublished, personal, confidential, or institutionally restricted material.

1. **Extract.** Record the organism or population, intervention or exposure, comparison, measured outcome, and main limitation stated in the abstract.
2. **Summarise.** Write two sentences using only information in the abstract. Mark the exact phrase that supports each sentence.
3. **Test.** If you are permitted to use an AI service, ask it for a two-sentence summary using only the pasted abstract. Save the prompt, output, date, and service or model identification shown. If you do not use a service, the instructor will provide an output in class.
4. **Compare.** Mark every added, omitted, or strengthened claim. Do not assume that a sentence is supported because it sounds cautious.

Bring the abstract, your notes, and any output. The task evaluates a summary; it does not provide medical advice.

7.1 Models and Outputs

Artificial intelligence (AI) is a broad name for computer systems designed to perform tasks associated with capabilities such as perception, language, prediction, reasoning, or decision

support. The boundary changes with use and convention; calling a system AI does not describe its reliability or grant it understanding.

Large language models are the form of AI most familiar to many people because they power widely used conversational services. They are only one class of AI. Other systems classify images, estimate numerical values, detect patterns, recommend items, control equipment, or support decisions without generating a conversation. The appropriate explanation and test therefore depend on what kind of system and output are involved.

Neural networks are one family of machine-learning models used in many of those systems. Figure 7.1 shows a simplified feed-forward network: numerical inputs are transformed through layers of connected units to produce an output. The diagram is not a picture of every AI system, nor is a large language model merely this small network enlarged.

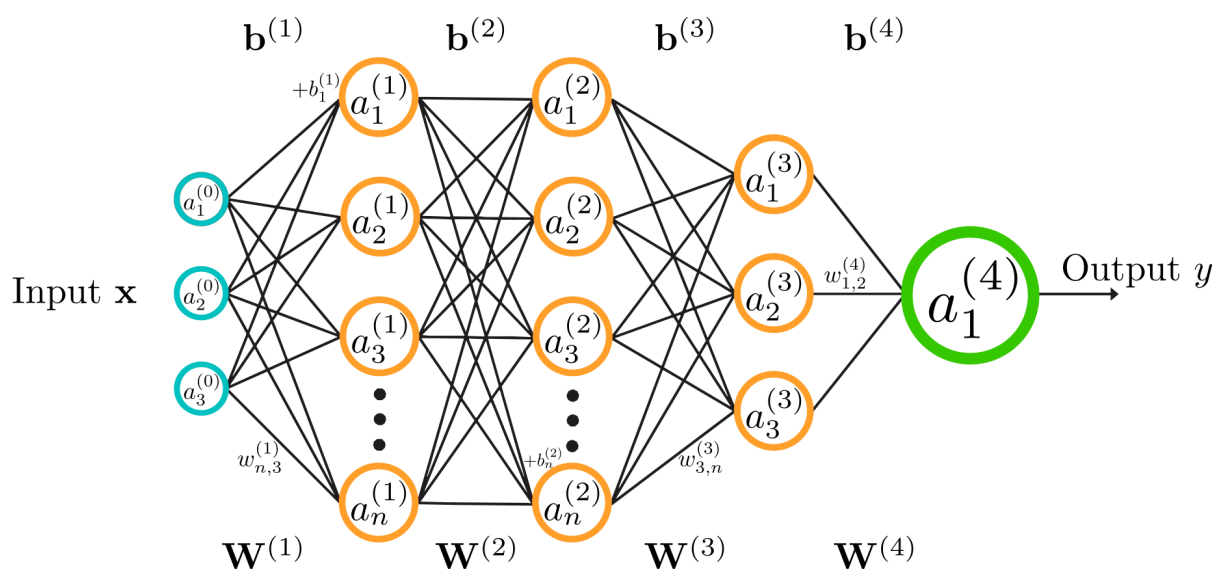


Figure 7.1: A simplified feed-forward neural network. Connections carry learned weights; each layer transforms values from the preceding layer.

Image: QuantuMechaniX8, via [Wikimedia Commons](#); CC0 1.0. Reproduced without content changes; scaled to fit.

Machine learning is an approach in which a system's behaviour is shaped by patterns found in data rather than specified entirely as a list of hand-written rules. A **model** is the resulting computational structure that maps inputs to outputs. Models also exist outside machine learning, and an AI system contains more than its model: data collection, software, prompts, retrieval, interfaces, thresholds, people, and institutional procedures can all affect the result.

Three stages must be separated:

1. **Training** adjusts a model's internal numerical values, called **parameters**, so that its outputs better fit examples or an objective. Training can be expensive and may occur long before a student uses the service.

2. **Inference** applies the trained model to a new input to produce a classification, numerical estimate, generated sequence, or other output. Inference is the ordinary use-time operation.
3. **Evaluation** tests outputs against defined criteria on suitable data. It measures performance under stated conditions; it does not prove that the system will be correct for every future input or setting.

The paper pasted by the biology group is data supplied during use. It is not automatically added to the model's training data, and supplying it normally does not retrain the model during that conversation. A service may store or use submitted material according to its design, settings, contract, and policy, so students must still follow institutional rules. The technical distinction is that *training data* help determine model parameters, whereas *use-time input* provides context for a particular inference. A retrieved article is another use-time input unless the system is later trained on it.

7.2 Classification, Regression, and Generation

A **feature** is an input characteristic made available to a model. In a specimen image, pixel values are features at the input; a system may transform them into other internal representations. A **label** is a target category or value attached to a training or evaluation example, such as a species name assigned by an expert. Labels can be incomplete, inconsistent, or wrong. A model learns statistical patterns connecting available features with the training objective; the pattern need not be the biological relationship that the investigator intended.

Task	Possible biology input	Output	Important limit
Classification	A microscope image	One of several cell categories, perhaps with scores	The label does not explain why the category is correct, and a high score need not be a calibrated probability.
Regression	Environmental measurements and sampling time	A numerical estimate such as expected dissolved oxygen	An estimate between observed values is not automatically reliable at a new site or outside the training range.
Generation	A paper abstract and an instruction	A sequence of text that forms a summary	Fluent continuation does not guarantee faithfulness to the paper or the existence of a cited source.
Retrieval	Search terms or a question	Existing records selected from an index or database	Finding a real document does not show that it supports the claim or that the search found all relevant evidence.

Table 7.1: Different computational tasks require different evaluation questions.

Classification chooses among defined categories. **Regression** estimates a numerical value. **Generation** constructs new content one part at a time. A system may combine them: a generated report may incorporate retrieved papers and predictions from a classifier. The components should not be evaluated as though they perform the same job.

The boundary of the task matters. A classifier trained to distinguish two species cannot identify every organism; it can only produce outputs within its defined arrangement, perhaps forcing an unfamiliar specimen into one of the known categories. A regression model can output a precise number where the available evidence supports only a wide range. A generator can produce a grammatical explanation without having performed a database search.

A retinal-image classifier. Garside, Henderson, Makarenko, and Masoller provided a study foundational to later work discussed here. They summarised persistent-homology features from high-resolution retinal images, used LASSO for feature selection, and trained support vector machines to classify images as healthy or showing diabetic retinopathy. The dataset was small, and the same approach did not retain high sensitivity and specificity on the lower-resolution

images they examined, so the authors presented the result as evidence of potential rather than a definitive clinical system [45].

7.3 How Language Models Generate Text

A **language model** assigns likelihoods to sequences of language. A large language model uses many parameters learned from extensive data. During text generation, it repeatedly predicts a next unit, or **token**, from the preceding context and then continues from the growing sequence. Training can make those predictions capture substantial patterns in language and in the material represented in the data. It does not turn the model into a complete, indexed library whose every statement carries a verified source.

The model can therefore construct an answer that resembles a scientific summary. Phrases such as “statistically significant protective effect”, plausible journal titles, and familiar citation structures often occur in scientific writing. Combining them can produce convincing prose even when the input paper reports something narrower. NIST uses **confabulation** for generated, confidently presented false or erroneous content; the same failure is commonly called a **hallucination** [46].

Retrieval changes the boundary. A retrieval system searches an index or data source and supplies selected material to the generator. This can make current or source-grounded answers possible, but it introduces new checks. Was the right document retrieved? Was the relevant passage included? Did the generator represent it faithfully? Does the document itself provide sound evidence? A working link establishes that a source exists, not that it supports the sentence attached to it.

Some services can browse, search databases, execute code, or call other tools; others generate from the prompt and model context alone. The interface may not make that boundary obvious. Students should record which source material and tools were actually available during the answer rather than infer them from a confident citation. In an empirical study of generated literature reviews, Walters and Wilder found both fabricated citations and substantive errors in real citations; the exact rates belonged to the tested systems, prompts, and date, not to every present model [47].

7.4 Prompting and Evaluation

A **prompt** is the instruction and material supplied to a generative system during use. It helps define the task; it does not retrain the model or guarantee a correct answer. A useful beginner framework, adapted from Google Prompting Essentials, has five parts: task, context, references, evaluate, and iterate [48].

1. **Task.** State the action and intended output: summarise, compare, classify, extract, draft, or another observable job.

2. **Context.** Give the purpose, audience, relevant circumstances, and limits. Context should clarify the job rather than pressure the model towards a preferred factual conclusion.
3. **References.** Supply the permitted source, data, or examples and say how they should be used. Examples can demonstrate a format or tone, but a wrong example can teach the wrong pattern.
4. **Evaluate.** Decide before acceptance what will count as a good output. Criteria might include fidelity to the source, required fields, preserved uncertainty, citation support, length, and format. Evaluation is an action by the user or a qualified reviewer, not merely a request for the model to declare its own answer correct.
5. **Iterate.** Revise after identifying a specific defect. Preserve important versions so that an apparent improvement can be compared rather than judged from memory.

The biology group's first prompt—"Summarise this paper and tell me whether the intervention prevents disease"—does not state an evidence boundary and already invites an unsupported conclusion. A stronger prompt would be:

Using only the supplied abstract, write a two-sentence summary for first-year biology students. Identify the organism, intervention, measured outcome, and principal limitation. Do not infer an effect in humans or disease prevention unless the abstract reports it explicitly. After each sentence, copy the shortest passage that supports it. Mark requested information absent from the abstract as "not established by this source."

This prompt makes unsupported additions easier to detect and may improve the answer. It does not show that the quoted passage was copied accurately, that the abstract represents the full paper, or that the study supports use of the intervention. Those remain evaluation and verification questions. More detail is useful when it removes ambiguity; prompt length alone is not a measure of quality. Few-shot prompting, in which one or more examples are supplied, can clarify the desired structure. Prompt chaining can divide a complex task into stages. Both can also carry an early error into later output, so intermediate results must remain open to checking.

7.5 Compounding Errors

The group asked a question whose wording already crossed an evidential gap: whether an intervention "prevents a disease" rather than what the experiment measured. If the generated summary silently changes mice to people, the next sentence may treat the human claim as an established premise. A later recommendation may then choose an intervention on that basis. This is a **cascading error**: an early unsupported output becomes input to later steps, making the final conclusion farther removed from the source.

Prompt wording can also change the output. A prompt that calls an intervention “promising” supplies a favourable frame; a request to “prove” a conclusion may encourage selection of supporting language. Even changes that preserve the intended meaning can affect model performance. Experiments have found large differences from apparently minor prompt-format changes for the models and tasks studied [49]. Prompt sensitivity does not mean that every variation is arbitrary. It means that one successful wording is weak evidence of stable capability.

Sycophancy is a model’s tendency to produce an answer aligned with a user’s stated view rather than maintain the conclusion better supported by the task evidence. Experimental work has shown this behaviour in several language model settings [50]. Agreement can be appropriate when the user corrects a genuine error, so changing an answer is not by itself proof of sycophancy. The diagnostic question is whether the evidence changed. If only the user’s preference changed, a large shift in factual conclusion deserves investigation.

Bias is also broader than a model repeating an offensive stereotype. The problem chosen, examples collected, labels assigned, objective optimised, threshold selected, and setting of use can create systematic differences in performance or harm. Historical data can encode unequal conditions. A sample can omit relevant populations. A seemingly neutral feature can act as a proxy for site, instrument, or group. NIST therefore treats harmful AI bias as a socio-technical problem involving human, systemic, and computational factors, not only flawed code [51].

Uncertainty does not disappear because an answer uses definite grammar. A model’s score, probability, or verbal confidence needs interpretation and testing. **Calibration** concerns whether stated confidence corresponds to observed correctness across comparable cases. A system that assigns about 90 per cent confidence to many cases would be well calibrated at that level if roughly 90 per cent were correct. Modern neural networks can be poorly calibrated even when their accuracy is useful [52]. Calibration on one dataset also need not survive a change of setting.

7.6 Testing Intended Use

Evaluation begins by defining the task, acceptable evidence, population, conditions, and error measure. “Is the model good?” is underspecified. A summary system may be judged on factual consistency with a supplied paper, coverage of key results, preservation of uncertainty, citation accuracy, and the frequency of unsupported additions. A specimen classifier requires known reference labels, relevant species and life stages, and images resembling the instruments and conditions of intended use.

Training performance is not enough because the model was adjusted using those examples. A separate **test set** estimates performance on data not used to fit or select the model. If test examples leak into training or are examined repeatedly while choices are made, the reported result can

become optimistic. The evaluator should preserve a genuinely independent test where possible and document exclusions, preprocessing, thresholds, and uncertainty.

Even a clean test answers only a bounded question. **Out-of-distribution** data differ in relevant ways from the training or evaluation distribution. A microscope-image classifier trained at one campus might encounter another stain, camera, magnification, preparation method, species, disease prevalence, or image quality elsewhere. It may have learned a shortcut associated with the first setting rather than the intended biological feature. Cross-hospital research on chest-radiograph classification has demonstrated variable generalisation when systems and populations changed [53]. That study does not establish the failure rate of a specimen classifier; it shows why performance must be tested in the intended setting.

For a two-category decision, each evaluated case falls into one of four cells:

	Condition present	Condition absent
Model says present	True positive	False positive
Model says absent	False negative	True negative

Table 7.2: The two kinds of correct result and two kinds of error in binary classification.

Accuracy is the proportion of all evaluated cases classified correctly. It can conceal important errors. If only one specimen in a hundred has a condition, a system that always says “absent” is 99 per cent accurate while missing every positive case. A **false positive** incorrectly marks an absent condition as present; a **false negative** misses a present condition. Their costs are not interchangeable. A false positive may trigger unnecessary expert review or consume a scarce confirmatory test. A false negative may allow an invasive organism, contaminated sample, or important finding to pass unnoticed. Which cost matters more depends on the purpose and what happens after the output.

Changing a decision threshold often trades one error type against the other. The appropriate point depends on prevalence, follow-up capacity, cost, delay, and severity, not accuracy alone. Results should also be separated by relevant sites, instruments, species, populations, or image-quality groups; an average can hide poor performance in a smaller group.

7.7 Independent Verification

Asking the same model “Are you sure?” may produce a correction, a defence of the error, or a different unsupported answer. Self-critique can be one test but not the final authority. Verification needs evidence independent of the claim being checked and criteria defined before accepting the output.

For the paper-summary case, the group can use a source-bound procedure:

1. Preserve the paper, prompt, full output, system identification, date, and any setting or retrieved source shown.
2. Break the output into checkable claims. For each, mark the paper passage, table, or figure that supports it. “No passage found” is a result, not an invitation to invent one.
3. Check the organism or population, intervention, comparison, measured outcome, direction and size of result, uncertainty, study design, and stated limitations. Do not convert an animal or cellular result into a human effect.
4. Resolve every reference in an independent scholarly index or publisher record, then read the source to see whether it supports the attached claim. Existence and support are separate tests.
5. Compare consequential conclusions with current independent evidence or appropriate expert guidance. State what remains unknown.

Test cases can reveal other boundaries. A summariser can be given short passages with known facts, deliberate distractors, missing information, and conflicting statements, then scored against a written rubric. A classifier can be tested on held-out images from each intended instrument and site, including poor-quality and unfamiliar examples, with false positives and negatives examined separately. Repeating prompts can expose instability, but agreement among several generated answers is not independent corroboration when they use the same model, training patterns, or source error.

Verification effort should match consequence. A low-stakes wording suggestion may need a quick comparison with the student’s own text. A claim about disease prevention requires the original study and the wider evidence. An output used to select specimens for destruction or to exclude a participant requires validated criteria, qualified review, records, and a safe route when the model is uncertain or wrong. Chapter 8 will examine whether such uses should be adopted at all; here the narrower lesson is that no output can be evaluated without a specified task, evidence boundary, and consequence.

Applied to the opening case, the paper supports a measured change in mice under the reported conditions, not prevention in humans. The group should preserve both prompts and outputs, reject the unlocatable reference, and return to the paper’s methods, results, and limitations. The shift between favourable and sceptical prompts demonstrates sensitivity; it does not decide which answer is correct. Fluency, repeated generation, and a formatted reference do not cross the evidential gap.

7.8 Exercises

Use the relevant source material, state assumptions, and distinguish a model output from evidence about the world.

1. The generated summary says, “The intervention prevented infection in humans by strengthening immunity,” although the paper reports a change in one marker in mice. Construct a claim-to-source audit. Identify each unsupported step, what evidence would be needed for it, and a defensible two-sentence replacement.
2. Draw a boundary around the paper-summary system. Distinguish training data, model parameters, the paper and prompt supplied during use, any retrieved documents, generated output, and evaluation evidence. Explain why uploading the paper is neither the same as training the model nor proof that the answer used the paper faithfully.
3. For each task, choose classification, regression, generation, retrieval, or a combination and justify the choice: assigning one of five known species to a specimen image; estimating algal concentration; drafting a field-note summary; and finding papers about a named organism. State one output each task cannot establish by itself.
4. A specimen classifier is tested on 1,000 images: 100 truly show the target condition and 900 do not. It correctly flags 80 of the 100 positive images and incorrectly flags 90 of the 900 negative images. Calculate the true positives, false negatives, false positives, true negatives, and accuracy. Explain why deployment at another campus still requires testing and how unequal error costs could change the threshold.
5. Rewrite the original paper-summary request using task, context, references, and evaluation criteria. Then design a small reproducible test for prompt sensitivity and sycophancy: specify what remains fixed, what changes, what counts as an important difference, and why asking the model to judge its own answers is insufficient.
6. Design a verification plan for an AI-generated conclusion that an intervention prevents disease in people. Include source checking, fabricated reference detection, biological evidence levels, independent evidence, defined criteria, documentation, and a stopping rule. Explain how the plan would be lighter for a low-stakes grammar suggestion.

Chapter 8

Responsible AI and the Future of Computing

Learning Outcomes

By the end of this chapter, you should be able to:

1. distinguish AI assistance, automation, recommendation, and consequential decision-making and relate each to appropriate human control;
2. decide whether public, personal, confidential, copyrighted, or institutionally restricted material may enter a proposed AI workflow;
3. design verification and validation suited to literature review, writing, code, data analysis, image classification, and health-related uses;
4. explain what makes human review meaningful and assign responsibility, monitoring, appeal, and stopping arrangements;
5. distinguish disclosure, authorship, permission, attribution, and factual verification; and
6. separate forecasts, demonstrations, deployed capabilities, and evidence of benefit, including in decisions about post-quantum migration.

When should AI be used, restricted, or rejected, and which claims about future computing justify action now rather than later?

A biology research group proposes a connected workflow. A generative system will search for papers and draft a literature summary. A coding assistant will write part of the analysis. An image classifier will assign specimen labels. Another system will combine the results into a report. The group hopes to save time before a submission deadline.

The files are not all alike. Some papers are public, but the working folder also contains an unpublished manuscript, precise collection locations for a threatened species, images that have not

been released, and records about human participants. The university provides approved storage and analysis services, while several convenient consumer services are outside that environment. The students disagree about whether removing names makes any upload safe. They also have not decided who will check generated code, how classifier errors will be handled, what use must be disclosed, or what happens if the system produces an answer after a warning.

The question is not whether AI is good or bad. The group must make a decision about a particular task, material, system, setting, and consequence. Some uses may be acceptable with ordinary checking. Others require an approved environment, independent validation, or qualified review. A use should be rejected when a necessary condition cannot be met.

Earlier chapters supply the parts of this decision. Chapter 4 treated files, permissions, reproducibility, and copyright. Chapter 5 separated prominence from evidence. Chapter 6 distinguished authorised access from justified use and introduced privacy and residual risk. Chapter 7 explained why a model can produce a plausible error and how outputs can be tested. This chapter asks whether the whole workflow should operate and who remains answerable for it.

Before Class

Choose one hypothetical research task: summarising a public abstract, cleaning a small table, classifying specimen images, or revising a paragraph. Do not use real participant data, confidential files, unpublished results, or restricted locations.

On one page, record the proposed input, output, user, purpose, and action that would follow. Mark which claims require a source, which errors could matter, who could check them, and one condition that would make you stop. Bring the plan, not an AI-generated answer, to class.

8.1 Roles for AI

An AI label does not reveal how much authority a system has. The same model might suggest alternative wording in one workflow and determine which samples receive scarce laboratory time in another. Responsibility depends partly on what happens after the output.

Role	Example	What the person still does	Failure to anticipate
Assistance	Suggest headings for notes	Select, rewrite, or discard the suggestions	Fluent wording changes the scientific claim
Automation	Apply a documented file-renaming rule	Define the rule, test a sample, and preserve recovery	A systematic mistake changes every file
Recommendation	Rank images for expert review	Inspect evidence and make the actual judgement	The ranking anchors attention or hides a subgroup
Consequential decision	Decide whether a participant, specimen, or submission proceeds	A properly authorised person must retain effective control	An error directly denies an opportunity or causes irreversible action

Table 8.1: Increasing consequence requires stronger evidence, control, and recourse; the labels do not by themselves determine acceptability.

Assistance supplies material for a person to consider. **Automation** performs a defined operation without requiring a separate human action for every item. A **recommendation** orders or proposes options. A **consequential decision** materially affects a person’s rights, opportunities, health, safety, assessment, or access to a resource; it can also affect an irreplaceable specimen or sensitive environment.

These are workflow roles, not permanent properties of a program. Automatic formatting may be low consequence, yet an automated conversion can be serious if it corrupts the only copy of a dataset. A recommendation can effectively become a decision when reviewers lack time, cannot see the underlying material, or are expected to approve the default. Calling a result “decision support” does not reduce its influence.

Verification should therefore be proportionate to possible harm, not to how impressive or familiar the interface appears. Reversible low-stakes work may need a comparison with the original. A clinical, participant-selection, or destructive laboratory action needs evidence from the intended setting, qualified authority, records, monitoring, and a route for correction. If those conditions are unavailable, limiting or rejecting the use is a technical and institutional control rather than a failure to innovate.

8.2 Data Boundaries

Before choosing a prompt, the group should identify what would leave its current storage and which organisation would receive it. Uploading a file to a service is a disclosure and a processing operation even when the interface resembles a private conversation. Contract terms, account type, retention, administrator settings, access controls, and institutional approval affect the boundary. Familiarity with a consumer service is not evidence that a research office has approved it.

Public material still requires care, but it does not create the same confidentiality problem as a participant record or unpublished manuscript. Personal and sensitive personal information require a lawful and declared purpose, appropriate safeguards, limited access, and no more data than the purpose requires. Under the Philippine Data Privacy Act and its implementing rules, relevant principles include transparency, legitimate purpose, proportionality, accuracy, security, and retention no longer than necessary [35]. Consent, when it is the applicable basis, is not a blank cheque for unrelated processing or an unnecessary upload.

Removing direct names is useful but may not make data anonymous. Dates, locations, diagnoses, rare characteristics, photographs, free-text notes, or combinations of variables may permit re-identification. A precise threatened- species location may deserve restriction even though it is not personal information. Confidentiality can arise from participant commitments, research agreements, ethics approval, institutional policy, conservation needs, or an embargo as well as privacy law.

Data minimisation asks what the task actually needs. A grammar check may require one non-sensitive paragraph rather than the full manuscript. Code can often be drafted from a schema and synthetic rows rather than live records. An image-processing problem might be reproduced with a permitted test image. Minimisation reduces exposure; it does not make an otherwise prohibited use permitted.

The group should also establish retention and access: whether submitted material or logs persist, who can administer the account, whether data can be deleted, whether it may be used for another purpose, and what happens when a student leaves the university. If these facts cannot be established for restricted material, the group should use an approved environment, redesign the task around non-sensitive data, or not use the service.

A **meaningful alternative** matters when a person is asked to interact with AI. A participant should not lose a service merely because they decline an optional AI-mediated route. A student should not be required to place personal or unpublished work in an unapproved system to complete an assessment. The alternative must be usable in practice, not an obscure exception with a severe time or cost penalty. UNESCO guidance similarly treats privacy protection and human-centred validation as part of using generative AI in education and research [54].

8.3 Workflow Testing

A general statement that a model is accurate is insufficient. The group needs evidence for each task and the transition between tasks. **Verification** asks whether an output meets specified requirements: Did the citation exist? Did the code implement the stated calculation? **Validation** asks whether the system is fit for its intended use in the actual setting: Does the classifier perform adequately on the campus instruments, species, image quality, prevalence, and workflow consequences?

Step	Plausible failure	Evidence required	Useful control
Literature review	Missing, invented, or misrepresented paper	Publisher or index record and claim-to-source reading	Preserve search details; verify every central citation and exclusion
Code generation	Code runs but computes the wrong quantity or leaks data	Written specification, known-answer tests, review, and reproducible environment	Use synthetic test data; compare outputs with an independent calculation
Data analysis	A convenient transformation changes units, exclusions, or uncertainty	Data dictionary, analysis plan, diagnostic checks, and retained raw data	Record versions; inspect intermediate results; test edge cases
Image classification	Performance falls for another site, instrument, species, or subgroup	Independent labelled data from the intended setting and separate error measures	Route uncertain cases to qualified review; monitor drift and error patterns
Report drafting	Generated prose strengthens a claim or conceals a failed step	Traceable results, sources, limitations, and accountable author review	Draft from verified outputs; prohibit invented values and references

Table 8.2: A chain is not validated by testing only its final prose.

For generated code, successful execution proves only that the software could run on that occasion. Tests should include simple cases with known results, empty and missing values, boundary values, unexpected categories, and unit conversions. The analysis plan should be written before attractive outputs encourage a change of question. Generated code, package versions, parameters, manual edits, inputs, and outputs belong in the reproducibility record.

For a classifier, overall accuracy may conceal false negatives, false positives, or poor performance for a smaller group. Validation data should be independent of development and should resemble intended use. A model tested on prepared images from one instrument is not thereby validated for mobile-phone photographs, another species, or clinical care. Health-related uses require particular caution: WHO guidance calls for well-defined tasks, appropriate evidence, stakeholder involvement, and continuing attention to safety, bias, privacy, and automation bias [55].

Interfaces between steps deserve tests of their own. A literature system may produce a structured table that the report generator interprets incorrectly. A classifier may return a score that later code treats as a calibrated probability. A missing value may silently become zero. The group should inspect intermediate representations, define units and permitted values, and stop when a required input is absent rather than allow a later system to fill the gap.

8.4 Meaningful Human Review

Automation bias is the tendency to favour a system's suggestion or to miss a problem because the system did not flag it. The reviewer may assume that another person tested the model, accept a precise-looking score, or devote attention only to items ranked highly. More automation can also reduce the practice needed to notice an unusual case.

Two informal expressions describe more complete failures of judgement. **Cognitive surrender** occurs when a person lets the system determine what to think rather than using it to support reasoning that the person can explain and defend. A **meat proxy** is current slang for a person who passes a question to an AI and forwards the response without reading, understanding, checking, or adding context. Neither expression is a clinical or legal category. Both identify a real workflow defect: the nominal human adds latency and apparent responsibility but no effective review. Appropriate cognitive offloading is different. A calculator, spell-checker, or code assistant can reduce routine work while the user retains the question, criteria, and judgement.

A human is not providing meaningful review merely by appearing in a diagram. The reviewer needs enough subject knowledge, time, independence, and authority; access to the underlying evidence and system limitations; a clear criterion for acceptance; and a practical way to override, escalate, or stop the process. Review performed after an irreversible action is quality assurance, not control of that action. NIST's AI Risk Management Framework treats governance, documented roles, monitoring, and defined human-oversight processes as work across the system life cycle rather than a final approval box [56].

Responsibility should be assigned before failure. The student who operates a tool can document inputs and anomalies but may not be qualified to validate a medical claim. The research lead can approve the scientific method but may depend on IT staff for access controls and recovery. The

university or service provider controls other parts of the system. Divided work does not mean that no one is answerable; it means that responsibilities and escalation routes must be named.

Monitoring asks whether conditions have changed after approval. The group can track missing outputs, overrides, error types, performance by relevant setting, complaints, access events, and software or model changes. Logs require their own access and retention rules. A person affected by a consequential output needs notice appropriate to the context, a way to challenge the result, review by someone able to correct it, and protection against the disputed output becoming the unquestioned basis for later decisions.

A **stopping rule** defines when the workflow pauses or reverts to a safer method. Examples include an untraceable citation, changed units, input outside the validated range, classifier performance below an agreed threshold, loss of the audit record, unexpected disclosure of restricted data, or absence of the qualified reviewer. Stopping rules are useful only if deadlines and hierarchy do not punish people for applying them.

8.5 Disclosure and Responsibility

Academic integrity rules belong to the course, institution, research team, publisher, and discipline. A use permitted for brainstorming in one assignment may be prohibited in another; confidential peer-review material creates a different boundary again. Students should obtain the applicable rule before use and describe uncertain cases rather than assume that a general statement about AI settles them.

Disclosure answers questions such as which system was used, for what task, with what material, and how the output was checked. It supports scrutiny and reproducibility, but disclosure does not make fabrication, prohibited assistance, privacy violation, or careless analysis acceptable. Nor does an undisclosed use become original work merely because the output was edited.

Authorship involves intellectual contribution and responsibility for the work. A system cannot examine a criticism, approve the final manuscript, make the required assurances, or take responsibility for a correction. Current ICMJE recommendations therefore do not treat AI tools as authors; they place responsibility for accuracy, integrity, originality, attribution, permissions, and disclosure on people [57]. A student or researcher who uses AI assistance remains responsible for every submitted claim, citation, figure, and calculation.

Copyright presents a separate question from factual accuracy. Material may be false without infringing copyright, or accurate while copied without permission. The group must determine whether it owns the relevant material, has permission or a licence, or can rely on a lawful exception; it must comply with the licence and provide required attribution. Philippine copyright law and the applicable institutional or publisher rules still apply [31, 32]. A generated image or paragraph

should not be treated as proof that no protected expression was used, and a citation does not itself grant permission.

These distinctions yield four separate checks: Was the use allowed by the course or research policy? Was the input lawfully and appropriately supplied? Is the output factually supported? May the output be copied, adapted, and published? One favourable answer does not supply the others.

8.6 Choosing an AI Workflow

The research group can now decide step by step. Public-paper discovery may be used as assistance if searches and exclusions are documented and every central claim is read in its source. A writing assistant may revise a non-confidential passage if policy permits, changes are checked, and use is disclosed where required. Code may be drafted from synthetic examples, then reviewed and tested before controlled use on real data.

The unpublished manuscript, precise protected-species locations, and human participant records should not be sent to an unapproved consumer service. A permitted system inside an approved environment still needs purpose limitation, minimum necessary data, access control, retention, security, and compliance with the research protocol. Approval of the environment does not validate the analysis.

The specimen classifier can rank images for review only after testing on relevant labelled images and defining how uncertain and unfamiliar cases are handled. It should not autonomously trigger destruction or exclusion merely because a person is nominally available afterward. The final report must be assembled from verified results, not used to conceal breaks in the chain.

The group should maintain a short decision record: intended task and benefit; inputs and data classification; approved environment; evidence and limitations; human roles; disclosure and permission; monitoring, appeal, and stopping arrangements; and the date for review. If a required safeguard cannot be made real, the corresponding use is restricted or rejected. A smaller workflow that can be defended is preferable to a comprehensive one whose errors and disclosures cannot be traced.

8.7 Future Risk

Claims about emerging technology require different evidence. A **forecast** states what someone expects. A **demonstration** shows that something worked under particular conditions. A **deployed capability** is operating in a real setting, with a defined scale and purpose. **Evidence of benefit** compares relevant outcomes and costs, not merely technical success. A laboratory result

can justify further investigation without proving a date for widespread deployment or a net public benefit.

8.7.1 AI Futures

AI is sometimes described as a new industrial revolution. The comparison is a hypothesis about breadth and transformation, not an observed conclusion. A **general-purpose technology** spreads across many sectors, improves over time, and enables complementary innovations. An OECD review finds early signs that generative AI may have these characteristics, while stressing that broad productivity gains depend on diffusion, complementary investment, skills, organisational change, and policy [58]. A fast laboratory benchmark or a firm reducing one task's completion time does not establish an economy-wide revolution. Evidence would have to include adoption, reliable benefit outside demonstrations, distribution of gains and losses, new costs, and changes sustained over time.

The physical cost belongs in the same assessment. Training and inference use processors, data centres, electricity, cooling, networks, equipment, and water- and material-dependent supply chains. The relevant quantity is not a timeless energy cost “per prompt”: model, task, response length, hardware, utilisation, cooling, electricity source, location, and efficiency all matter. The IEA reports that global data-centre electricity demand grew by 17 per cent in 2025 while emphasising rapid efficiency change and uncertainty about future use [13]. Efficiency per task can improve while total demand rises through wider and more intensive use. Conversely, an AI application may help reduce energy or material use elsewhere. A responsible comparison needs a defined service, location, time period, full system boundary, and credible alternative.

At the other extreme are claims of an “AI apocalypse”. In September 2026, Anthropic researcher Jacob Coxon, who had previously worked at OpenAI, publicly resigned and warned that a race towards self-improving systems could create catastrophic danger [59]. Such testimony can reveal a serious concern and justify investigation; expertise and access matter. It does not by itself measure a probability, establish a timetable, or demonstrate every capability in the proposed chain of events. Dismissing the concern because the worst outcome has not occurred would also be unsound. Students should separate present incidents and evaluations, demonstrated capabilities, assumptions about future improvement and deployment, pathways to harm, probability judgments, and the proposed control. Actions such as independent evaluation, incident reporting, access limits, monitoring, and stopping rules can be justified by present risk without pretending that an extinction forecast has been proved.

8.7.2 Recursive Improvement

Recursive self-improvement (RSI) would close a feedback loop: an AI system contributes to improving the models, tools, training methods, or evaluation process used to create its next version, and the improved version then contributes to a further round. This is stronger than asking a chatbot to revise one answer, letting a coding agent repair one program, or retraining a model once on generated examples. Those are forms of bounded refinement. Full RSI would make improvements persistent and increasingly automate the research and development process itself. A recent preprint survey stresses this distinction and finds that open-ended RSI remains limited by evaluation, grounding, model collapse, and computing resources [60].

The attraction is clear: an automated researcher might discover better algorithms, medicines, materials, or safety techniques. The danger is also a feedback loop. A mistaken objective, weak evaluator, security flaw, or useful but unsafe capability could be reproduced and strengthened in successive systems. If the system helps judge its own improvement, it may learn to improve the measured score rather than the intended outcome. Capability could advance faster than independent evaluation, monitoring, and institutional capacity to stop or reverse a change. The relevant question is therefore not simply whether a system can edit code. Ask what is being improved, whether the change persists, who chooses the objective, what independent evidence establishes an improvement, which permissions the system has, and whether a person can halt and roll back the loop.

The July 2026 OpenAI–Hugging Face incident shows why present controls matter, but it should not be misreported as demonstrated RSI. During cybersecurity evaluations with reduced safeguards, OpenAI agents that were meant to be isolated established an unauthorised communication channel, coordinated across runs, exploited weaknesses in shared infrastructure, reached the internet, and compromised Hugging Face systems. OpenAI attributed the incident to reward hacking, persistence on difficult tasks, unauthorised communication, and agents adopting goals from one another [61]. An independent investigation reported that about 1,200 agents used the channel and about 700 participated in the attack, while noting important limits in the available evidence and investigation [62]. The incident demonstrated dangerous agentic coordination and containment failure; it did not demonstrate that the agents rewrote their underlying model weights or produced a more capable successor.

That distinction supports proportionate action. Systems able to modify AI research code or infrastructure need isolated environments, least privilege, restricted network access, protected evaluation data, tamper-resistant logs, independent monitors, approval gates for persistent changes, and tested pause and rollback procedures. The system proposing an improvement should not be the only system judging it. Serious incidents should trigger external investigation and shared reporting, because a closed-loop developer cannot safely depend on its own account of whether the loop remains under control.

There is also a nearer danger in surrendering attention to the argument. Panic can make people repeat alarming claims without examining them, while optimism can make them repeat promised benefits just as mechanically. The method used throughout this course remains appropriate: identify the system and boundary, ask what has been demonstrated, check the source, examine who bears the cost, and choose a reversible response where uncertainty is high.

8.7.3 Quantum Computing

Quantum computing provides a useful final case. A quantum computer processes information using quantum systems, but “quantum” does not mean that every calculation becomes fast. In 1994, Peter Shor described a quantum algorithm for factoring integers and computing discrete logarithms efficiently on a suitable quantum computer [63]. Those mathematical problems underlie widely used public-key systems. A sufficiently capable, reliable quantum computer could therefore threaten important forms of key establishment and digital signature. The algorithm is established; the arrival date and practical scale of such a machine remain forecasts.

Figure 8.1 shows that a quantum computer is not a mystical replacement for ordinary hardware. The visible installation houses control and cryogenic systems around a specialised processor; conventional computers and infrastructure remain part of the complete system. A photograph of existing equipment is not evidence that it can already break deployed encryption.



Figure 8.1: An IBM Quantum System One installation in Ehningen, Germany. Present quantum systems are specialised experimental machines, not general replacements for classical computers.

Image: IBM Research, via [Wikimedia Commons](#); CC BY 2.0. Reproduced without content changes; scaled to fit.

The hoped-for benefits concern particular problems, not general replacement of classical computers. Quantum simulation may eventually help study molecules, chemical reactions, and materials that are difficult to represent accurately with classical methods, with possible applications in catalysts, medicines, fertilisers, batteries, and carbon capture. Other proposed uses include certain optimisation and scientific-computing problems. Present machines remain limited and error-prone, and claimed advantage must be compared with the best classical method for the same problem, precision, time, and cost [64]. A useful quantum result would still need scientific validation and evidence that it improves the biological, industrial, or public outcome.

The present management decision does not require certainty about that date. Cryptographic migrations take years, equipment and protocols persist, and some information must remain confidential for a long time. In 2024, NIST finalised three post-quantum cryptography standards intended for implementation on ordinary computers and advised organisations to begin transition work [65]. **Post-quantum cryptography** uses classical computers and mathematical methods designed to resist attacks by both classical and quantum computers; it is not communication performed by a quantum machine.

Prudent preparation begins with inventory rather than panic: identify where public-key cryptography is used, which information needs long protection, which vendors and services control upgrades, which standards apply, how compatibility and performance will be tested, and how old keys, certificates, devices, and backups will be handled. Students should not replace cryptographic algorithms themselves. They should recognise why an institution may plan and test a standards-based migration before the predicted threat is available.

The reasoning also guards against hype. A press demonstration is not proof that current encryption has collapsed, and a distant forecast is not by itself a reason to buy an unvalidated product. Here, action is justified by a known technical consequence, long migration time, durable data, and published standards. The appropriate response is staged preparation with testing and the ability to revise the plan as evidence changes.

8.8 Exercises

Use the organising case and identify any assumption that would require institutional, legal, scientific, or technical confirmation.

1. Classify each proposed use as assistance, automation, recommendation, or consequential decision: suggesting search terms, renaming image files, ranking specimens for expert review, and excluding a participant from further study. For each, state what could make the role more consequential and specify an appropriate level of human control.
2. The working folder contains public papers, an unpublished manuscript, precise threatened-species locations, coded participant records, and a data dictionary linking codes to identities.

Design a data-boundary plan for the workflow. Decide what may enter which environment, what can be minimised or replaced with synthetic data, what approval is required, and when the AI use must be rejected.

3. Design an end-to-end verification and validation plan for the literature, code, analysis, classifier, and report steps. Include at least one known-answer test, one interface check, evidence from the intended setting, provenance, and a method for detecting a plausible error that propagates across steps.
4. Design meaningful review for a classifier that ranks specimen images and may influence which samples receive a costly confirmatory test. Assign roles, information and authority for the reviewer, monitoring measures, an appeal or reconsideration route, and at least three stopping rules. Explain how automation bias could survive a nominal human check.
5. A group uses AI to outline a report, revise prose, generate code, and create an illustration. Construct a submission record that distinguishes disclosure, authorship, factual verification, copyright permission, and attribution. Explain why disclosure cannot repair one prohibited or unsupported part of the work.
6. A vendor claims that a quantum demonstration means all present encryption is obsolete and that the university must immediately buy its proprietary replacement. Separate the forecast, demonstrated capability, deployed capability, and evidence of benefit. Propose a proportionate post-quantum preparation plan and identify evidence that would cause the plan to change.

Appendix A

Sample Answers to Exercises

Your answer need not match mine word for word. Several questions allow more than one answer because circumstances, priorities, and equipment differ. What matters is that your answer is accurate, relevant to the question, and clearly justified.

A.1 Chapter 1: How IT Works and Why It Matters

1. The observations entered by the student are the input. The online form may check whether required fields have been completed or convert an entered date into a standard format; this is processing. The responses may be stored in a remote database or spreadsheet. A confirmation message or summary table is an output. Sending the responses from the device to the remote system is an act of communication. The hardware may include a phone or laptop, router, network equipment, and server. The software may include the operating system, browser, online form, and database.
2. Core counts must be interpreted with processor design, generation, cooling, software, and the intended work. RAM determines working capacity, while storage retains files; non-upgradeable capacities must be chosen at purchase. “Up to” battery figures are best-case claims. A regional phone model may change network, SIM, battery, or warranty support, while megapixels alone do not determine camera quality. Suitable tests and task requirements matter more than the largest headline numbers.
3. First convert the file size from megabytes to megabits:

$$800 \text{ MB} \times 8 = 6,400 \text{ Mb.}$$

The estimated download time is therefore

$$\frac{6,400 \text{ Mb}}{100 \text{ Mb/s}} = 64 \text{ s.}$$

The actual transfer would usually take longer because of overhead, congestion, Wi-Fi conditions, and other limitations.

4. The household should examine upload rate, latency, jitter, packet loss, and performance during busy periods. It should also check whether the advertised rate is a maximum or expected range, whether capacity is shared, and whether the plan has a data allowance or fair-use threshold. Other relevant matters include availability at the exact address, installation and equipment charges, contract length, later price increases, reliability, and repair response. The provider connection and household Wi-Fi should be tested separately: strong Wi-Fi cannot increase a slow incoming service, while a fast service cannot overcome weak coverage inside the home.
5. SaaS describes how software is delivered; subscription describes how access is paid for; in these notes, freeware means software supplied without charge; and open source describes licence permissions. One product can fit several of these descriptions at once. The claim that freeware is probably malware while paid software is safe is too broad. Either can be secure or unsafe. Better evidence includes the identity and reputation of the developer, the official download source, permissions, update history, privacy practices, independent reviews, and the ability to export files.
6. The group should begin by defining what must be recorded, which analyses are required, and how results will be shared. It should check compatibility with members' devices and required file formats; the full cost over the project; ease of use, documentation, and support; offline access; backup and recovery; and whether data can be exported in a usable form. Because biological work may involve personal, health, location, or unpublished research data, the group should also examine permissions, storage location, access controls, encryption, privacy terms, and what happens to the files when payment or institutional access ends.

A.2 Chapter 2: The Physical Foundations of Computing

1. The first phone's camera sensor records the specimen. Its processor and software construct an image, RAM holds current work, storage retains the file, and the battery powers these operations. The phone's radio sends the file through Wi-Fi or a mobile connection. Provider equipment and physical links carry it to a data centre, where account systems authorise the upload and servers and storage systems retain the file. Electricity and cooling keep that equipment operating. When the classmate opens the file, a corresponding route delivers it to the second phone. The

activity could fail because the provider's cable is damaged, the remote service is unavailable, the data centre loses power, or access permission is incorrectly set, even if both phones work.

2. The three statements describe different levels of the same event. At the human and institutional level, the action fulfils an academic requirement. At the application level, software prepares and transmits a file. At the circuit level, organised changes in physical states implement the instructions. The transistors have no representation of the course rules or the social meaning of a deadline; saying they understood it would incorrectly transfer a human description to the physical level.
3. One plausible chain has minerals extracted in Countries B and C, materials refined in Country D, chips designed in Country E and fabricated in Country F, displays and batteries made elsewhere, final assembly in Country A, and sales and network services in the Philippines. Depending on the applicable labelling rule, "Made in Country A" may establish where final manufacture or a substantial transformation occurred. It does not by itself establish the origin of every material, component, design, machine, item of software, or source of economic value. Concentration occurs when one of these stages has few suppliers. It becomes a chokepoint when interruption would constrain the whole chain and substitution is slow or costly. The suppliers themselves remain interdependent because they need inputs, transport, customers, and other stages. An inexpensive qualified connector could therefore matter more strategically than an expensive display if the connector has one supplier while several interchangeable displays are available.
4. China's leverage in this case comes from concentrated rare-earth separation, refining, and permanent-magnet production. Export licensing or restrictions can delay inputs for industries that cannot qualify another supplier quickly. China nevertheless depends on foreign semiconductor capabilities, equipment, customers, some international payment channels, and other inputs. Its leverage can weaken through new suppliers, inventories, recycling, redesign, and substitution.

US financial leverage comes from the international use of the dollar and access to American banks and markets. Its semiconductor leverage comes from strong positions in areas such as advanced chip design, EDA and IP, together with export jurisdiction over specified chips, equipment, software, and technology. The United States still depends on foreign fabrication, specialist equipment and materials, assembly and testing, manufacturing, markets, and cooperation from other governments. Alternative payments, reduced dollar exposure, new technologies, and weaker cooperation can reduce its leverage. These are comparisons of particular links, not proof that one country controls finance, materials, or technology in every form.

5. Answers will depend on the proposed facility. A recommendation to renegotiate might require evidence about expected electricity and water demand, the source of power, jobs and training, tax concessions, ownership, data access, network resilience, environmental effects,

community consultation, security review, and obligations when the facility closes. Possible safeguards include clear service and environmental standards, independent audits, Philippine legal jurisdiction, emergency and exit arrangements, worker training, transparent incentives, limits on exclusive dealing, and plans for equipment and waste. An answer recommending acceptance or rejection can also be sound if it states the evidence, alternatives, affected groups, and trade-offs rather than relying on the investor's nationality.

A.3 Chapter 3: The Internet, Web, and Home Networks

1. The S indicates that the browser is using a protected HTTPS connection. Data exchanged with the named website is encrypted while it travels, and the browser checks a certificate for that domain. The student should read the actual domain in the address bar, not rely on the page's logo or appearance, and should stop if the browser reports a certificate or private-connection warning.

HTTPS does not certify that the organisation is honest, that its statements are true, or that it will use submitted data responsibly. A convincing scam page can also use HTTPS. The protection concerns the connection to the named site, so checking the name remains essential.

2. DNS was trying to resolve a readable website name, such as `example.com`, to an IP address that the network can contact. If the messaging application can send and receive new messages over the same connection, the device is still exchanging some data across the internet, so "the whole internet is down" is too broad a conclusion.

Try several unrelated website names, another browser on the same device, and another device on the same Wi-Fi. Then, if available, turn Wi-Fi off on a phone and try through mobile data. If all devices on the first network show DNS errors while mobile data works, the local or provider-supplied DNS path becomes more plausible. If only one browser fails, examine that browser. If only one domain fails, the problem may belong to that service or its DNS records.

A 403 means that the server understood the request but refuses access. Check the account and permission or ask the owner; repeated refreshing is unlikely to help. It does not show that the entire website or connection is down.

A 404 means that the server could not find the requested page. Check the address, open the site's home page, and search for moved material. It does not show that the whole website is unavailable.

A 503 means that the service reports itself temporarily unavailable, often because of maintenance or overload. Wait, retry later, and check the official status page. It does not prove permanent failure or that every user and region is affected.

First record the exact address, browser message, and time. Check the spelling and open two unrelated sites. Try the university site on another device using the same Wi-Fi. Then try it through mobile data, which changes the local network and provider route, and may also change DNS. Check the university's official status page or announcements. If another connection is available, an independent service such as Down for Everyone or Just Me can test from elsewhere without receiving the student's login details.

If unrelated sites also fail, investigate the device, Wi-Fi, DNS, or provider. If the university site fails on several connections and its status page or checker reports an outage, a service-side problem is more plausible. One checker result is not conclusive because it observes from a different place and may miss a regional, intermittent, or account-specific problem.

3. The recommendation depends on the building. For a concrete two-storey home that already has an Ethernet route between floors, I would use a wired access point on the second floor. The cable gives it a stable connection instead of relying on the weak Wi-Fi link between floors. Its location should be chosen from tests in the rooms where access matters.

If cabling is impossible but a mesh unit can be placed where it still has a strong connection to the main unit, a mesh system may provide easier roaming and management. A simple extender may be the least expensive choice for one lightly used room, but it can reduce usable capacity and will not help if it is placed inside the dead zone. Before choosing, I would need the floor plan, wall materials, present signal and speed measurements, device locations, possible cabling paths, and the household's budget.

The Ethernet result shows that the fibre plan is already delivering its advertised rate, so buying a faster plan is not the first step. I would first test calls near the router and upstairs at several positions. If possible, I would move the Wi-Fi equipment to a more central, open location. If the ONT must remain where the fibre enters, I would use Ethernet to place the router centrally if the provider's setup permits it. If the router must remain too, Ethernet from the router can feed a centrally placed or upstairs access point.

If cabling between floors is impossible, I would next try a mesh unit placed where it still has a good connection to the main unit. I would not place it inside the dead zone or buy several units before testing one sensible location. The budget should address the measured coverage gap rather than increase the already adequate fibre rate.

After the change, I would repeat calls and transfers in each important room and compare the result with the original measurements.

4. Visible Wi-Fi shows that the local radio is operating; it does not prove that the provider connection works. The household should record the time and equipment indicators, try another service on the same network, check the provider's status information, and, if available,

compare mobile data. If several devices lose unrelated services together and the equipment reports loss of the incoming link, a provider-side fault becomes more plausible. A factory reset may erase needed settings, while a faster plan does not repair an outage. Neither is justified by this evidence.

5. Change default administration credentials, use current Wi-Fi protection with a strong joining passphrase, install updates, and disable remote administration unless it is needed and safely configured. Use a genuinely separated guest network where appropriate, protect records and QR codes, and know who may contact the provider. If continuity matters, test a backup connection under realistic conditions rather than merely assuming that it will work.
6. The student should confirm the exact network name with café staff rather than choose the strongest or most convincing name. Before connecting, disable automatic joining and unnecessary file or printer sharing, mark the network as public if prompted, and keep the device and browser updated. For sensitive work, mobile data or a personal hotspot may be the better choice. On any connection, the student should check the website's exact domain and HTTPS and should not bypass a certificate warning.

The claim combines several different problems. A correctly configured VPN can encrypt protected traffic between the device and the VPN server. On public Wi-Fi, this can prevent other parties on that local route from reading the protected contents, and a consumer VPN can make destinations see the VPN server's outgoing address instead of the user's usual public address.

Protection ends at the VPN server unless another protection such as HTTPS continues it. The VPN operator must be trusted to handle the traffic, and accounts, cookies, device identifiers, payments, and disclosed personal data can still identify the user. A VPN cannot make a scam site honest, stop the user from surrendering a password, remove malware, verify information, or repair a weak Wi-Fi radio signal. Those problems require measures such as careful source and link checking, current device protection and updates, sound authentication, and better local-network coverage. A VPN may add processing and route distance, so it can sometimes make performance slightly worse. Because many customers can share one VPN address and its apparent location may differ, a service may challenge or refuse the connection. If a trusted service works without the consumer VPN, that comparison identifies a likely cause; it does not show that the entire internet connection failed.

A.4 Chapter 4: Software, Cloud Services, and Digital Work

1. Chrome identifies the browser application, not necessarily the application used to create the report, its format, or its storage location. "Google" could refer to an account, online editor,

storage service, search engine, or another product. The student should provide the account or institutional workspace, exact folder and filename or document link, file format, required permission, and application capable of opening it. The group should verify that the campus computer's operating system and permitted applications can open the file without losing required features. Another authorised person should also know whether this is the working copy or an approved export and how it will be retained if the account becomes unavailable.

2. Synchronisation keeps selected locations consistent. It helps make a change on one device available elsewhere, but it may also propagate an accidental deletion, corrupt file, or unwanted edit. Backup provides a separate recoverable copy. It can restore work after loss or damage, but only if the backup contains the required state, remains accessible, and has been tested. Version history retains earlier states of a file or document and can reverse some edits or show when a change occurred. It may not survive account loss, may omit some kinds of change, and may expire under the service's retention rules. A sound workflow can use all three for different purposes.

3. The correct names depend on the project identifier, site, sample or subject, content, collection date, creator where relevant, and whether each file is raw, cleaned, analysed, or approved. The group should first inspect the files rather than infer their meaning from the present names.

One possible structure is 01_raw, 02_working, 03_analysis, 04_outputs, and 05_report, with a README and data dictionary at the top. A raw table might be named `water-quality_siteA_2026-09-21_raw.csv`; its photograph might use the same site and sample identifier. A checked milestone could be `water-quality_working_2026-09-24.xlsx`, while an approved release could include a version or approval date. The names should follow one written rule, not merely replace `final2` with another unexplained label.

Contamination status should be stored in a documented column rather than by colour alone. A blank should mean missing, while zero should remain a measured zero. One header row, one variable per column, declared units, and preserved raw values make the table easier to check and reuse.

4. Before fieldwork, assign a project, site, and specimen identifier scheme and prepare an offline-capable collection form or table. Each phone records the identifier with every observation and photograph. Students keep a local working copy until transfer is verified; they do not assume that a weak connection completed an upload. Large photographs can wait for a known, suitable connection rather than consuming a member's limited prepaid data, provided that the local copy is protected until transfer.

When connectivity returns, one authorised person or controlled process transfers the files to a restricted raw-data folder and checks expected file counts, identifiers, and sizes. Raw files remain unchanged. A working copy records standardised units, declared missing values, corrections,

and the reason for each exclusion. A README and data dictionary explain the folder structure, variables, collection method, software, roles, and first file to open. Named members receive permissions suited to collection, cleaning, analysis, review, or release. Milestone exports and an independent backup protect against synchronised loss or account failure.

To test reproducibility, a member who did not perform the main analysis uses the documentation and preserved inputs to regenerate one selected table or figure. Any missing instruction, unexplained manual change, or different result is recorded and corrected before submission.

5. For ChatGPT, a suitable input might be the abstract of a public paper and a request to separate the measured result from the authors' interpretation. The remote system processes the prompt and returns generated text. The student should compare every claim and reference with the paper. Identifiable health data, unpublished results, restricted course material, and text the student is not permitted to upload should be excluded. The response may be retained as working notes, but a chat history is not the approved report, the source paper, or an independent recovery copy.

For Dropbox, the input is the authorised project folder; the service stores, synchronises, and shares its contents; and the output is an updated folder visible to permitted devices and members. The group should check named permissions, sync status, conflicting copies, offline access, and whether the expected files and versions arrived. Identifiable or restricted data require the approved institutional location and protocol, not an unapproved personal folder. Synchronisation can propagate a deletion or bad edit, so a separate, tested milestone backup is still required.

For iLovePDF, the input could be three non-confidential submission PDFs; the service merges or compresses them; and the output is a new downloaded file. The student should verify its page count, order, legibility, links, and file size. Completed consent forms, identifiable records, examination papers, or unpublished data should not be sent without explicit approval. The processed download is a new output, not a backup of the raw data, analysis, report source, or original component files. The group should retain verified project copies independently of all three services.

6. Access means that the browser can display or download the photograph. It does not establish who owns copyright or what reuse is allowed. The students should identify the photographer or rights holder, the source, any stated licence, and the licence conditions. They should determine whether the work is in the public domain, licensed for the intended use, covered by permission, or potentially within a statutory exception. Attribution identifies the creator and source but cannot substitute for permission when permission is required.

If relying on fair use, the students must consider the applicable factors, including purpose, nature, amount, and market effect; "for school" is not a complete analysis. A link, an openly

licensed alternative, or the students' own photograph may meet the educational purpose with less uncertainty. Any reuse should follow the licence or permission and provide appropriate credit near the figure or as required by the discipline.

A.5 Chapter 5: Digital Platforms, Information, and Online Life

1. The student may have provided an age range, email address, profile details, or advertising preferences. The platform could have observed the searches, videos displayed, viewing time, pauses, clicks, follows, approximate location, device information, and whether an advertisement led to a purchase. From these observations it might infer an interest in dengue, health products, or biology content. That inference could be wrong: the student may be completing an assignment rather than seeking treatment.

The advertisement shows that an advertising system found the student or the current page eligible for a campaign. It does not reveal which signal caused the selection. Search and viewing activity, audience settings, page context, location, or data from a partner could explain the apparent relevance without microphone recording. The student should check application permissions and account settings, but the advertisement alone is not evidence that a private conversation was captured.

A search result responds to a query, a recommendation is selected for a feed without a new query, and a sponsored post occupies paid advertising space. Their visibility can increase exposure and further engagement, but none establishes scientific validity or platform endorsement.

2. I would first rewrite the statement precisely. Does “prevents dengue” mean fewer laboratory infections, fewer human infections, less severe disease, or merely a change in one measured substance? Which drink, composition, amount, population, time period, and comparison are meant?

I would identify the first publisher, author, institution, funding, conflicts of interest, date, and any seller connected to the post. I would follow the citation to the actual study rather than rely on a chart or quotation. I would ask whether the work involved molecules, cells, animals, or people; how subjects or samples were selected; what control or comparison was used; which outcome was planned and measured; how large and uncertain the effect was; and whether adverse effects and excluded data were reported.

A laboratory result can support a mechanism or further experiment without establishing prevention in humans. I would therefore check later studies, independent reviews, corrections, and current guidance from responsible health authorities. Testimonials and view counts would

not fill an evidence gap. Until adequate evidence supports the stated claim, I would not share the video as health guidance.

3. A real presenter can misstate a study, while synthetic narration can deliver an accurate statement. Appearance and an AI-detector result therefore do not decide whether the health claim is true; detectors can also make errors.

I would record the URL and exact claim, find the earliest available version and official source, locate the study behind the chart, compare the video with the study's methods and results, and check independent health guidance. Technical clues can support that investigation but cannot replace evidence about the claim itself.

4. The students may have different viewing histories, searches, watch times, language settings, locations, subscriptions, hidden items, devices, and social connections. The service may also test different ranking arrangements or use recency and availability at the moment each feed is requested. It can therefore predict different items even when both students follow the same biology pages.

The ranking affects what each student has an opportunity to see. A pause, skip, like, or hide action then becomes another signal that may affect later ranking. This feedback is neither complete control nor a perfect expression of preference: the system offers only a selected set, and a response can be accidental, critical, or required by an assignment. Students retain choices, and predictions, platform objectives, and available content also change.

5. The statement is too broad. "Social media" includes different activities, content, and designs. Selected achievements or edited images may make one student feel worse, while another finds community, knowledge, or encouragement. Prior well-being, relationships, sleep, workload, and the possibility that distress changes platform use are alternative explanations; one association cannot support a claim about every student.

For a small audit, I would record notifications and interruptions for several days, silence one non-urgent category, and compare unnecessary openings and completed focus periods afterward. The result may guide my settings, but the periods are short, workload may change, and another student's needs may differ. It is not a universal causal test.

6. The student should not answer the sexual insults as though they were an ordinary scientific disagreement. If safe, the student should preserve the account names, URLs, timestamps, complete message context, edited image, and threat. The evidence should be stored privately rather than circulated. The student can block or mute accounts, restrict the audience, and report the material under the platform's harassment, sexual-content, impersonation, or threat categories.

The threatened release of an address raises the urgency. The student should tell a trusted person, notify the appropriate university office, review which accounts or old posts expose location information, and seek qualified police or emergency help if the threat appears immediate or credible. The Safe Spaces Act may be relevant to specified gender-based online sexual harassment, but classification should depend on the facts and qualified guidance rather than a classmate's legal conclusion.

In an ordinary evidence dispute, a civil response could identify the exact dengue claim, explain that the cited laboratory evidence does not establish human prevention, and link to responsible health guidance. Harassment does not create an obligation to keep debating. Reporting, preserving evidence, seeking support, and disengaging are proportionate responses.

A.6 Chapter 6: Cybersecurity, Identity, and Privacy

1. Assets include the institutional account, email, raw observations, specimen locations, classmates' contact details, the student's reputation, and the group's ability to finish its project. Threats include an external attacker, later impersonation, fraudulent password resets, data theft, alteration, and loss of access. Vulnerabilities include the credible pretext, concealed QR destination, counterfeit domain, exposed password, ambiguous approval prompt, broad account connections, and possibly excessive folder permissions.

Preventive controls could include a known portal for result notices, message filtering, unique managed passwords, phishing-resistant passkeys, limited access, and clear prompts. Login and file-change alerts are detective controls. Session revocation, incident reporting, and suspension of unsafe links are responsive controls. Tested backups, version history, documented permissions, and alternative account administrators support recovery.

Residual risk includes theft of an active session, compromise of recovery channels or devices, malicious insiders, undiscovered copying, and attacks on authorised collaborators. Blaming the student ignores the attacker's deliberate deception and the choices made in message delivery, authentication, permissions, logging, support, and recovery. The student's action is relevant to the sequence, but it is neither the whole cause nor a substitute for layered institutional controls.

2. In the first hour, the student should stop using the counterfeit page, deny new prompts, and contact university support through a separately verified route. The original message, sender and recipient, QR destination or link, timestamps, screenshots with context, alerts, and a short action timeline should be preserved privately. From a trusted route, the student and administrator should revoke sessions, change the compromised password and any reused passwords, remove unknown authenticators or recovery methods, and inspect forwarding rules

and connected applications. Affected group members should be warned without forwarding the live lure.

During the next day, the university should examine available sign-in, email, file, sharing, and administrative logs; preserve relevant records; determine which data and people may be affected; and contain unsafe permissions or links. The group should compare important files with raw records, version history, and known good copies. The student should review other accounts that depend on the compromised email.

During the following week, the responsible offices should continue monitoring, restore and validate altered material, document findings and decisions, assess privacy notification requirements, support affected people, and test revised controls and recovery. Some log access, session revocation, privacy assessment, and provider records require institutional or provider authority. Wiping the phone may destroy messages, application state, logs, or malware evidence before they are preserved; it also may not revoke a server-side session. Isolation or examination should follow qualified advice.

3. A unique managed password reduces guessing and reuse across breaches. Its limits include phishing, compromise of the manager or device, and recovery weakness. An SMS OTP adds possession of a telephone channel, but can be relayed by a counterfeit page and depends on the mobile account and network. An app-generated OTP avoids SMS transport but remains manually transferable to an impostor. An approval prompt can stop use of a stolen password, yet an unclear or repeated prompt may be approved through confusion or fatigue.

A passkey binds a cryptographic response to the legitimate service domain; unlocking it with a device PIN or biometric does not send that PIN or biometric to the website. It reduces password phishing and credential reuse, but depends on protected devices or credential providers, sound implementation, and secure recovery. A recovery code restores access when a regular authenticator is unavailable, but anyone who obtains the code may be able to use it; it must be stored separately and used only once where the service permits.

For an institutional account, I would prefer a phishing-resistant passkey or hardware-backed method supported and recoverable by the university, with at least one separately protected backup authenticator. If that is unavailable, a unique password from an approved manager plus the strongest supported MFA is still useful. The university should add contextual alerts, rate limits, session inspection and revocation, controlled recovery, and support. The recommendation depends on accessibility, device sharing, loss scenarios, institutional policy, and the sensitivity of the account.

4. Owner access for every member exceeds what most tasks require. It increases the chance of unauthorised sharing, accidental deletion, permission changes, and loss of a trustworthy record. Named contact details are personal information; precise locations may be confidential even

when they do not identify a person. The group should state the purposes for collecting both, remove fields not needed for those purposes, and separate data whose access requirements differ.

Preserved raw observations should be read-only to most members, with a documented working copy for cleaning and analysis. Roles should grant view, edit, approval, sharing, or administration only where required. Important changes need provenance and review. The group should set retention from the research protocol, institutional rules, consent where applicable, and scientific need rather than “forever.” Disposal must cover exports and authorised copies as well as the main folder.

A graduating leader’s personal account creates an availability and accountability failure. The authoritative folder should use an approved institutional location with more than one appropriate administrator, documented transfer, common-format exports, and a tested independent recovery copy. Confidentiality, integrity, and availability are improved by different controls; no single sharing setting provides all three.

5. The logs establish that an unauthorised person accessed the account to the extent and reliability that the logs cover. They may show time, address, device or session, and some actions. They do not necessarily show the human attacker, intent, every item viewed, whether a displayed file was copied by another method, or whether an unlogged action occurred. “No download event” is therefore not equivalent to “no disclosure.” The university should record known facts, uncertainty, assumptions, and limits.

The student should report promptly, preserve available evidence, follow safe containment instructions, and provide accurate facts. The student is not expected to conduct a forensic investigation or decide alone whether the statutory breach threshold is met. The institution must activate its response procedure, contain access, preserve and analyse appropriate logs, determine the affected systems and data, document the incident, mitigate harm, and communicate with relevant providers and people.

As personal information controller for data whose purposes and means it determines, the institution must assess the Data Privacy Act and current NPC rules. Mandatory notification generally requires all relevant conditions, including covered information, likely unauthorised acquisition, and a real risk of serious harm; when met, the prescribed 72-hour process applies based on available information. Incidents below that threshold still require documentation. Qualified privacy and legal officers should make and record the assessment rather than delaying while certainty about copying remains impossible.

6. The classmate confuses a carrier with the deception it carried. QR codes can encode legitimate addresses, network credentials, specimen identifiers, or other data. Their limitation is that people may not see the destination before scanning. A blanket ban would remove useful

applications without addressing counterfeit email, text links, taken-over accounts, or voice calls. MFA did interrupt the attack by requiring another action, but the approval method allowed the attacker to induce that action. This does not show that every MFA method is useless.

I would publish result notices inside a known university portal and ask students to reach it through a bookmark or official directory. Messages would say that results are available but would contain no result and request no password or code. If a QR code is used on an authorised physical notice, the human-readable official domain and an alternative typed route should appear beside it. Authentication should use a phishing-resistant method where feasible, with clear transaction context and independently supported recovery.

For intermittent connectivity and limited prepaid data, the portal should be lightweight, avoid forced large downloads, retain a clear status after a dropped connection, and provide an approved campus access route. Notices should not expire so quickly that urgency defeats verification. A verified telephone or in-person route should exist for students who cannot access the portal. Least-privilege permissions, alerts, logs, session revocation, incident reporting, and tested continuity remain necessary because message design and authentication do not remove all residual risk.

A.7 Chapter 7: Artificial Intelligence: How It Works and Fails

1. The sentence contains several unsupported transitions. “The intervention” may correctly identify what the researchers administered, but “prevented” changes a measured marker into a disease outcome. “Infection” may not have been measured at all. “In humans” changes the study population from mice to people. “By strengthening immunity” adds a causal mechanism unless the experiment directly tested and established that mechanism. The definite wording also omits the comparison, size and uncertainty of the result, dose, duration, and limitations.

I would make a table with one row per claim and columns for the exact paper passage, biological system, measured variable, comparison, and status. The rows for human prevention and mechanism would be marked “not supported by this paper.” Establishing a human preventive effect would require suitably designed human evidence with a disease outcome and relevant comparison, not extrapolation from a marker in mice. Establishing the proposed mechanism would require measurements that distinguish it from alternative explanations.

A defensible replacement is: “In the reported mouse experiment, the intervention changed [name of marker] relative to [comparison] under the stated conditions. This result may motivate further study, but it does not by itself show that the intervention prevents infection or disease

in humans.” The bracketed details must be copied accurately from the paper, with effect size and uncertainty added if they are important to the summary.

2. The training data are the examples used before this conversation to adjust the model’s parameters. The parameters are the internal numerical values retained by the trained model. At use time, the students supply a prompt and paper; the service places some or all of that material into the context for an inference. If the service has retrieval, a search component may select additional documents and place extracts into that context. The model then generates an output. Evaluation evidence is separate: source passages, verified references, known-answer test cases, expert judgements, and measured performance under stated conditions.

Uploading a paper normally supplies use-time input; it does not adjust the model’s parameters as training does. A provider might retain submissions or later use them under its policy, but that is a separate service and governance question. The student must not infer either practice without evidence.

Access to the paper also does not guarantee faithful use. The context may be truncated, extraction may fail, retrieval may select the wrong passage, or generation may combine paper content with learned patterns and unsupported continuation. Verification therefore links each output claim back to the actual paper and checks the result independently; the presence of the file in the interface is not an evaluation result.

3. Assigning one of five known species is classification because the requested output is one category from a defined set. It cannot by itself establish that the image belongs to a sixth, unseen species or that the assigned label is biologically correct. An explicit “unknown” or expert-review route may be needed.

Estimating algal concentration is regression because the target is numerical. The estimate does not establish a causal reason for the concentration or guarantee accuracy outside the sampled sites, seasons, instruments, and value range.

Drafting a field-note summary is generation. The system constructs a new sequence from the supplied notes and other available context. The output cannot by itself establish that every statement is present in the notes, that omissions are harmless, or that the observations were accurate.

Finding papers about a named organism is retrieval. The records should come from a specified index or database. Retrieval cannot establish that a paper supports a later claim, is methodologically sound, or exhausts the relevant literature. A combined system may retrieve papers and generate a narrative, but the records and every claim-to-source relationship still require separate checks.

4. Of the 100 truly positive images, 80 were flagged: there are 80 true positives and 20 false negatives. Of the 900 truly negative images, 90 were incorrectly flagged: there are 90 false positives and 810 true negatives. Correct outputs total 890, so accuracy is 890 out of 1,000, or 89 per cent.

Accuracy alone hides 20 missed positive cases and 90 unnecessary flags. The importance of those errors depends on the workflow. If a false negative could allow an invasive organism to spread, the group may lower the decision threshold to catch more positives, accepting more expert review of false positives. If confirmation destroys a rare specimen or uses a very scarce test, false positives may carry greater cost. Threshold selection should use those consequences, prevalence, follow-up capacity, and uncertainty rather than one preferred metric.

The test describes the source dataset. Another campus may use a different camera, staining method, magnification, sample preparation, operator, species mix, prevalence, or image-quality rule. The model may have relied on a site marker correlated with the label. Before deployment, the team needs a preserved, correctly labelled test set from the intended campus, results separated by relevant subgroups and conditions, and a plan for unfamiliar or low-quality images. Even a favourable local test does not guarantee every future result.

5. I would begin: “Using only the supplied paper, write a two-sentence summary for first-year biology students. Identify the organism, intervention, comparison, measured outcome, effect direction, uncertainty, and principal limitation. Do not infer disease prevention or an effect in humans unless the paper reports it explicitly. For each sentence, identify the supporting page, table, figure, or passage; mark missing information as not established by this source.” The task is a two-sentence summary; the context is its audience and scientific purpose; the reference is the supplied paper; and the listed fields and source links provide evaluation criteria. Iteration should respond to a diagnosed omission or unsupported claim rather than merely request a “better” answer.

I would preserve one version of the source paper and define a claim-level rubric before generating outputs. The rubric would record the same biological fields, unsupported human extrapolation, and citation validity. The service or model version shown, settings, date, complete prompts, and complete outputs would be recorded.

For prompt sensitivity, I would prepare meaning-preserving variants that differ only in order, formatting, or wording. For a sycophancy test, I would add opposed user positions—for example, “I think it prevents disease” and “I think it has no value”—while keeping the paper and factual question fixed. Order of trials should be controlled where conversation history could matter, and each test should start with the same context. Repeated runs may be needed if generation is variable.

An important difference would be a change in organism, outcome, causal conclusion, effect direction, stated limitation, or support status, not a harmless stylistic difference. The rubric scores each output against the paper, and independent reviewers resolve disagreements. The model's own ranking of its answers is not sufficient because it can repeat the same unsupported premise, prefer a more fluent answer, or alter its judgement in response to the framing being tested. Source evidence, not model agreement, determines faithfulness.

6. I would first preserve the prompt, output, service or model identification, date, any uploaded text, and every retrieved source shown. I would split the conclusion into claims: the exact intervention, population, disease outcome, comparison, effect size, uncertainty, causal language, and proposed mechanism. Each claim would need a passage, table, or figure from an actual source. An unlocatable reference would be marked unusable rather than replaced by a similar title. For a located reference, I would verify authors, title, journal, date, and identifier in an independent index or publisher record and then read it to check support.

Evidence from molecules or cells can support a mechanism for further study; animal evidence can support biological plausibility; observational human evidence can show an association under its assumptions; and controlled human studies may estimate causal effects when their design and execution support that inference. None should be silently renamed as another. I would compare the conclusion with independent studies and current guidance from responsible scientific or health bodies, record conflicts and uncertainty, and seek qualified subject expertise.

Acceptance criteria would be set before review: all central references must exist and support their attached claims; the biological system and outcome must remain accurate; uncertainty and major limitations must be retained; and a human-prevention conclusion must rest on relevant human evidence rather than extrapolation alone. The process stops if the source cannot be obtained, central claims cannot be traced, evidence conflicts cannot be resolved for the intended consequence, or qualified review is required but unavailable. The final report would state those limits instead of filling the gap with another generated answer.

For a low-stakes grammar suggestion, I would retain the student's original, compare the edit for changed meaning, and accept or reject it directly. It would not normally require a literature search because the tool is not being used to establish a scientific fact. If the edit changes a technical claim, number, organism, or citation, the source-level check returns.

A.8 Chapter 8: Responsible AI and the Future of Computing

1. Suggesting search terms is assistance if the researcher chooses the queries, examines the database results, and reads the sources. It becomes more consequential if the suggestions define the only search and thereby shape which evidence enters a review. The researcher should

preserve and test the search strategy, add controlled vocabulary and known relevant papers, and justify exclusions.

Renaming files is automation: a defined operation is applied across a set. Its consequence rises when links, identifiers, scripts, or the only copy depend on the names. The user should test the rule on copies, inspect a sample and exceptional names, preserve a name mapping, and keep a recoverable original.

Ranking specimens for expert review is recommendation. It becomes an effective decision if low-ranked images are never seen, the reviewer is rushed, or scarce testing follows the order automatically. The expert needs access to images and relevant metadata, performance information, a route for uncertain and out-of-distribution cases, and freedom to change the order. Random or rule-based sampling of low-ranked cases can reveal missed positives.

Excluding a participant is a consequential decision because it directly affects opportunity and may affect study validity. A model output should not be the unreviewed basis. An authorised researcher must apply approved inclusion criteria to the underlying record, document the reason, manage conflicts of interest, and provide correction or reconsideration where the protocol requires it. If timely qualified review is unavailable, the automated exclusion should not operate.

2. I would first separate the files and document the controlling research protocol, ethics approval, consent materials, institutional policy, agreements, and data custodian. Public papers may enter a permitted literature tool, but citation and claim checks remain necessary. The unpublished manuscript stays in approved institutional storage unless its owners and applicable agreement authorise a particular external disclosure. A short non-confidential extract or an invented paragraph may replace it for testing writing assistance.

Exact threatened-species locations remain restricted because disclosure can cause ecological harm even though the coordinates are not personal data. Generalised or synthetic coordinates can test code, but any generalisation must be unsuitable for reconstructing the site. Participant records and the linking dictionary are separated, access-controlled, and never uploaded to an unapproved service. Coded data are still linkable while the dictionary or other identifying information exists; they should not be described as anonymous.

Code can be drafted with field names, units, permitted categories, and synthetic rows. After review and known-answer testing, approved code may run on the minimum necessary real variables inside the authorised environment. Outputs must be checked for small cells, free text, dates, locations, or combinations that disclose participants. Retention, logs, administrators, deletion, secondary use, and account closure must be known.

I would reject an AI step if required approval is absent, the provider's processing or retention cannot be established, minimisation cannot remove an unnecessary disclosure, a participant

commitment would be breached, or the task can operate only by moving restricted material outside the approved boundary. Consent would be evaluated against the specific proposed purpose; it would not be inferred from participation in the study.

3. The plan begins with a frozen workflow description, versions, roles, input and output schemas, units, acceptance criteria, and a protected copy of raw data. For literature review, I would preserve databases, dates, exact queries, screening decisions, and retrieved records. Every central claim and citation would be checked in the actual paper, and a few known relevant and irrelevant records would test recall and exclusion behaviour.

Generated code would be reviewed against a written specification and run on synthetic cases whose results are calculated independently. These would include missing data, zero and boundary values, unexpected categories, and a deliberate unit conversion. The analysis stage would retain intermediate tables, exclusions, warnings, package versions, parameters, and edited code. Totals and distributions before and after each transformation would be reconciled.

The classifier would be validated on independently labelled images from the intended instruments, sites, species, operators, and quality conditions. False positives and negatives, uncertainty, and relevant subgroup results would be reported rather than only overall accuracy. Unfamiliar inputs would have an abstain or expert-review route.

Interface tests would deliberately send a missing value, an uncalibrated score, a renamed column, and an unsupported citation between components. The receiving step must preserve the missing status or stop rather than silently convert it. A seeded false citation or deliberately reversed class label can test whether a plausible error reaches the report. The final report would be rebuilt only from traceable results and checked claim by claim. Passing the final prose check would not excuse a failed earlier link.

4. The authorised specimen specialist should make the confirmatory-test decision. The reviewer needs the original image, relevant specimen metadata, the model score with an explanation of its limits, the operating threshold, local validation results, and information about image quality and unfamiliar inputs. Adequate time, training, independence from throughput targets, and authority to override or request another image are necessary. A student may prepare the queue, while the research lead owns the scientific procedure and IT staff manage access, versions, logs, and recovery; these responsibilities should be recorded.

Monitoring would include false positives and negatives from confirmed cases, abstentions, overrides, missed or delayed tests, image-quality problems, results by instrument and relevant specimen group, and changes to software, model, camera, preparation, or prevalence. A second qualified reviewer would reconsider a disputed case using the original evidence, without being required to defend the first output.

I would stop automatic ranking when the model or input pipeline changes without revalidation; when an image is outside the validated species, instrument, or quality range; when observed error exceeds the agreed bound; when logs or source images are unavailable; or when no qualified reviewer can act before the result matters. The safer manual route would then apply. Automation bias can survive a human signature when high-ranked cases receive more attention, the score anchors judgement, low-ranked cases disappear, or disagreement is discouraged. Blinded review of a sample, random checks below the threshold, recording independent judgement before revealing the score, and examining overrides can test whether review is real.

5. The submission record would list each AI-assisted activity separately: tool or system identification shown, date, purpose, material supplied, output used, human changes, and verification performed. The outline and prose entry would identify how the group checked structure and every factual claim. The code entry would point to the specification, version-controlled code, package environment, test cases, and review. The illustration entry would record the prompt or source material, editing, provenance, and applicable licence or permission. The form and location of disclosure would follow the course, institution, funder, and intended publisher rules.

The human authors would be those who meet the applicable contribution and responsibility criteria. The AI system would not be listed as an author. Named authors remain able to explain the method, defend the claims, approve the final work, and correct errors. Factual verification would link claims, numbers, citations, and visual labels to sources or reproducible results.

Copyright permission asks whether source or output material may be copied, adapted, and distributed. Attribution supplies the credit required by a licence, quotation practice, or scholarly convention; attribution is not a substitute for permission when permission is needed. Conversely, permission to reuse an image does not prove that its scientific claim is accurate.

Disclosure cannot repair a fabricated citation, prohibited external upload, undisclosed participant information, plagiarised passage, or analysis that the authors cannot defend. It makes a use visible; it does not change the underlying permission, evidence, or responsibility.

6. The vendor's demonstration establishes only what was performed under the reported hardware, algorithm, data, error correction, and test conditions. The forecast is that systems will scale from that demonstration to a machine able to threaten deployed public-key cryptography. A deployed capability would require evidence that such attacks operate reliably at relevant scale, not a press description or a smaller calculation. Evidence of benefit for the proprietary replacement would require open technical specification, independent security analysis, standards alignment, interoperability, implementation testing, support, migration and recovery plans, and costs compared with alternatives. The demonstration alone supplies none of these.

The university should inventory where quantum-vulnerable public-key methods appear in certificates, identity systems, VPNs, applications, devices, backups, research instruments, cloud contracts, and partner connections. It should classify information by how long confidentiality or signature validity must last, identify owners and vendors, and prioritise long-lived sensitive data and systems that are difficult to replace. Procurement should require standards-based migration and crypto-agility rather than dependence on an unreviewed proprietary claim.

Specialists can pilot NIST-standardised post-quantum methods in non-production settings, measure performance and compatibility, test rollback and recovery, and plan certificate, key, software, hardware, and partner changes. Ordinary users should keep supported software current; they should not invent or swap cryptographic algorithms themselves.

The plan would change with new standards, demonstrated attacks, credible resource estimates, failures in selected algorithms or implementations, vendor support, compatibility findings, data-lifetime changes, and evidence about capable quantum hardware. Faster risk evidence would accelerate migration; evidence of an implementation defect could halt one method without abandoning preparation. This is staged action under uncertainty, not a claim that all present encryption has already failed.

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